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METHOD FOR ADAPTIVE CONTROL SCHEMES FOR SURGICAL NETWORK CONTROL AND INTERACTION

Abstract

A method for adaptive control of surgical network control and interaction is disclosed. The surgical network includes a surgical feedback system. The surgical feedback system includes a surgical instrument, a data source, and a surgical hub configured to communicably couple to the data source and the surgical instrument. The surgical hub includes a control circuit. The method includes receiving, by the control circuit, information related to devices communicatively coupled to the surgical network; and adaptively controlling, by the control circuit, the surgical network based on the received information.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 18/237,224, titled METHOD FOR ADAPTIVE CONTROL SCHEMES FOR SURGICAL NETWORK CONTROL AND INTERACTION, filed Aug. 23, 2023, now U.S. Patent Application Publication No. 2024/0023966, which is a continuation application claiming priority under 35 U.S.C. § 120 to Ser. No. 17/676,611, titled METHOD FOR ADAPTIVE CONTROL SCHEMES FOR SURGICAL NETWORK CONTROL AND INTERACTION, filed Feb. 21, 2022, which issued on Oct. 17, 2023, now U.S. Pat. No. 11,786,251, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 16/209,465, titled METHOD FOR ADAPTIVE CONTROL SCHEMES FOR SURGICAL NETWORK CONTROL AND INTERACTION, filed Dec. 4, 2018, which issued on Apr. 19, 2022 as U.S. Pat. No. 11,304,699, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/773,778, titled METHOD FOR ADAPTIVE CONTROL SCHEMES FOR SURGICAL NETWORK CONTROL AND INTERACTION, filed Nov. 30, 2018, U.S. Provisional Patent Application No. 62/773,728, titled METHOD FOR SITUATIONAL AWARENESS FOR SURGICAL NETWORK OR SURGICAL NETWORK CONNECTED DEVICE CAPABLE OF ADJUSTING FUNCTION BASED ON A SENSED SITUATION OR USAGE, filed Nov. 30, 2018, U.S. Provisional Patent Application No. 62/773,741, titled METHOD FOR FACILITY DATA COLLECTION AND INTERPRETATION, filed Nov. 30, 2018, U.S. Provisional Patent Application No. 62/773,742, titled METHOD FOR CIRCULAR STAPLER CONTROL ALGORITHM ADJUSTMENT BASED ON SITUATIONAL AWARENESS, filed Nov. 30, 2018, U.S. Provisional Patent Application No. 62/750,529, titled METHOD FOR OPERATING A POWERED ARTICULATING MULTI-CLIP APPLIER, filed Oct. 25, 2018, to U.S. Provisional Patent Application No. 62/750,539, titled SURGICAL CLIP APPLIER, filed Oct. 25, 2018, U.S. Provisional Patent Application No. 62/750,555, titled SURGICAL CLIP APPLIER, filed Oct. 25, 2018, U.S. Provisional Patent Application No. 62/729,183, titled CONTROL FOR A SURGICAL NETWORK OR SURGICAL NETWORK CONNECTED DEVICE THAT ADJUSTS ITS FUNCTION BASED ON A SENSED SITUATION OR USAGE, filed Sep. 10, 2018, U.S. Provisional Patent Application No. 62/729,177, titled AUTOMATED DATA SCALING, ALIGNMENT, AND ORGANIZING BASED ON PREDEFINED PARAMETERS WITHIN A SURGICAL NETWORK BEFORE TRANSMISSION, filed Sep. 10, 2018, U.S. Provisional Patent

Application No. 62/729,176, titled INDIRECT COMMAND AND CONTROL OF A FIRST OPERATING ROOM SYSTEM THROUGH THE USE OF A SECOND OPERATING ROOM SYSTEM WITHIN A STERILE FIELD WHERE THE SECOND OPERATING ROOM SYSTEM HAS PRIMARY AND SECONDARY OPERATING MODES, filed Sep. 10, 2018, U.S. Provisional Patent Application No. 62/729,185, titled POWERED STAPLING DEVICE THAT IS CAPABLE OF ADJUSTING FORCE, ADVANCEMENT SPEED, AND OVERALL STROKE OF CUTTING MEMBER OF THE DEVICE BASED ON SENSED PARAMETER OF FIRING OR CLAMPING, filed Sep. 10, 2018, U.S. Provisional Patent Application No. 62/729,184, titled POWERED SURGICAL TOOL WITH A PREDEFINED ADJUSTABLE CONTROL ALGORITHM FOR CONTROLLING AT LEAST ONE END EFFECTOR PARAMETER AND A MEANS FOR LIMITING THE ADJUSTMENT, filed Sep. 10, 2018, U.S. Provisional Patent Application No. 62/729,182, titled SENSING THE PATIENT POSITION AND CONTACT UTILIZING THE MONO-POLAR RETURN PAD ELECTRODE TO PROVIDE SITUATIONAL AWARENESS TO THE HUB, filed Sep. 10, 2018, U.S. Provisional Patent Application No. 62/729,191, titled SURGICAL NETWORK RECOMMENDATIONS FROM REAL TIME ANALYSIS OF PROCEDURE VARIABLES AGAINST A BASELINE HIGHLIGHTING DIFFERENCES FROM THE OPTIMAL SOLUTION, filed Sep. 10, 2018, U.S. Provisional Patent Application No. 62/729,195, titled ULTRASONIC ENERGY DEVICE WHICH VARIES PRESSURE APPLIED BY CLAMP ARM TO PROVIDE THRESHOLD CONTROL PRESSURE AT A CUT PROGRESSION LOCATION, filed Sep. 10, 2018, U.S. Provisional Patent Application No. 62/729,186, titled WIRELESS PAIRING OF A SURGICAL DEVICE WITH ANOTHER DEVICE WITHIN A STERILE SURGICAL FIELD BASED ON THE USAGE AND SITUATIONAL AWARENESS OF DEVICES, filed Sep. 10, 2018, U.S. Provisional Patent Application No. 62/721,995, titled CONTROLLING AN ULTRASONIC SURGICAL INSTRUMENT ACCORDING TO TISSUE LOCATION, filed Aug. 23, 2018, U.S. Provisional Patent Application No. 62/721,998, titled SITUATIONAL AWARENESS OF ELECTROSURGICAL SYSTEMS, filed Aug. 23, 2018, U.S. Provisional Patent Application No. 62/721,999, titled INTERRUPTION OF ENERGY DUE TO INADVERTENT CAPACITIVE COUPLING, filed Aug. 23, 2018, U.S. Provisional Patent Application No. 62/721,994, titled BIPOLAR COMBINATION DEVICE THAT AUTOMATICALLY ADJUSTS PRESSURE BASED ON ENERGY MODALITY, filed Aug. 23, 2018, U.S. Provisional Patent Application No. 62/721,996, titled RADIO FREQUENCY ENERGY DEVICE FOR DELIVERING COMBINED ELECTRICAL SIGNALS, filed Aug. 23, 2018, U.S. Provisional Patent Application No. 62/692,747, titled SMART ACTIVATION OF AN ENERGY DEVICE BY ANOTHER DEVICE, filed on Jun. 30, 2018, U.S. Provisional Patent Application No. 62/692,748, titled SMART ENERGY ARCHITECTURE, filed on Jun. 30, 2018, U.S. Provisional Patent Application No. 62/692,768, titled SMART ENERGY DEVICES, filed on Jun. 30, 2018, U.S. Provisional Patent Application No. 62/691,228, titled METHOD OF USING REINFORCED FLEX CIRCUITS WITH MULTIPLE SENSORS WITH ELECTROSURGICAL DEVICES, filed Jun. 28, 2018, to U.S. Provisional Patent Application No. 62/691,227, titled CONTROLLING A SURGICAL INSTRUMENT ACCORDING TO SENSED CLOSURE PARAMETERS, filed Jun. 28, 2018, to U.S. Provisional Patent Application No. 62/691,230, titled SURGICAL INSTRUMENT HAVING A FLEXIBLE ELECTRODE, filed Jun. 28, 2018, to U.S. Provisional Patent Application No. 62/691,219, titled SURGICAL EVACUATION SENSING AND MOTOR CONTROL, filed Jun. 28, 2018, to U.S. Provisional Patent Application No. 62/691,257, titled COMMUNICATION OF SMOKE EVACUATION SYSTEM PARAMETERS TO HUB OR CLOUD IN SMOKE EVACUATION MODULE FOR INTERACTIVE SURGICAL PLATFORM, filed Jun. 28, 2018, to U.S. Provisional Patent Application No. 62/691,262, titled SURGICAL EVACUATION SYSTEM WITH A COMMUNICATION CIRCUIT FOR COMMUNICATION BETWEEN A FILTER AND A SMOKE EVACUATION DEVICE, filed Jun. 28, 2018, and to U.S. Provisional Patent

Application No. 62/691,251, titled DUAL IN-SERIES LARGE AND SMALL DROPLET FILTERS, filed Jun. 28, 2018, U.S. Provisional Patent Application No. 62/665,129, titled SURGICAL SUTURING SYSTEMS, filed May 1, 2018, U.S. Provisional Patent Application No. 62/665,139, titled SURGICAL INSTRUMENTS COMPRISING CONTROL SYSTEMS, filed May 1, 2018, to U.S. Provisional Patent Application No. 62/665,177, titled SURGICAL INSTRUMENTS COMPRISING HANDLE ARRANGEMENTS, filed May 1, 2018, U.S. Provisional Patent Application No. 62/665,128, titled MODULAR SURGICAL INSTRUMENTS, filed May 1, 2018, U.S. Provisional Patent Application No. 62/665,192, titled SURGICAL DISSECTORS, filed May 1, 2018, U.S. Provisional Patent Application No. 62/665,134, titled SURGICAL CLIP APPLIER, filed May 1, 2018, U.S. Provisional Patent Application No. 62/659,900, titled METHOD OF HUB COMMUNICATION, filed on Apr. 19, 2018, U.S. Provisional Patent Application No. 62/650,898, filed on Mar. 30, 2018, titled CAPACITIVE COUPLED RETURN PATH PAD WITH SEPARABLE ARRAY ELEMENTS, U.S. Provisional Patent Application No. 62/650,887, titled SURGICAL SYSTEMS WITH OPTIMIZED SENSING CAPABILITIES, filed Mar. 30, 2018, U.S. Provisional Patent Application No. 62/650,882, titled SMOKE EVACUATION MODULE FOR INTERACTIVE SURGICAL PLATFORM, filed Mar. 30, 2018, U.S. Provisional Patent Application No. 62/650,877, titled SURGICAL SMOKE EVACUATION SENSING AND CONTROLS, filed Mar. 30, 2018, U.S. Provisional Patent Application No. 62/649,302, titled INTERACTIVE SURGICAL SYSTEMS WITH ENCRYPTED COMMUNICATION CAPABILITIES, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,294, titled DATA STRIPPING METHOD TO INTERROGATE PATIENT RECORDS AND CREATE ANONYMIZED RECORD, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,300, titled SURGICAL HUB SITUATIONAL AWARENESS, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,309, titled SURGICAL HUB SPATIAL AWARENESS TO DETERMINE DEVICES IN OPERATING THEATER, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,310, titled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,291, titled USE OF LASER LIGHT AND RED-GREEN-BLUE COLORATION TO DETERMINE PROPERTIES OF BACK SCATTERED LIGHT, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,296, titled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL DEVICES, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,333, titled CLOUD-BASED MEDICAL ANALYTICS FOR CUSTOMIZATION AND RECOMMENDATIONS TO A USER, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,327, titled CLOUD-BASED MEDICAL ANALYTICS FOR SECURITY AND AUTHENTICATION TRENDS AND REACTIVE MEASURES, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,315, titled DATA HANDLING AND PRIORITIZATION IN A CLOUD ANALYTICS NETWORK, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,313, titled CLOUD INTERFACE FOR COUPLED SURGICAL DEVICES, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,320, titled DRIVE ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,307, titled AUTOMATIC TOOL ADJUSTMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/649,323, titled SENSING ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, filed Mar. 28, 2018, U.S. Provisional Patent Application No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, U.S. Provisional Patent Application No. 62/611,340, titled CLOUD-BASED MEDICAL ANALYTICS, filed Dec. 28, 2017, and U.S. Provisional Patent Application No. 62/611,339, titled ROBOT ASSISTED SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of each of which is herein incorporated by reference in its entirety.

BACKGROUND

[0002] The present disclosure relates to various surgical systems. Surgical procedures are typically performed in surgical operating theaters or rooms in a healthcare facility such as, for example, a hospital. A sterile field is typically created around the patient. The sterile field may include the scrubbed team members, who are properly attired, and all furniture and fixtures in the area. Various surgical devices and systems are utilized in performance of a surgical procedure.

[0003] Furthermore, in the Digital and Information Age, medical systems and facilities are often slower to implement systems or procedures utilizing newer and improved technologies due to patient safety and a general desire for maintaining traditional practices. However, often times medical systems and facilities may lack communication and shared knowledge with other neighboring or similarly situated facilities as a result. To improve patient practices, it would be desirable to find ways to help interconnect medical systems and facilities better.

SUMMARY

[0004] In one aspect the present disclosure provides a method for adaptive control of surgical network control and interaction. The surgical network comprises a surgical feedback system. The surgical feedback system comprises a surgical instrument, a data source, and a surgical hub configured to communicably couple to the data source and the surgical instrument. The surgical hub comprises a control circuit. The method comprises receiving, by the control circuit, information related to devices communicatively coupled to the surgical network; and adaptively controlling, by the control circuit, the surgical network based on the received information.

[0005] In another aspect the present disclosure provides a method for adaptive feedback and control of a surgical system. The surgical system comprises a surgical instrument, a data source, and a surgical hub configured to communicably couple to the data source and the surgical instrument. The surgical hub comprises a control circuit. The method comprises: receiving, by the control circuit, information related to devices communicatively coupled to the surgical system; and adaptively adjusting, by the control circuit, an operating parameter of a device communicatively coupled the surgical system based on the received communicated recommendation.

[0006] In another aspect the present disclosure provides a method for adaptively controlling a surgical network based on validating data purportedly generated in a surgical procedure. The surgical network comprises a medical hub, at least one remote server communicatively coupled to the medical hub, and a medical instrument communicatively coupled to the medical hub. The system is configured to access the data, validate the data to determine if the data is validly generated by the surgical procedure, determine that the data contains at least one flaw or error, and improve data integrity by preventing the at least one flaw or error from being integrated into a larger dataset associated with the at least one remote server. The method comprises: receiving, by the server, information related to a surgical procedure from a device communicatively coupled to the surgical network; validating, by the server, the received information; and adaptively adjusting, by the server, the surgical network based on the received information.

Description

FIGURES

[0007] The various aspects described herein, both as to organization and methods of operation, together with further objects and advantages thereof, may best be understood by reference to the following description, taken in conjunction with the accompanying drawings as follows.

[0008] FIG. 1 is a block diagram of a computer-implemented interactive surgical system, in accordance with at least one aspect of the present disclosure.

[0009] FIG. 2 is a surgical system being used to perform a surgical procedure in an operating room, in accordance with at least one aspect of the present disclosure.

[0010] FIG. 3 is a surgical hub paired with a visualization system, a robotic system, and an intelligent instrument, in accordance with at least one aspect of the present disclosure.

[0011] FIG. 4 is a partial perspective view of a surgical hub enclosure, and of a combo generator module slidably receivable in a drawer of the surgical hub enclosure, in accordance with at least one aspect of the present disclosure.

[0012] FIG. 5 is a perspective view of a combo generator module with bipolar, ultrasonic, and monopolar contacts and a smoke evacuation component, in accordance with at least one aspect of the present disclosure.

[0013] FIG. 6 illustrates individual power bus attachments for a plurality of lateral docking ports of a lateral modular housing configured to receive a plurality of modules, in accordance with at least one aspect of the present disclosure.

[0014] FIG. 7 illustrates a vertical modular housing configured to receive a plurality of modules, in accordance with at least one aspect of the present disclosure.

[0015] FIG. 8 illustrates a surgical data network comprising a modular communication hub configured to connect modular devices located in one or more operating theaters of a healthcare facility, or any room in a healthcare facility specially equipped for surgical operations, to the cloud, in accordance with at least one aspect of the present disclosure.

[0016] FIG. 9 illustrates a computer-implemented interactive surgical system, in accordance with at least one aspect of the present disclosure.

[0017] FIG. 10 illustrates a surgical hub comprising a plurality of modules coupled to the modular control tower, in accordance with at least one aspect of the present disclosure.

[0018] FIG. 11 illustrates one aspect of a Universal Serial Bus (USB) network hub device, in accordance with at least one aspect of the present disclosure.

[0019] FIG. 12 is a block diagram of a cloud computing system comprising a plurality of smart surgical instruments coupled to surgical hubs that may connect to the cloud component of the cloud computing system, in accordance with at least one aspect of the present disclosure.

[0020] FIG. 13 is a functional module architecture of a cloud computing system, in accordance with at least one aspect of the present disclosure.

[0021] FIG. 14 illustrates a diagram of a situationally aware surgical system, in accordance with at least one aspect of the present disclosure.

[0022] FIG. 15 is a timeline depicting situational awareness of a surgical hub, in accordance with at least one aspect of the present disclosure.

[0023] FIG. 16 is a block diagram of a system for automated scaling, organizing, fusing and aligning of disparate data sets, in accordance with at least one aspect of the present disclosure.

[0024] FIG. 17 is a set of graphs including a first graph depicting measured blood pressure verse time, a second graph depicting fused blood pressure verse time, and a third graph depicting blood pressure verse time for different sample rates, in accordance with at least one aspect of the present disclosure.

[0025] FIG. 18 is a graph depicting blood pressure relative to high and low thresholds, in accordance with at least one aspect of the present disclosure.

[0026] FIG. 19 is a graph depicting ultrasonic system frequency verse time, in accordance with at least one aspect of the present disclosure.

[0027] FIG. 20 is a graph depicting expected blood pressure for different vessel types, in accordance with at least one aspect of the present disclosure.

[0028] FIG. 21 is a block diagram depicting layered contextual information, in accordance with at least one aspect of the present disclosure.

[0029] FIG. 22 is a block diagram depicting instrument functional settings, in accordance with at least one aspect of the present disclosure.

[0030] FIG. 23 is a graph depicting force to fire (FTF) and firing velocity for patients having different complication risks.

[0031] FIG. **24** illustrates a surgical device including a user interface and a surgical stapling end effector, in accordance with at least one aspect of the present disclosure.

[0032] FIG. **25** is a schematic diagram of various components of the surgical device of FIG. **24**.

[0033] FIG. **26** is a logic flow diagram of a process depicting a control program or a logic configuration for displaying interpreted information based on external data, in accordance with at least one aspect of the present disclosure.

[0034] FIG. **27** is a logic flow diagram of a process depicting a control program or a logic configuration for displaying interpreted information based on external data, in accordance with at least one aspect of the present disclosure.

[0035] FIG. **28** illustrates a surgical device including a user interface and a surgical stapling end effector, in accordance with at least one aspect of the present disclosure.

[0036] FIG. **29** is a logic flow diagram of a process depicting a control program or a logic configuration for adjusting a parameter setting of the surgical device of FIG. **28**, in accordance with at least one aspect of the present disclosure.

[0037] FIG. **30** is a logic flow diagram of a process depicting a control program or a logic configuration for adjusting a parameter setting of the surgical device of FIG. **28**, in accordance with at least one aspect of the present disclosure.

[0038] FIG. **31** illustrates a surgical device including a user interface and a surgical stapling end effector, in accordance with at least one aspect of the present disclosure.

[0039] FIG. **32** is a logic flow diagram of a process depicting a control program or a logic configuration for automatically adjusting a field of view of a medical imaging device with respect to a detected critical structure, in accordance with at least one aspect of the present disclosure.

[0040] FIG. **33** is a logic flow diagram of a process depicting a control program or a logic configuration for obtaining user permission to automatically adjust a field of view of a medical imaging device with respect to a critical structure, in accordance with at least one aspect of the present disclosure.

[0041] FIG. **34** is a graph depicting positive and negative outcomes for illustrative uses of a surgical device according to a calculated threshold, in accordance with at least one aspect of the present disclosure.

[0042] FIG. **35** is a block diagram depicting surgical procedures having various characteristics grouped according to outcomes, in accordance with at least one aspect of the present disclosure.

[0043] FIG. **36** is a diagram depicting a validation analysis, in accordance with at least one aspect of the present disclosure.

[0044] FIG. **37** is a block diagram depicting a validation analysis, in accordance with at least one aspect of the present disclosure.

[0045] FIG. **38** is a flowchart depicting an algorithm for a surgical system, in accordance with at least one aspect of the present disclosure.

[0046] FIG. **39** is a logic flow diagram of a process depicting a control program or a logic configuration for adjusting surgical hub responses, in accordance with at least one aspect of the present disclosure.

[0047] FIG. **40** is a timeline of an illustrative surgical procedure and the corresponding information inferred by a surgical hub during the procedure, in accordance with at least one aspect of the present disclosure.

[0048] FIG. **41** is a logic flow diagram of a process depicting a control program or a logic configuration for selecting operational modes of a surgical hub, in accordance with at least one aspect of the present disclosure.

[0049] FIG. **42** is a logic flow diagram of a process depicting a control program or a logic configuration for determining whether a surgical procedure is underway, in accordance with at least one aspect of the present disclosure.

[0050] FIG. **43** is a logic flow diagram of a process depicting a control program or a logic

configuration for determining whether surgery is in progress, in accordance with at least one aspect of the present disclosure.

[0051] FIG. **44** is a logic flow diagram of a process depicting a control program or a logic configuration for responding to sensed parameters, in accordance with at least one aspect of the present disclosure.

[0052] FIG. **45** is a logic flow diagram of a process depicting a control program or a logic configuration for adjusting operational parameters of a surgical stapler in the event of a detected security fault, in accordance with at least one aspect of the present disclosure.

[0053] FIG. **46** also depicts various examples of responding to sensed parameters based on a determined situational parameter, in accordance with at least one aspect of the present disclosure.

[0054] FIG. **47** is a logic flow diagram of a process depicting a control program or a logic configuration for assessing operational fitness of a modular device, in accordance with at least one aspect of the present disclosure.

[0055] FIG. **48** is a logic flow diagram of a process depicting a control program or a logic configuration for generating suitable responses to unauthorized interactions with a modular device, in accordance with at least one aspect of the present disclosure.

[0056] FIG. **49** is schematic diagram illustrating multiple components of a modular device, in accordance with at least one aspect of the present disclosure.

[0057] FIG. **50** is a logic flow diagram of a process depicting a control program or a logic configuration for generating suitable responses to unauthorized interactions with a modular device, in accordance with at least one aspect of the present disclosure.

[0058] FIG. **51** is a circuit diagram of a circuit for calculating parameters of an antenna, in accordance with at least one aspect of the present disclosure.

[0059] FIG. **52** is a diagram featuring a compensating circuit for adjusting the applied power, in accordance with at least one aspect of the present disclosure.

[0060] FIG. **53** is a logic flow diagram of a process for peaking applied power, in accordance with at least one aspect of the present disclosure.

[0061] FIG. **54** is an illustration of a set of graphs and depicting generator power level and patient continuity over time, respectively, in accordance with at least one aspect of the present disclosure.

[0062] FIG. **55** is an illustration of a patient situated on a monopolar return pad during a surgical procedure, in accordance with at least one aspect of the present disclosure.

[0063] FIG. **56** is a block diagram of a system for controlling a power level applied by a monopolar instrument, in accordance with at least one aspect of the present disclosure.

[0064] FIG. **57** shows an illustration of a probe approaching a nerve.

[0065] FIG. **58** shows an illustration of a probe directly touching a nerve at location.

[0066] FIG. **59** is a diagram of a graphical user interface (GUI) for controlling various device parameters in accordance with at least one aspect of the present disclosure.

[0067] FIG. **60** is a graphical user interface for controlling adaptive parameters of a surgical device in accordance with at least one aspect of the present disclosure.

[0068] FIG. **61** is a flowchart of a control circuit in accordance with at least one aspect of the present disclosure.

[0069] FIG. **62** is a block diagram depicting a surgical system in accordance with at least one aspect of the present disclosure.

DESCRIPTION

[0070] Applicant of the present application owns the following U.S. patent applications, filed on Dec. 4, 2018, the disclosure of each of which is herein incorporated by reference in its entirety:

[0071] U.S. patent application Ser. No. 16/209,385, titled METHOD OF HUB

COMMUNICATION, PROCESSING, STORAGE AND DISPLAY, now U.S. Patent Application

Publication No. 2019/0200844; [0072] U.S. patent application Ser. No. 16/209,395, titled

METHOD OF HUB COMMUNICATION, now U.S. Patent Application Publication No.

2019/0201136; [0073] U.S. patent application Ser. No. 16/209,403, titled METHOD OF CLOUD BASED DATA ANALYTICS FOR USE WITH THE HUB, now U.S. Patent Application Publication No. 2019/0206569; [0074] U.S. patent application Ser. No. 16/209,407, titled METHOD OF ROBOTIC HUB COMMUNICATION, DETECTION, AND CONTROL, now U.S. Patent Application Publication No. 2019/0201137; [0075] U.S. patent application Ser. No. 16/209,416, titled METHOD OF HUB COMMUNICATION, PROCESSING, DISPLAY, AND CLOUD ANALYTICS, now U.S. Patent Application Publication No. 2019/0206562; [0076] U.S. patent application Ser. No. 16/209,423, titled METHOD OF COMPRESSING TISSUE WITHIN A STAPLING DEVICE AND SIMULTANEOUSLY DISPLAYING THE LOCATION OF THE TISSUE WITHIN THE JAWS, now U.S. Patent Application Publication No. 2019/0200981; [0077] U.S. patent application Ser. No. 16/209,427, titled METHOD OF USING REINFORCED FLEXIBLE CIRCUITS WITH MULTIPLE SENSORS TO OPTIMIZE PERFORMANCE OF RADIO FREQUENCY DEVICES, now U.S. Patent Application Publication No. 2019/0208641; [0078] U.S. patent application Ser. No. 16/209,433, titled METHOD OF SENSING PARTICULATE FROM SMOKE EVACUATED FROM A PATIENT, ADJUSTING THE PUMP SPEED BASED ON THE SENSED INFORMATION, AND COMMUNICATING THE FUNCTIONAL PARAMETERS OF THE SYSTEM TO THE HUB, now U.S. Patent Application Publication No. 2019/0201594; [0079] U.S. patent application Ser. No. 16/209,447, titled METHOD FOR SMOKE EVACUATION FOR SURGICAL HUB, now U.S. Patent Application Publication No. 2019/0201045; [0080] U.S. patent application Ser. No. 16/209,453, titled METHOD FOR CONTROLLING SMART ENERGY DEVICES, now U.S. Patent Application Publication No. 2019/0201046; [0081] U.S. patent application Ser. No. 16/209,458, titled METHOD FOR SMART ENERGY DEVICE INFRASTRUCTURE, now U.S. Patent Application Publication No. 2019/0201047; [0082] U.S. patent application Ser. No. 16/209,478, titled METHOD FOR SITUATIONAL AWARENESS FOR SURGICAL NETWORK OR SURGICAL NETWORK CONNECTED DEVICE CAPABLE OF ADJUSTING FUNCTION BASED ON A SENSED SITUATION OR USAGE, now U.S. Patent Application Publication No. 2019/0104919; [0083] U.S. patent application Ser. No. 16/209,490, titled METHOD FOR FACILITY DATA COLLECTION AND INTERPRETATION, now U.S. Patent Application Publication No. 2019/0206564; and [0084] U.S. patent application Ser. No. 16/209,491, titled METHOD FOR CIRCULAR STAPLER CONTROL ALGORITHM ADJUSTMENT BASED ON SITUATIONAL AWARENESS, now U.S. Pat. No. 11,109,866.

[0085] Applicant of the present application owns the following U.S. patent applications, filed on Nov. 6, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: [0086] U.S. patent application Ser. No. 16/182,224, titled SURGICAL NETWORK, INSTRUMENT, AND CLOUD RESPONSES BASED ON VALIDATION OF RECEIVED DATASET AND AUTHENTICATION OF ITS SOURCE AND INTEGRITY; [0087] U.S. patent application Ser. No. 16/182,230, titled SURGICAL SYSTEM FOR PRESENTING INFORMATION INTERPRETED FROM EXTERNAL DATA; [0088] U.S. patent application Ser. No. 16/182,233, titled SURGICAL SYSTEMS WITH AUTONOMOUSLY ADJUSTABLE CONTROL PROGRAMS; [0089] U.S. patent application Ser. No. 16/182,239, titled ADJUSTMENT OF DEVICE CONTROL PROGRAMS BASED ON STRATIFIED CONTEXTUAL DATA IN ADDITION TO THE DATA; [0090] U.S. patent application Ser. No. 16/182,243, titled SURGICAL HUB AND MODULAR DEVICE RESPONSE ADJUSTMENT BASED ON SITUATIONAL AWARENESS; [0091] U.S. patent application Ser. No. 16/182,248, titled DETECTION AND ESCALATION OF SECURITY RESPONSES OF SURGICAL INSTRUMENTS TO INCREASING SEVERITY THREATS; [0092] U.S. patent application Ser. No. 16/182,251, titled INTERACTIVE SURGICAL SYSTEM; [0093] U.S. patent application Ser. No. 16/182,260, titled AUTOMATED DATA SCALING, ALIGNMENT, AND ORGANIZING BASED ON PREDEFINED PARAMETERS WITHIN SURGICAL NETWORKS; [0094] U.S.

patent application Ser. No. 16/182,267, titled SENSING THE PATIENT POSITION AND CONTACT UTILIZING THE MONO-POLAR RETURN PAD ELECTRODE TO PROVIDE SITUATIONAL AWARENESS TO THE HUB; [0095] U.S. patent application Ser. No. 16/182,249, titled POWERED SURGICAL TOOL WITH PREDEFINED ADJUSTABLE CONTROL ALGORITHM FOR CONTROLLING END EFFECTOR PARAMETER; [0096] U.S. patent application Ser. No. 16/182,246, titled ADJUSTMENTS BASED ON AIRBORNE PARTICLE PROPERTIES; [0097] U.S. patent application Ser. No. 16/182,256, titled ADJUSTMENT OF A SURGICAL DEVICE FUNCTION BASED ON SITUATIONAL AWARENESS; [0098] U.S. patent application Ser. No. 16/182,242, titled REAL-TIME ANALYSIS OF COMPREHENSIVE COST OF ALL INSTRUMENTATION USED IN SURGERY UTILIZING DATA FLUIDITY TO TRACK INSTRUMENTS THROUGH STOCKING AND IN-HOUSE PROCESSES; [0099] U.S. patent application Ser. No. 16/182,255, titled USAGE AND TECHNIQUE ANALYSIS OF SURGEON/STAFF PERFORMANCE AGAINST A BASELINE TO OPTIMIZE DEVICE UTILIZATION AND PERFORMANCE FOR BOTH CURRENT AND FUTURE PROCEDURES; [0100] U.S. patent application Ser. No. 16/182,269, titled IMAGE CAPTURING OF THE AREAS OUTSIDE THE ABDOMEN TO IMPROVE PLACEMENT AND CONTROL OF A SURGICAL DEVICE IN USE; [0101] U.S. patent application Ser. No. 16/182,278, titled COMMUNICATION OF DATA WHERE A SURGICAL NETWORK IS USING CONTEXT OF THE DATA AND REQUIREMENTS OF A RECEIVING SYSTEM/USER TO INFLUENCE INCLUSION OR LINKAGE OF DATA AND METADATA TO ESTABLISH CONTINUITY; [0102] U.S. patent application Ser. No. 16/182,290, titled SURGICAL NETWORK RECOMMENDATIONS FROM REAL TIME ANALYSIS OF PROCEDURE VARIABLES AGAINST A BASELINE HIGHLIGHTING DIFFERENCES FROM THE OPTIMAL SOLUTION; [0103] U.S. patent application Ser. No. 16/182,232, titled CONTROL OF A SURGICAL SYSTEM THROUGH A SURGICAL BARRIER; [0104] U.S. patent application Ser. No. 16/182,227, titled SURGICAL NETWORK DETERMINATION OF PRIORITIZATION OF COMMUNICATION, INTERACTION, OR PROCESSING BASED ON SYSTEM OR DEVICE NEEDS; [0105] U.S. patent application Ser. No. 16/182,231, titled WIRELESS PAIRING OF A SURGICAL DEVICE WITH ANOTHER DEVICE WITHIN A STERILE SURGICAL FIELD BASED ON THE USAGE AND SITUATIONAL AWARENESS OF DEVICES; [0106] U.S. patent application Ser. No. 16/182,229, titled ADJUSTMENT OF STAPLE HEIGHT OF AT LEAST ONE ROW OF STAPLES BASED ON THE SENSED TISSUE THICKNESS OR FORCE IN CLOSING; [0107] U.S. patent application Ser. No. 16/182,234, titled STAPLING DEVICE WITH BOTH COMPULSORY AND DISCRETIONARY LOCKOUTS BASED ON SENSED PARAMETERS; [0108] U.S. patent application Ser. No. 16/182,240, titled POWERED STAPLING DEVICE CONFIGURED TO ADJUST FORCE, ADVANCEMENT SPEED, AND OVERALL STROKE OF CUTTING MEMBER BASED ON SENSED PARAMETER OF FIRING OR CLAMPING; [0109] U.S. patent application Ser. No. 16/182,235, titled VARIATION OF RADIO FREQUENCY AND ULTRASONIC POWER LEVEL IN COOPERATION WITH VARYING CLAMP ARM PRESSURE TO ACHIEVE PREDEFINED HEAT FLUX OR POWER APPLIED TO TISSUE; and [0110] U.S. patent application Ser. No. 16/182,238, titled ULTRASONIC ENERGY DEVICE WHICH VARIES PRESSURE APPLIED BY CLAMP ARM TO PROVIDE THRESHOLD CONTROL PRESSURE AT A CUT PROGRESSION LOCATION.

[0111] Applicant of the present application owns the following U.S. patent applications that were filed on Oct. 26, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: [0112] U.S. patent application Ser. No. 16/172,303, titled METHOD FOR OPERATING A POWERED ARTICULATING MULTI-CLIP APPLIER; [0113] U.S. patent application Ser. No. 16/172,130, titled CLIP APPLIER COMPRISING INTERCHANGEABLE CLIP RELOADS; [0114] U.S. patent application Ser. No. 16/172,066, titled CLIP APPLIER COMPRISING A MOVABLE CLIP MAGAZINE; [0115] U.S. patent application Ser. No. 16/172,078, titled CLIP

APPLIER COMPRISING A ROTATABLY CLIP MAGAZINE; [0116] U.S. patent application Ser. No. 16/172,087, titled CLIP APPLIER COMPRISING CLIP ADVANCING SYSTEMS; [0117] U.S. patent application Ser. No. 16/172,094, titled CLIP APPLIER COMPRISING A CLIP CRIMPING SYSTEM; [0118] U.S. patent application Ser. No. 16/172,128, titled CLIP APPLIER COMPRISING A RECIPROCATING CLIP ADVANCING MEMBER; [0119] U.S. patent application Ser. No. 16/172,168, titled CLIP APPLIER COMPRISING A MOTOR CONTROLLER; [0120] U.S. patent application Ser. No. 16/172,164, titled SURGICAL SYSTEM COMPRISING A SURGICAL TOOL AND A SURGICAL HUB; [0121] U.S. patent application Ser. No. 16/172,328, titled METHOD OF HUB COMMUNICATION WITH SURGICAL INSTRUMENT SYSTEMS; [0122] U.S. patent application Ser. No. 16/172,280, titled METHOD FOR PRODUCING A SURGICAL INSTRUMENT COMPRISING A SMART ELECTRICAL SYSTEM; [0123] U.S. patent application Ser. No. 16/172,219, titled METHOD OF HUB COMMUNICATION WITH SURGICAL INSTRUMENT SYSTEMS; [0124] U.S. patent application Ser. No. 16/172,248, titled METHOD OF HUB COMMUNICATION WITH SURGICAL INSTRUMENT SYSTEMS; [0125] U.S. patent application Ser. No. 16/172,198, titled METHOD OF HUB COMMUNICATION WITH SURGICAL INSTRUMENT SYSTEMS; and [0126] U.S. patent application Ser. No. 16/172,155, titled METHOD OF HUB COMMUNICATION WITH SURGICAL INSTRUMENT SYSTEMS.

[0127] Applicant of the present application owns the following U.S. patent applications, filed on Aug. 28, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: [0128] U.S. patent application Ser. No. 16/115,214, titled ESTIMATING STATE OF ULTRASONIC END EFFECTOR AND CONTROL SYSTEM THEREFOR; [0129] U.S. patent application Ser. No. 16/115,205, titled TEMPERATURE CONTROL OF ULTRASONIC END EFFECTOR AND CONTROL SYSTEM THEREFOR; [0130] U.S. patent application Ser. No. 16/115,233, titled RADIO FREQUENCY ENERGY DEVICE FOR DELIVERING COMBINED ELECTRICAL SIGNALS; [0131] U.S. patent application Ser. No. 16/115,208, titled CONTROLLING AN ULTRASONIC SURGICAL INSTRUMENT ACCORDING TO TISSUE LOCATION; [0132] U.S. patent application Ser. No. 16/115,220, titled CONTROLLING ACTIVATION OF AN ULTRASONIC SURGICAL INSTRUMENT ACCORDING TO THE PRESENCE OF TISSUE; [0133] U.S. patent application Ser. No. 16/115,232, titled DETERMINING TISSUE COMPOSITION VIA AN ULTRASONIC SYSTEM; [0134] U.S. patent application Ser. No. 16/115,239, titled DETERMINING THE STATE OF AN ULTRASONIC ELECTROMECHANICAL SYSTEM ACCORDING TO FREQUENCY SHIFT; [0135] U.S. patent application Ser. No. 16/115,247, titled DETERMINING THE STATE OF AN ULTRASONIC END EFFECTOR; [0136] U.S. patent application Ser. No. 16/115,211, titled SITUATIONAL AWARENESS OF ELECTROSURGICAL SYSTEMS; [0137] U.S. patent application Ser. No. 16/115,226, titled MECHANISMS FOR CONTROLLING DIFFERENT ELECTROMECHANICAL SYSTEMS OF AN ELECTROSURGICAL INSTRUMENT; [0138] U.S. patent application Ser. No. 16/115,240, titled DETECTION OF END EFFECTOR EMERSION IN LIQUID; [0139] U.S. patent application Ser. No. 16/115,249, titled INTERRUPTION OF ENERGY DUE TO INADVERTENT CAPACITIVE COUPLING; [0140] U.S. patent application Ser. No. 16/115,256, titled INCREASING RADIO FREQUENCY TO CREATE PAD-LESS MONOPOLAR LOOP; [0141] U.S. patent application Ser. No. 16/115,223, titled BIPOLAR COMBINATION DEVICE THAT AUTOMATICALLY ADJUSTS PRESSURE BASED ON ENERGY MODALITY; and [0142] U.S. patent application Ser. No. 16/115,238, titled ACTIVATION OF ENERGY DEVICES.

[0143] Applicant of the present application owns the following U.S. patent applications, filed on Aug. 24, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: [0144] U.S. patent application Ser. No. 16/112,129, titled SURGICAL SUTURING INSTRUMENT CONFIGURED TO MANIPULATE TISSUE USING MECHANICAL AND

ELECTRICAL POWER; [0145] U.S. patent application Ser. No. 16/112,155, titled SURGICAL SUTURING INSTRUMENT COMPRISING A CAPTURE WIDTH WHICH IS LARGER THAN TROCAR DIAMETER; [0146] U.S. patent application Ser. No. 16/112,168, titled SURGICAL SUTURING INSTRUMENT COMPRISING A NON-CIRCULAR NEEDLE; [0147] U.S. patent application Ser. No. 16/112,180, titled ELECTRICAL POWER OUTPUT CONTROL BASED ON MECHANICAL FORCES; [0148] U.S. patent application Ser. No. 16/112,193, titled REACTIVE ALGORITHM FOR SURGICAL SYSTEM; [0149] U.S. patent application Ser. No. 16/112,099, titled SURGICAL INSTRUMENT COMPRISING AN ADAPTIVE ELECTRICAL SYSTEM; [0150] U.S. patent application Ser. No. 16/112,112, titled CONTROL SYSTEM ARRANGEMENTS FOR A MODULAR SURGICAL INSTRUMENT; [0151] U.S. patent application Ser. No. 16/112,119, titled ADAPTIVE CONTROL PROGRAMS FOR A SURGICAL SYSTEM COMPRISING MORE THAN ONE TYPE OF CARTRIDGE; [0152] U.S. patent application Ser. No. 16/112,097, titled SURGICAL INSTRUMENT SYSTEMS COMPRISING BATTERY ARRANGEMENTS; [0153] U.S. patent application Ser. No. 16/112,109, titled SURGICAL INSTRUMENT SYSTEMS COMPRISING HANDLE ARRANGEMENTS; [0154] U.S. patent application Ser. No. 16/112,114, titled SURGICAL INSTRUMENT SYSTEMS COMPRISING FEEDBACK MECHANISMS; [0155] U.S. patent application Ser. No. 16/112,117, titled SURGICAL INSTRUMENT SYSTEMS COMPRISING LOCKOUT MECHANISMS; [0156] U.S. patent application Ser. No. 16/112,095, titled SURGICAL INSTRUMENTS COMPRISING A LOCKABLE END EFFECTOR SOCKET; [0157] U.S. patent application Ser. No. 16/112,121, titled SURGICAL INSTRUMENTS COMPRISING A SHIFTING MECHANISM; [0158] U.S. patent application Ser. No. 16/112,151, titled SURGICAL INSTRUMENTS COMPRISING A SYSTEM FOR ARTICULATION AND ROTATION COMPENSATION; [0159] U.S. patent application Ser. No. 16/112,154, titled SURGICAL INSTRUMENTS COMPRISING A BIASED SHIFTING MECHANISM; [0160] U.S. patent application Ser. No. 16/112,226, titled SURGICAL INSTRUMENTS COMPRISING AN ARTICULATION DRIVE THAT PROVIDES FOR HIGH ARTICULATION ANGLES; [0161] U.S. patent application Ser. No. 16/112,062, titled SURGICAL DISSECTORS AND MANUFACTURING TECHNIQUES; [0162] U.S. patent application Ser. No. 16/112,098, titled SURGICAL DISSECTORS CONFIGURED TO APPLY MECHANICAL AND ELECTRICAL ENERGY; [0163] U.S. patent application Ser. No. 16/112,237, titled SURGICAL CLIP APPLIER CONFIGURED TO STORE CLIPS IN A STORED STATE; [0164] U.S. patent application Ser. No. 16/112,245, titled SURGICAL CLIP APPLIER COMPRISING AN EMPTY CLIP CARTRIDGE LOCKOUT; [0165] U.S. patent application Ser. No. 16/112,249, titled SURGICAL CLIP APPLIER COMPRISING AN AUTOMATIC CLIP FEEDING SYSTEM; [0166] U.S. patent application Ser. No. 16/112,253, titled SURGICAL CLIP APPLIER COMPRISING ADAPTIVE FIRING CONTROL; and [0167] U.S. patent application Ser. No. 16/112,257, titled SURGICAL CLIP APPLIER COMPRISING ADAPTIVE CONTROL IN RESPONSE TO A STRAIN GAUGE CIRCUIT.

[0168] Applicant of the present application owns the following U.S. patent applications, filed on Jun. 29, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: [0169] U.S. patent application Ser. No. 16/024,090, titled CAPACITIVE COUPLED RETURN PATH PAD WITH SEPARABLE ARRAY ELEMENTS; [0170] U.S. patent application Ser. No. 16/024,057, titled CONTROLLING A SURGICAL INSTRUMENT ACCORDING TO SENSED CLOSURE PARAMETERS; [0171] U.S. patent application Ser. No. 16/024,067, titled SYSTEMS FOR ADJUSTING END EFFECTOR PARAMETERS BASED ON PERIOPERATIVE INFORMATION; [0172] U.S. patent application Ser. No. 16/024,075, titled SAFETY SYSTEMS FOR SMART POWERED SURGICAL STAPLING; [0173] U.S. patent application Ser. No. 16/024,083, titled SAFETY SYSTEMS FOR SMART POWERED SURGICAL STAPLING; [0174] U.S. patent application Ser. No. 16/024,094, titled SURGICAL SYSTEMS FOR

DETECTING END EFFECTOR TISSUE DISTRIBUTION IRREGULARITIES; [0175] U.S. patent application Ser. No. 16/024,138, titled SYSTEMS FOR DETECTING PROXIMITY OF SURGICAL END EFFECTOR TO CANCEROUS TISSUE; [0176] U.S. patent application Ser. No. 16/024,150, titled SURGICAL INSTRUMENT CARTRIDGE SENSOR ASSEMBLIES; [0177] U.S. patent application Ser. No. 16/024,160, titled VARIABLE OUTPUT CARTRIDGE SENSOR ASSEMBLY; [0178] U.S. patent application Ser. No. 16/024,124, titled SURGICAL INSTRUMENT HAVING A FLEXIBLE ELECTRODE; [0179] U.S. patent application Ser. No. 16/024,132, titled SURGICAL INSTRUMENT HAVING A FLEXIBLE CIRCUIT; [0180] U.S. patent application Ser. No. 16/024,141, titled SURGICAL INSTRUMENT WITH A TISSUE MARKING ASSEMBLY; [0181] U.S. patent application Ser. No. 16/024,162, titled SURGICAL SYSTEMS WITH PRIORITIZED DATA TRANSMISSION CAPABILITIES; [0182] U.S. patent application Ser. No. 16/024,066, titled SURGICAL EVACUATION SENSING AND MOTOR CONTROL; [0183] U.S. patent application Ser. No. 16/024,096, titled SURGICAL EVACUATION SENSOR ARRANGEMENTS; [0184] U.S. patent application Ser. No. 16/024,116, titled SURGICAL EVACUATION FLOW PATHS; [0185] U.S. patent application Ser. No. 16/024,149, titled SURGICAL EVACUATION SENSING AND GENERATOR CONTROL; [0186] U.S. patent application Ser. No. 16/024,180, titled SURGICAL EVACUATION SENSING AND DISPLAY; [0187] U.S. patent application Ser. No. 16/024,245, titled COMMUNICATION OF SMOKE EVACUATION SYSTEM PARAMETERS TO HUB OR CLOUD IN SMOKE EVACUATION MODULE FOR INTERACTIVE SURGICAL PLATFORM; [0188] U.S. patent application Ser. No. 16/024,258, titled SMOKE EVACUATION SYSTEM INCLUDING A SEGMENTED CONTROL CIRCUIT FOR INTERACTIVE SURGICAL PLATFORM; [0189] U.S. patent application Ser. No. 16/024,265, titled SURGICAL EVACUATION SYSTEM WITH A COMMUNICATION CIRCUIT FOR COMMUNICATION BETWEEN A FILTER AND A SMOKE EVACUATION DEVICE; and [0190] U.S. patent application Ser. No. 16/024,273, titled DUAL IN-SERIES LARGE AND SMALL DROPLET FILTERS.

[0191] Applicant of the present application owns the following U.S. patent applications, filed on Mar. 29, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: [0192] U.S. patent application Ser. No. 15/940,641, titled INTERACTIVE SURGICAL SYSTEMS WITH ENCRYPTED COMMUNICATION CAPABILITIES; [0193] U.S. patent application Ser. No. 15/940,648, titled INTERACTIVE SURGICAL SYSTEMS WITH CONDITION HANDLING OF DEVICES AND DATA CAPABILITIES; [0194] U.S. patent application Ser. No. 15/940,656, titled SURGICAL HUB COORDINATION OF CONTROL AND COMMUNICATION OF OPERATING ROOM DEVICES; [0195] U.S. patent application Ser. No. 15/940,666, titled SPATIAL AWARENESS OF SURGICAL HUBS IN OPERATING ROOMS; [0196] U.S. patent application Ser. No. 15/940,670, titled COOPERATIVE UTILIZATION OF DATA DERIVED FROM SECONDARY SOURCES BY INTELLIGENT SURGICAL HUBS; [0197] U.S. patent application Ser. No. 15/940,677, titled SURGICAL HUB CONTROL ARRANGEMENTS; [0198] U.S. patent application Ser. No. 15/940,632, titled DATA STRIPPING METHOD TO INTERROGATE PATIENT RECORDS AND CREATE ANONYMIZED RECORD; [0199] U.S. patent application Ser. No. 15/940,640, titled COMMUNICATION HUB AND STORAGE DEVICE FOR STORING PARAMETERS AND STATUS OF A SURGICAL DEVICE TO BE SHARED WITH CLOUD BASED ANALYTICS SYSTEMS; [0200] U.S. patent application Ser. No. 15/940,645, titled SELF DESCRIBING DATA PACKETS GENERATED AT AN ISSUING INSTRUMENT; [0201] U.S. patent application Ser. No. 15/940,649, titled DATA PAIRING TO INTERCONNECT A DEVICE MEASURED PARAMETER WITH AN OUTCOME; [0202] U.S. patent application Ser. No. 15/940,654, titled SURGICAL HUB SITUATIONAL AWARENESS; [0203] U.S. patent application Ser. No. 15/940,663, titled SURGICAL SYSTEM DISTRIBUTED PROCESSING; [0204] U.S. patent application Ser. No. 15/940,668, titled AGGREGATION AND REPORTING OF SURGICAL HUB DATA; [0205] U.S. patent application Ser. No. 15/940,671,

titled SURGICAL HUBSPATIAL AWARENESS TO DETERMINE DEVICES IN OPERATING THEATER; [0206] U.S. patent application Ser. No. 15/940,686, titled DISPLAY OF ALIGNMENT OF STAPLE CARTRIDGE TO PRIOR LINEAR STAPLE LINE; [0207] U.S. patent application Ser. No. 15/940,700, titled STERILE FIELD INTERACTIVE CONTROL DISPLAYS; [0208] U.S. patent application Ser. No. 15/940,629, titled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS; [0209] U.S. patent application Ser. No. 15/940,704, titled USE OF LASER LIGHT AND RED-GREEN-BLUE COLORATION TO DETERMINE PROPERTIES OF BACK SCATTERED LIGHT; [0210] U.S. patent application Ser. No. 15/940,722, titled CHARACTERIZATION OF TISSUE IRREGULARITIES THROUGH THE USE OF MONO-CHROMATIC LIGHT REFRACTIVITY; [0211] U.S. patent application Ser. No. 15/940,742, titled DUAL CMOS ARRAY IMAGING; [0212] U.S. patent application Ser. No. 15/940,636, titled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL DEVICES; [0213] U.S. patent application Ser. No. 15/940,653, titled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL HUBS; [0214] U.S. patent application Ser. No. 15/940,660, titled CLOUD-BASED MEDICAL ANALYTICS FOR CUSTOMIZATION AND RECOMMENDATIONS TO A USER; [0215] U.S. patent application Ser. No. 15/940,679, titled CLOUD-BASED MEDICAL ANALYTICS FOR LINKING OF LOCAL USAGE TRENDS WITH THE RESOURCE ACQUISITION BEHAVIORS OF LARGER DATA SET; [0216] U.S. patent application Ser. No. 15/940,694, titled CLOUD-BASED MEDICAL ANALYTICS FOR MEDICAL FACILITY SEGMENTED INDIVIDUALIZATION OF INSTRUMENT FUNCTION; [0217] U.S. patent application Ser. No. 15/940,634, titled CLOUD-BASED MEDICAL ANALYTICS FOR SECURITY AND AUTHENTICATION TRENDS AND REACTIVE MEASURES; [0218] U.S. patent application Ser. No. 15/940,706, titled DATA HANDLING AND PRIORITIZATION IN A CLOUD ANALYTICS NETWORK; [0219] U.S. patent application Ser. No. 15/940,675, titled CLOUD INTERFACE FOR COUPLED SURGICAL DEVICES; [0220] U.S. patent application Ser. No. 15/940,627, titled DRIVE ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; [0221] U.S. patent application Ser. No. 15/940,637, titled COMMUNICATION ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; [0222] U.S. patent application Ser. No. 15/940,642, titled CONTROLS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; [0223] U.S. patent application Ser. No. 15/940,676, titled AUTOMATIC TOOL ADJUSTMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; [0224] U.S. patent application Ser. No. 15/940,680, titled CONTROLLERS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; [0225] U.S. patent application Ser. No. 15/940,683, titled COOPERATIVE SURGICAL ACTIONS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; [0226] U.S. patent application Ser. No. 15/940,690, titled DISPLAY ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; and [0227] U.S. patent application Ser. No. 15/940,711, titled SENSING ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS.

[0228] Applicant of the present application owns the following U.S. Provisional Patent Applications, filed on Mar. 8, 2018, the disclosure of each of which is herein incorporated by reference in its entirety: [0229] U.S. Provisional Patent Application No. 62/640,417, titled TEMPERATURE CONTROL IN ULTRASONIC DEVICE AND CONTROL SYSTEM THEREFOR; and [0230] U.S. Provisional Patent Application No. 62/640,415, titled ESTIMATING STATE OF ULTRASONIC END EFFECTOR AND CONTROL SYSTEM THEREFOR.

[0231] Before explaining various aspects of surgical devices and generators in detail, it should be noted that the illustrative examples are not limited in application or use to the details of construction and arrangement of parts illustrated in the accompanying drawings and description. The illustrative examples may be implemented or incorporated in other aspects, variations and modifications, and may be practiced or carried out in various ways. Further, unless otherwise indicated, the terms and expressions employed herein have been chosen for the purpose of describing the illustrative examples for the convenience of the reader and are not for the purpose of

limitation thereof. Also, it will be appreciated that one or more of the following-described aspects, expressions of aspects, and/or examples, can be combined with any one or more of the other following-described aspects, expressions of aspects and/or examples.

Surgical Hubs

[0232] Referring to FIG. 1, a computer-implemented interactive surgical system **100** includes one or more surgical systems **102** and a cloud-based system (e.g., the cloud **104** that may include a remote server **113** coupled to a storage device **105**). Each surgical system **102** includes at least one surgical hub **106** in communication with the cloud **104** that may include a remote server **113**. In one example, as illustrated in FIG. 1, the surgical system **102** includes a visualization system **108**, a robotic system **110**, and a handheld intelligent surgical instrument **112**, which are configured to communicate with one another and/or the hub **106**. In some aspects, a surgical system **102** may include an M number of hubs **106**, an N number of visualization systems **108**, an O number of robotic systems **110**, and a P number of handheld intelligent surgical instruments **112**, where M, N, O, and P are integers greater than or equal to one.

[0233] FIG. 2 depicts an example of a surgical system **102** being used to perform a surgical procedure on a patient who is lying down on an operating table **114** in a surgical operating room **116**. A robotic system **110** is used in the surgical procedure as a part of the surgical system **102**. The robotic system **110** includes a surgeon's console **118**, a patient side cart **120** (surgical robot), and a surgical robotic hub **122**. The patient side cart **120** can manipulate at least one removably coupled surgical tool **117** through a minimally invasive incision in the body of the patient while the surgeon views the surgical site through the surgeon's console **118**. An image of the surgical site can be obtained by a medical imaging device **124**, which can be manipulated by the patient side cart **120** to orient the imaging device **124**. The robotic hub **122** can be used to process the images of the surgical site for subsequent display to the surgeon through the surgeon's console **118**.

[0234] Other types of robotic systems can be readily adapted for use with the surgical system **102**. Various examples of robotic systems and surgical tools that are suitable for use with the present disclosure are described in U.S. Provisional Patent Application Ser. No. 62/611,339, titled ROBOT ASSISTED SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

[0235] Various examples of cloud-based analytics that are performed by the cloud **104**, and are suitable for use with the present disclosure, are described in U.S. Provisional Patent Application Ser. No. 62/611,340, titled CLOUD-BASED MEDICAL ANALYTICS, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

[0236] In various aspects, the imaging device **124** includes at least one image sensor and one or more optical components. Suitable image sensors include, but are not limited to, Charge-Coupled Device (CCD) sensors and Complementary Metal-Oxide Semiconductor (CMOS) sensors.

[0237] The optical components of the imaging device **124** may include one or more illumination sources and/or one or more lenses. The one or more illumination sources may be directed to illuminate portions of the surgical field. The one or more image sensors may receive light reflected or refracted from the surgical field, including light reflected or refracted from tissue and/or surgical instruments.

[0238] The one or more illumination sources may be configured to radiate electromagnetic energy in the visible spectrum as well as the invisible spectrum. The visible spectrum, sometimes referred to as the optical spectrum or luminous spectrum, is that portion of the electromagnetic spectrum that is visible to (i.e., can be detected by) the human eye and may be referred to as visible light or simply light. A typical human eye will respond to wavelengths in air that are from about 380 nm to about 750 nm.

[0239] The invisible spectrum (i.e., the non-luminous spectrum) is that portion of the electromagnetic spectrum that lies below and above the visible spectrum (i.e., wavelengths below about 380 nm and above about 750 nm). The invisible spectrum is not detectable by the human eye.

Wavelengths greater than about 750 nm are longer than the red visible spectrum, and they become invisible infrared (IR), microwave, and radio electromagnetic radiation. Wavelengths less than about 380 nm are shorter than the violet spectrum, and they become invisible ultraviolet, x-ray, and gamma ray electromagnetic radiation.

[0240] In various aspects, the imaging device **124** is configured for use in a minimally invasive procedure. Examples of imaging devices suitable for use with the present disclosure include, but not limited to, an arthroscope, angioscope, bronchoscope, choledochoscope, colonoscope, cytoscope, duodenoscope, enteroscope, esophagogastro-duodenoscope (gastroscope), endoscope, laryngoscope, nasopharyngo-neproscope, sigmoidoscope, thoracoscope, and ureterscope.

[0241] In one aspect, the imaging device employs multi-spectrum monitoring to discriminate topography and underlying structures. A multi-spectral image is one that captures image data within specific wavelength ranges across the electromagnetic spectrum. The wavelengths may be separated by filters or by the use of instruments that are sensitive to particular wavelengths, including light from frequencies beyond the visible light range, e.g., IR and ultraviolet. Spectral imaging can allow extraction of additional information the human eye fails to capture with its receptors for red, green, and blue. The use of multi-spectral imaging is described in greater detail under the heading “Advanced Imaging Acquisition Module” in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety. Multi-spectrum monitoring can be a useful tool in relocating a surgical field after a surgical task is completed to perform one or more of the previously described tests on the treated tissue.

[0242] It is axiomatic that strict sterilization of the operating room and surgical equipment is required during any surgery. The strict hygiene and sterilization conditions required in a “surgical theater,” i.e., an operating or treatment room, necessitate the highest possible sterility of all medical devices and equipment. Part of that sterilization process is the need to sterilize anything that comes in contact with the patient or penetrates the sterile field, including the imaging device **124** and its attachments and components. It will be appreciated that the sterile field may be considered a specified area, such as within a tray or on a sterile towel, that is considered free of microorganisms, or the sterile field may be considered an area, immediately around a patient, who has been prepared for a surgical procedure. The sterile field may include the scrubbed team members, who are properly attired, and all furniture and fixtures in the area.

[0243] In various aspects, the visualization system **108** includes one or more imaging sensors, one or more image-processing units, one or more storage arrays, and one or more displays that are strategically arranged with respect to the sterile field, as illustrated in FIG. 2. In one aspect, the visualization system **108** includes an interface for HL7, PACS, and EMR. Various components of the visualization system **108** are described under the heading “Advanced Imaging Acquisition Module” in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

[0244] As illustrated in FIG. 2, a primary display **119** is positioned in the sterile field to be visible to an operator at the operating table **114**. In addition, a visualization tower **111** is positioned outside the sterile field. The visualization tower **111** includes a first non-sterile display **107** and a second non-sterile display **109**, which face away from each other. The visualization system **108**, guided by the hub **106**, is configured to utilize the displays **107**, **109**, and **119** to coordinate information flow to operators inside and outside the sterile field. For example, the hub **106** may cause the visualization system **108** to display a snapshot of a surgical site, as recorded by an imaging device **124**, on a non-sterile display **107** or **109**, while maintaining a live feed of the surgical site on the primary display **119**. The snapshot on the non-sterile display **107** or **109** can permit a non-sterile operator to perform a diagnostic step relevant to the surgical procedure, for example.

[0245] In one aspect, the hub **106** is also configured to route a diagnostic input or feedback entered

by a non-sterile operator at the visualization tower **111** to the primary display **119** within the sterile field, where it can be viewed by a sterile operator at the operating table. In one example, the input can be in the form of a modification to the snapshot displayed on the non-sterile display **107** or **109**, which can be routed to the primary display **119** by the hub **106**.

[0246] Referring to FIG. 2, a surgical instrument **112** is being used in the surgical procedure as part of the surgical system **102**. The hub **106** is also configured to coordinate information flow to a display of the surgical instrument **112**. For example, coordinate information flow is further described in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety. A diagnostic input or feedback entered by a non-sterile operator at the visualization tower **111** can be routed by the hub **106** to the surgical instrument display **115** within the sterile field, where it can be viewed by the operator of the surgical instrument **112**. Example surgical instruments that are suitable for use with the surgical system **102** are described under the heading “Surgical Instrument Hardware” in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety, for example.

[0247] Referring now to FIG. 3, a hub **106** is depicted in communication with a visualization system **108**, a robotic system **110**, and a handheld intelligent surgical instrument **112**. The hub **106** includes a hub display **135**, an imaging module **138**, a generator module **140** (which can include a monopolar generator **142**, a bipolar generator **144**, and/or an ultrasonic generator **143**), a communication module **130**, a processor module **132**, and a storage array **134**. In certain aspects, as illustrated in FIG. 3, the hub **106** further includes a smoke evacuation module **126**, a suction/irrigation module **128**, and/or an OR mapping module **133**.

[0248] During a surgical procedure, energy application to tissue, for sealing and/or cutting, is generally associated with smoke evacuation, suction of excess fluid, and/or irrigation of the tissue. Fluid, power, and/or data lines from different sources are often entangled during the surgical procedure. Valuable time can be lost addressing this issue during a surgical procedure. Detangling the lines may necessitate disconnecting the lines from their respective modules, which may require resetting the modules. The hub modular enclosure **136** offers a unified environment for managing the power, data, and fluid lines, which reduces the frequency of entanglement between such lines.

[0249] Aspects of the present disclosure present a surgical hub for use in a surgical procedure that involves energy application to tissue at a surgical site. The surgical hub includes a hub enclosure and a combo generator module slidably receivable in a docking station of the hub enclosure. The docking station includes data and power contacts. The combo generator module includes two or more of an ultrasonic energy generator component, a bipolar RF energy generator component, and a monopolar RF energy generator component that are housed in a single unit. In one aspect, the combo generator module also includes a smoke evacuation component, at least one energy delivery cable for connecting the combo generator module to a surgical instrument, at least one smoke evacuation component configured to evacuate smoke, fluid, and/or particulates generated by the application of therapeutic energy to the tissue, and a fluid line extending from the remote surgical site to the smoke evacuation component.

[0250] In one aspect, the fluid line is a first fluid line and a second fluid line extends from the remote surgical site to a suction and irrigation module slidably received in the hub enclosure. In one aspect, the hub enclosure comprises a fluid interface.

[0251] Certain surgical procedures may require the application of more than one energy type to the tissue. One energy type may be more beneficial for cutting the tissue, while another different energy type may be more beneficial for sealing the tissue. For example, a bipolar generator can be used to seal the tissue while an ultrasonic generator can be used to cut the sealed tissue. Aspects of the present disclosure present a solution where a hub modular enclosure **136** is configured to accommodate different generators, and facilitate an interactive communication therebetween. One

of the advantages of the hub modular enclosure **136** is enabling the quick removal and/or replacement of various modules.

[0252] Aspects of the present disclosure present a modular surgical enclosure for use in a surgical procedure that involves energy application to tissue. The modular surgical enclosure includes a first energy-generator module, configured to generate a first energy for application to the tissue, and a first docking station comprising a first docking port that includes first data and power contacts, wherein the first energy-generator module is slidably movable into an electrical engagement with the power and data contacts and wherein the first energy-generator module is slidably movable out of the electrical engagement with the first power and data contacts,

[0253] Further to the above, the modular surgical enclosure also includes a second energy-generator module configured to generate a second energy, different than the first energy, for application to the tissue, and a second docking station comprising a second docking port that includes second data and power contacts, wherein the second energy-generator module is slidably movable into an electrical engagement with the power and data contacts, and wherein the second energy-generator module is slidably movable out of the electrical engagement with the second power and data contacts.

[0254] In addition, the modular surgical enclosure also includes a communication bus between the first docking port and the second docking port, configured to facilitate communication between the first energy-generator module and the second energy-generator module.

[0255] Referring to FIGS. **3-7**, aspects of the present disclosure are presented for a hub modular enclosure **136** that allows the modular integration of a generator module **140**, a smoke evacuation module **126**, and a suction/irrigation module **128**. The hub modular enclosure **136** further facilitates interactive communication between the modules **140**, **126**, **128**. As illustrated in FIG. **5**, the generator module **140** can be a generator module with integrated monopolar, bipolar, and ultrasonic components supported in a single housing unit **139** slidably insertable into the hub modular enclosure **136**. As illustrated in FIG. **5**, the generator module **140** can be configured to connect to a monopolar device **146**, a bipolar device **147**, and an ultrasonic device **148**. Alternatively, the generator module **140** may comprise a series of monopolar, bipolar, and/or ultrasonic generator modules that interact through the hub modular enclosure **136**. The hub modular enclosure **136** can be configured to facilitate the insertion of multiple generators and interactive communication between the generators docked into the hub modular enclosure **136** so that the generators would act as a single generator.

[0256] In one aspect, the hub modular enclosure **136** comprises a modular power and communication backplane **149** with external and wireless communication headers to enable the removable attachment of the modules **140**, **126**, **128** and interactive communication therebetween.

[0257] In one aspect, the hub modular enclosure **136** includes docking stations, or drawers, **151**, herein also referred to as drawers, which are configured to slidably receive the modules **140**, **126**, **128**. FIG. **4** illustrates a partial perspective view of a surgical hub enclosure **136**, and a combo generator module **145** slidably receivable in a docking station **151** of the surgical hub enclosure **136**. A docking port **152** with power and data contacts on a rear side of the combo generator module **145** is configured to engage a corresponding docking port **150** with power and data contacts of a corresponding docking station **151** of the hub modular enclosure **136** as the combo generator module **145** is slid into position within the corresponding docking station **151** of the hub module enclosure **136**. In one aspect, the combo generator module **145** includes a bipolar, ultrasonic, and monopolar module and a smoke evacuation module integrated together into a single housing unit **139**, as illustrated in FIG. **5**.

[0258] In various aspects, the smoke evacuation module **126** includes a fluid line **154** that conveys captured/collected smoke and/or fluid away from a surgical site and to, for example, the smoke evacuation module **126**. Vacuum suction originating from the smoke evacuation module **126** can draw the smoke into an opening of a utility conduit at the surgical site. The utility conduit, coupled

to the fluid line, can be in the form of a flexible tube terminating at the smoke evacuation module **126**. The utility conduit and the fluid line define a fluid path extending toward the smoke evacuation module **126** that is received in the hub enclosure **136**.

[0259] In various aspects, the suction/irrigation module **128** is coupled to a surgical tool comprising an aspiration fluid line and a suction fluid line. In one example, the aspiration and suction fluid lines are in the form of flexible tubes extending from the surgical site toward the suction/irrigation module **128**. One or more drive systems can be configured to cause irrigation and aspiration of fluids to and from the surgical site.

[0260] In one aspect, the surgical tool includes a shaft having an end effector at a distal end thereof and at least one energy treatment associated with the end effector, an aspiration tube, and an irrigation tube. The aspiration tube can have an inlet port at a distal end thereof and the aspiration tube extends through the shaft. Similarly, an irrigation tube can extend through the shaft and can have an inlet port in proximity to the energy deliver implement. The energy deliver implement is configured to deliver ultrasonic and/or RF energy to the surgical site and is coupled to the generator module **140** by a cable extending initially through the shaft.

[0261] The irrigation tube can be in fluid communication with a fluid source, and the aspiration tube can be in fluid communication with a vacuum source. The fluid source and/or the vacuum source can be housed in the suction/irrigation module **128**. In one example, the fluid source and/or the vacuum source can be housed in the hub enclosure **136** separately from the suction/irrigation module **128**. In such example, a fluid interface can be configured to connect the suction/irrigation module **128** to the fluid source and/or the vacuum source.

[0262] In one aspect, the modules **140**, **126**, **128** and/or their corresponding docking stations on the hub modular enclosure **136** may include alignment features that are configured to align the docking ports of the modules into engagement with their counterparts in the docking stations of the hub modular enclosure **136**. For example, as illustrated in FIG. 4, the combo generator module **145** includes side brackets **155** that are configured to slidably engage with corresponding brackets **156** of the corresponding docking station **151** of the hub modular enclosure **136**. The brackets cooperate to guide the docking port contacts of the combo generator module **145** into an electrical engagement with the docking port contacts of the hub modular enclosure **136**.

[0263] In some aspects, the drawers **151** of the hub modular enclosure **136** are the same, or substantially the same size, and the modules are adjusted in size to be received in the drawers **151**. For example, the side brackets **155** and/or **156** can be larger or smaller depending on the size of the module. In other aspects, the drawers **151** are different in size and are each designed to accommodate a particular module.

[0264] Furthermore, the contacts of a particular module can be keyed for engagement with the contacts of a particular drawer to avoid inserting a module into a drawer with mismatching contacts.

[0265] As illustrated in FIG. 4, the docking port **150** of one drawer **151** can be coupled to the docking port **150** of another drawer **151** through a communications link **157** to facilitate an interactive communication between the modules housed in the hub modular enclosure **136**. The docking ports **150** of the hub modular enclosure **136** may alternatively, or additionally, facilitate a wireless interactive communication between the modules housed in the hub modular enclosure **136**. Any suitable wireless communication can be employed, such as for example Air Titan-Bluetooth.

[0266] FIG. 6 illustrates individual power bus attachments for a plurality of lateral docking ports of a lateral modular housing **160** configured to receive a plurality of modules of a surgical hub **206**. The lateral modular housing **160** is configured to laterally receive and interconnect the modules **161**. The modules **161** are slidably inserted into docking stations **162** of lateral modular housing **160**, which includes a backplane for interconnecting the modules **161**. As illustrated in FIG. 6, the modules **161** are arranged laterally in the lateral modular housing **160**. Alternatively, the modules **161** may be arranged vertically in a lateral modular housing.

[0267] FIG. 7 illustrates a vertical modular housing **164** configured to receive a plurality of modules **165** of the surgical hub **106**. The modules **165** are slidably inserted into docking stations, or drawers, **167** of vertical modular housing **164**, which includes a backplane for interconnecting the modules **165**. Although the drawers **167** of the vertical modular housing **164** are arranged vertically, in certain instances, a vertical modular housing **164** may include drawers that are arranged laterally. Furthermore, the modules **165** may interact with one another through the docking ports of the vertical modular housing **164**. In the example of FIG. 7, a display **177** is provided for displaying data relevant to the operation of the modules **165**. In addition, the vertical modular housing **164** includes a master module **178** housing a plurality of sub-modules that are slidably received in the master module **178**.

[0268] In various aspects, the imaging module **138** comprises an integrated video processor and a modular light source and is adapted for use with various imaging devices. In one aspect, the imaging device is comprised of a modular housing that can be assembled with a light source module and a camera module. The housing can be a disposable housing. In at least one example, the disposable housing is removably coupled to a reusable controller, a light source module, and a camera module. The light source module and/or the camera module can be selectively chosen depending on the type of surgical procedure. In one aspect, the camera module comprises a CCD sensor. In another aspect, the camera module comprises a CMOS sensor. In another aspect, the camera module is configured for scanned beam imaging. Likewise, the light source module can be configured to deliver a white light or a different light, depending on the surgical procedure.

[0269] During a surgical procedure, removing a surgical device from the surgical field and replacing it with another surgical device that includes a different camera or a different light source can be inefficient. Temporarily losing sight of the surgical field may lead to undesirable consequences. The module imaging device of the present disclosure is configured to permit the replacement of a light source module or a camera module midstream during a surgical procedure, without having to remove the imaging device from the surgical field.

[0270] In one aspect, the imaging device comprises a tubular housing that includes a plurality of channels. A first channel is configured to slidably receive the camera module, which can be configured for a snap-fit engagement with the first channel. A second channel is configured to slidably receive the light source module, which can be configured for a snap-fit engagement with the second channel. In another example, the camera module and/or the light source module can be rotated into a final position within their respective channels. A threaded engagement can be employed in lieu of the snap-fit engagement.

[0271] In various examples, multiple imaging devices are placed at different positions in the surgical field to provide multiple views. The imaging module **138** can be configured to switch between the imaging devices to provide an optimal view. In various aspects, the imaging module **138** can be configured to integrate the images from the different imaging device.

[0272] Various image processors and imaging devices suitable for use with the present disclosure are described in U.S. Pat. No. 7,995,045, titled COMBINED SBI AND CONVENTIONAL IMAGE PROCESSOR, which issued on Aug. 9, 2011, which is herein incorporated by reference in its entirety. In addition, U.S. Pat. No. 7,982,776, titled SBI MOTION ARTIFACT REMOVAL APPARATUS AND METHOD, which issued on Jul. 19, 2011, which is herein incorporated by reference in its entirety, describes various systems for removing motion artifacts from image data. Such systems can be integrated with the imaging module **138**. Furthermore, U.S. Patent Application Publication No. 2011/0306840, titled CONTROLLABLE MAGNETIC SOURCE TO FIXTURE INTRACORPOREAL APPARATUS, which published on Dec. 15, 2011, and U.S. Patent Application Publication No. 2014/0243597, titled SYSTEM FOR PERFORMING A MINIMALLY INVASIVE SURGICAL PROCEDURE, which published on Aug. 28, 2014, each of which is herein incorporated by reference in its entirety.

[0273] FIG. 8 illustrates a surgical data network **201** comprising a modular communication hub

203 configured to connect modular devices located in one or more operating theaters of a healthcare facility, or any room in a healthcare facility specially equipped for surgical operations, to a cloud-based system (e.g., the cloud **204** that may include a remote server **213** coupled to a storage device **205**). In one aspect, the modular communication hub **203** comprises a network hub **207** and/or a network switch **209** in communication with a network router. The modular communication hub **203** also can be coupled to a local computer system **210** to provide local computer processing and data manipulation. The surgical data network **201** may be configured as passive, intelligent, or switching. A passive surgical data network serves as a conduit for the data, enabling it to go from one device (or segment) to another and to the cloud computing resources. An intelligent surgical data network includes additional features to enable the traffic passing through the surgical data network to be monitored and to configure each port in the network hub **207** or network switch **209**. An intelligent surgical data network may be referred to as a manageable hub or switch. A switching hub reads the destination address of each packet and then forwards the packet to the correct port. [0274] Modular devices **1a-1n** located in the operating theater may be coupled to the modular communication hub **203**. The network hub **207** and/or the network switch **209** may be coupled to a network router **211** to connect the devices **1a-1n** to the cloud **204** or the local computer system **210**. Data associated with the devices **1a-1n** may be transferred to cloud-based computers via the router for remote data processing and manipulation. Data associated with the devices **1a-1n** may also be transferred to the local computer system **210** for local data processing and manipulation. Modular devices **2a-2m** located in the same operating theater also may be coupled to a network switch **209**. The network switch **209** may be coupled to the network hub **207** and/or the network router **211** to connect to the devices **2a-2m** to the cloud **204**. Data associated with the devices **2a-2m** may be transferred to the cloud **204** via the network router **211** for data processing and manipulation. Data associated with the devices **2a-2m** may also be transferred to the local computer system **210** for local data processing and manipulation.

[0275] It will be appreciated that the surgical data network **201** may be expanded by interconnecting multiple network hubs **207** and/or multiple network switches **209** with multiple network routers **211**. The modular communication hub **203** may be contained in a modular control tower configured to receive multiple devices **1a-1n/2a-2m**. The local computer system **210** also may be contained in a modular control tower. The modular communication hub **203** is connected to a display **212** to display images obtained by some of the devices **1a-1n/2a-2m**, for example during surgical procedures. In various aspects, the devices **1a-1n/2a-2m** may include, for example, various modules such as an imaging module **138** coupled to an endoscope, a generator module **140** coupled to an energy-based surgical device, a smoke evacuation module **126**, a suction/irrigation module **128**, a communication module **130**, a processor module **132**, a storage array **134**, a surgical device coupled to a display, and/or a non-contact sensor module, among other modular devices that may be connected to the modular communication hub **203** of the surgical data network **201**.

[0276] In one aspect, the surgical data network **201** may comprise a combination of network hub(s), network switch(es), and network router(s) connecting the devices **1a-1n/2a-2m** to the cloud. Any one of or all of the devices **1a-1n/2a-2m** coupled to the network hub or network switch may collect data in real time and transfer the data to cloud computers for data processing and manipulation. It will be appreciated that cloud computing relies on sharing computing resources rather than having local servers or personal devices to handle software applications. The word “cloud” may be used as a metaphor for “the Internet,” although the term is not limited as such. Accordingly, the term “cloud computing” may be used herein to refer to “a type of Internet-based computing,” where different services—such as servers, storage, and applications—are delivered to the modular communication hub **203** and/or computer system **210** located in the surgical theater (e.g., a fixed, mobile, temporary, or field operating room or space) and to devices connected to the modular communication hub **203** and/or computer system **210** through the Internet. The cloud infrastructure may be maintained by a cloud service provider. In this context, the cloud service

provider may be the entity that coordinates the usage and control of the devices **1a-1n/2a-2m** located in one or more operating theaters. The cloud computing services can perform a large number of calculations based on the data gathered by smart surgical instruments, robots, and other computerized devices located in the operating theater. The hub hardware enables multiple devices or connections to be connected to a computer that communicates with the cloud computing resources and storage.

[0277] Applying cloud computer data processing techniques on the data collected by the devices **1a-1n/2a-2m**, the surgical data network provides improved surgical outcomes, reduced costs, and improved patient satisfaction. At least some of the devices **1a-1n/2a-2m** may be employed to view tissue states to assess leaks or perfusion of sealed tissue after a tissue sealing and cutting procedure. At least some of the devices **1a-1n/2a-2m** may be employed to identify pathology, such as the effects of diseases, using the cloud-based computing to examine data including images of samples of body tissue for diagnostic purposes. This includes localization and margin confirmation of tissue and phenotypes. At least some of the devices **1a-1n/2a-2m** may be employed to identify anatomical structures of the body using a variety of sensors integrated with imaging devices and techniques such as overlaying images captured by multiple imaging devices. The data gathered by the devices **1a-1n/2a-2m**, including image data, may be transferred to the cloud **204** or the local computer system **210** or both for data processing and manipulation including image processing and manipulation. The data may be analyzed to improve surgical procedure outcomes by determining if further treatment, such as the application of endoscopic intervention, emerging technologies, a targeted radiation, targeted intervention, and precise robotics to tissue-specific sites and conditions, may be pursued. Such data analysis may further employ outcome analytics processing, and using standardized approaches may provide beneficial feedback to either confirm surgical treatments and the behavior of the surgeon or suggest modifications to surgical treatments and the behavior of the surgeon.

[0278] In one implementation, the operating theater devices **1a-1n** may be connected to the modular communication hub **203** over a wired channel or a wireless channel depending on the configuration of the devices **1a-1n** to a network hub. The network hub **207** may be implemented, in one aspect, as a local network broadcast device that works on the physical layer of the Open System Interconnection (OSI) model. The network hub provides connectivity to the devices **1a-1n** located in the same operating theater network. The network hub **207** collects data in the form of packets and sends them to the router in half duplex mode. The network hub **207** does not store any media access control/Internet Protocol (MAC/IP) to transfer the device data. Only one of the devices **1a-1n** can send data at a time through the network hub **207**. The network hub **207** has no routing tables or intelligence regarding where to send information and broadcasts all network data across each connection and to a remote server **213** (FIG. 9) over the cloud **204**. The network hub **207** can detect basic network errors such as collisions, but having all information broadcast to multiple ports can be a security risk and cause bottlenecks.

[0279] In another implementation, the operating theater devices **2a-2m** may be connected to a network switch **209** over a wired channel or a wireless channel. The network switch **209** works in the data link layer of the OSI model. The network switch **209** is a multicast device for connecting the devices **2a-2m** located in the same operating theater to the network. The network switch **209** sends data in the form of frames to the network router **211** and works in full duplex mode. Multiple devices **2a-2m** can send data at the same time through the network switch **209**. The network switch **209** stores and uses MAC addresses of the devices **2a-2m** to transfer data.

[0280] The network hub **207** and/or the network switch **209** are coupled to the network router **211** for connection to the cloud **204**. The network router **211** works in the network layer of the OSI model. The network router **211** creates a route for transmitting data packets received from the network hub **207** and/or network switch **211** to cloud-based computer resources for further processing and manipulation of the data collected by any one of or all the devices **1a-1n/2a-2m**.

The network router **211** may be employed to connect two or more different networks located in different locations, such as, for example, different operating theaters of the same healthcare facility or different networks located in different operating theaters of different healthcare facilities. The network router **211** sends data in the form of packets to the cloud **204** and works in full duplex mode. Multiple devices can send data at the same time. The network router **211** uses IP addresses to transfer data.

[0281] In one example, the network hub **207** may be implemented as a USB hub, which allows multiple USB devices to be connected to a host computer. The USB hub may expand a single USB port into several tiers so that there are more ports available to connect devices to the host system computer. The network hub **207** may include wired or wireless capabilities to receive information over a wired channel or a wireless channel. In one aspect, a wireless USB short-range, high-bandwidth wireless radio communication protocol may be employed for communication between the devices **1a-1n** and devices **2a-2m** located in the operating theater.

[0282] In other examples, the operating theater devices **1a-1n/2a-2m** may communicate to the modular communication hub **203** via Bluetooth wireless technology standard for exchanging data over short distances (using short-wavelength UHF radio waves in the ISM band from 2.4 to 2.485 GHz) from fixed and mobile devices and building personal area networks (PANs). In other aspects, the operating theater devices **1a-1n/2a-2m** may communicate to the modular communication hub **203** via a number of wireless or wired communication standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long-term evolution (LTE), and Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, and Ethernet derivatives thereof, as well as any other wireless and wired protocols that are designated as 3G, 4G, 5G, and beyond. The computing module may include a plurality of communication modules. For instance, a first communication module may be dedicated to shorter-range wireless communications such as Wi-Fi and Bluetooth, and a second communication module may be dedicated to longer-range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

[0283] The modular communication hub **203** may serve as a central connection for one or all of the operating theater devices **1a-1n/2a-2m** and handles a data type known as frames. Frames carry the data generated by the devices **1a-1n/2a-2m**. When a frame is received by the modular communication hub **203**, it is amplified and transmitted to the network router **211**, which transfers the data to the cloud computing resources by using a number of wireless or wired communication standards or protocols, as described herein.

[0284] The modular communication hub **203** can be used as a standalone device or be connected to compatible network hubs and network switches to form a larger network. The modular communication hub **203** is generally easy to install, configure, and maintain, making it a good option for networking the operating theater devices **1a-1n/2a-2m**.

[0285] FIG. **9** illustrates a computer-implemented interactive surgical system **200**. The computer-implemented interactive surgical system **200** is similar in many respects to the computer-implemented interactive surgical system **100**. For example, the computer-implemented interactive surgical system **200** includes one or more surgical systems **202**, which are similar in many respects to the surgical systems **102**. Each surgical system **202** includes at least one surgical hub **206** in communication with a cloud **204** that may include a remote server **213**. In one aspect, the computer-implemented interactive surgical system **200** comprises a modular control tower **236** connected to multiple operating theater devices such as, for example, intelligent surgical instruments, robots, and other computerized devices located in the operating theater. As shown in FIG. **10**, the modular control tower **236** comprises a modular communication hub **203** coupled to a computer system **210**. As illustrated in the example of FIG. **9**, the modular control tower **236** is coupled to an imaging module **238** that is coupled to an endoscope **239**, a generator module **240** that is coupled to an energy device **241**, a smoke evacuator module **226**, a suction/irrigation module

228, a communication module **230**, a processor module **232**, a storage array **234**, a smart device/instrument **235** optionally coupled to a display **237**, and a non-contact sensor module **242**. The operating theater devices are coupled to cloud computing resources and data storage via the modular control tower **236**. A robot hub **222** also may be connected to the modular control tower **236** and to the cloud computing resources. The devices/instruments **235**, visualization systems **208**, among others, may be coupled to the modular control tower **236** via wired or wireless communication standards or protocols, as described herein. The modular control tower **236** may be coupled to a hub display **215** (e.g., monitor, screen) to display and overlay images received from the imaging module, device/instrument display, and/or other visualization systems **208**. The hub display also may display data received from devices connected to the modular control tower in conjunction with images and overlaid images.

[0286] FIG. **10** illustrates a surgical hub **206** comprising a plurality of modules coupled to the modular control tower **236**. The modular control tower **236** comprises a modular communication hub **203**, e.g., a network connectivity device, and a computer system **210** to provide local processing, visualization, and imaging, for example. As shown in FIG. **10**, the modular communication hub **203** may be connected in a tiered configuration to expand the number of modules (e.g., devices) that may be connected to the modular communication hub **203** and transfer data associated with the modules to the computer system **210**, cloud computing resources, or both. As shown in FIG. **10**, each of the network hubs/switches in the modular communication hub **203** includes three downstream ports and one upstream port. The upstream network hub/switch is connected to a processor to provide a communication connection to the cloud computing resources and a local display **217**. Communication to the cloud **204** may be made either through a wired or a wireless communication channel.

[0287] The surgical hub **206** employs a non-contact sensor module **242** to measure the dimensions of the operating theater and generate a map of the surgical theater using either ultrasonic or laser-type non-contact measurement devices. An ultrasound-based non-contact sensor module scans the operating theater by transmitting a burst of ultrasound and receiving the echo when it bounces off the perimeter walls of an operating theater as described under the heading “Surgical Hub Spatial Awareness Within an Operating Room” in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, which is herein incorporated by reference in its entirety, in which the sensor module is configured to determine the size of the operating theater and to adjust Bluetooth-pairing distance limits. A laser-based non-contact sensor module scans the operating theater by transmitting laser light pulses, receiving laser light pulses that bounce off the perimeter walls of the operating theater, and comparing the phase of the transmitted pulse to the received pulse to determine the size of the operating theater and to adjust Bluetooth pairing distance limits, for example.

[0288] The computer system **210** comprises a processor **244** and a network interface **245**. The processor **244** is coupled to a communication module **247**, storage **248**, memory **249**, non-volatile memory **250**, and input/output interface **251** via a system bus. The system bus can be any of several types of bus structure(s) including the memory bus or memory controller, a peripheral bus or external bus, and/or a local bus using any variety of available bus architectures including, but not limited to, 9-bit bus, Industrial Standard Architecture (ISA), Micro-Charmel Architecture (MSA), Extended ISA (EISA), Intelligent Drive Electronics (IDE), VESA Local Bus (VLB), Peripheral Component Interconnect (PCI), USB, Advanced Graphics Port (AGP), Personal Computer Memory Card International Association bus (PCMCIA), Small Computer Systems Interface (SCSI), or any other proprietary bus.

[0289] The processor **244** may be any single-core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In one aspect, the processor may be an LM4F230H5QR ARM Cortex-M4F Processor Core, available from Texas Instruments, for example, comprising an on-chip memory of 256 KB single-cycle flash memory, or other non-

volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle serial random access memory (SRAM), an internal read-only memory (ROM) loaded with StellarisWare® software, a 2 KB electrically erasable programmable read-only memory (EEPROM), and/or one or more pulse width modulation (PWM) modules, one or more quadrature encoder inputs (QEI) analogs, one or more 12-bit analog-to-digital converters (ADCs) with **12** analog input channels, details of which are available for the product datasheet.

[0290] In one aspect, the processor **244** may comprise a safety controller comprising two controller-based families such as TMS570 and RM4x, known under the trade name Hercules ARM Cortex R4, also by Texas Instruments. The safety controller may be configured specifically for IEC 61508 and ISO 26262 safety critical applications, among others, to provide advanced integrated safety features while delivering scalable performance, connectivity, and memory options.

[0291] The system memory includes volatile memory and non-volatile memory. The basic input/output system (BIOS), containing the basic routines to transfer information between elements within the computer system, such as during start-up, is stored in non-volatile memory. For example, the non-volatile memory can include ROM, programmable ROM (PROM), electrically programmable ROM (EPROM), EEPROM, or flash memory. Volatile memory includes random-access memory (RAM), which acts as external cache memory. Moreover, RAM is available in many forms such as SRAM, dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), and direct Rambus RAM (DRRAM).

[0292] The computer system **210** also includes removable/non-removable, volatile/non-volatile computer storage media, such as for example disk storage. The disk storage includes, but is not limited to, devices like a magnetic disk drive, floppy disk drive, tape drive, Jaz drive, Zip drive, LS-60 drive, flash memory card, or memory stick. In addition, the disk storage can include storage media separately or in combination with other storage media including, but not limited to, an optical disc drive such as a compact disc ROM device (CD-ROM), compact disc recordable drive (CD-R Drive), compact disc rewritable drive (CD-RW Drive), or a digital versatile disc ROM drive (DVD-ROM). To facilitate the connection of the disk storage devices to the system bus, a removable or non-removable interface may be employed.

[0293] It is to be appreciated that the computer system **210** includes software that acts as an intermediary between users and the basic computer resources described in a suitable operating environment. Such software includes an operating system. The operating system, which can be stored on the disk storage, acts to control and allocate resources of the computer system. System applications take advantage of the management of resources by the operating system through program modules and program data stored either in the system memory or on the disk storage. It is to be appreciated that various components described herein can be implemented with various operating systems or combinations of operating systems.

[0294] A user enters commands or information into the computer system **210** through input device(s) coupled to the I/O interface **251**. The input devices include, but are not limited to, a pointing device such as a mouse, trackball, stylus, touch pad, keyboard, microphone, joystick, game pad, satellite dish, scanner, TV tuner card, digital camera, digital video camera, web camera, and the like. These and other input devices connect to the processor through the system bus via interface port(s). The interface port(s) include, for example, a serial port, a parallel port, a game port, and a USB. The output device(s) use some of the same types of ports as input device(s). Thus, for example, a USB port may be used to provide input to the computer system and to output information from the computer system to an output device. An output adapter is provided to illustrate that there are some output devices like monitors, displays, speakers, and printers, among other output devices that require special adapters. The output adapters include, by way of illustration and not limitation, video and sound cards that provide a means of connection between the output device and the system bus. It should be noted that other devices and/or systems of

devices, such as remote computer(s), provide both input and output capabilities.

[0295] The computer system **210** can operate in a networked environment using logical connections to one or more remote computers, such as cloud computer(s), or local computers. The remote cloud computer(s) can be a personal computer, server, router, network PC, workstation, microprocessor-based appliance, peer device, or other common network node, and the like, and typically includes many or all of the elements described relative to the computer system. For purposes of brevity, only a memory storage device is illustrated with the remote computer(s). The remote computer(s) is logically connected to the computer system through a network interface and then physically connected via a communication connection. The network interface encompasses communication networks such as local area networks (LANs) and wide area networks (WANs). LAN technologies include Fiber Distributed Data Interface (FDDI), Copper Distributed Data Interface (CDDI), Ethernet/IEEE 802.3, Token Ring/IEEE 802.5 and the like. WAN technologies include, but are not limited to, point-to-point links, circuit-switching networks like Integrated Services Digital Networks (ISDN) and variations thereon, packet-switching networks, and Digital Subscriber Lines (DSL).

[0296] In various aspects, the computer system **210** of FIG. **10**, the imaging module **238** and/or visualization system **208**, and/or the processor module **232** of FIGS. **9-10**, may comprise an image processor, image-processing engine, media processor, or any specialized digital signal processor (DSP) used for the processing of digital images. The image processor may employ parallel computing with single instruction, multiple data (SIMD) or multiple instruction, multiple data (MIMD) technologies to increase speed and efficiency. The digital image-processing engine can perform a range of tasks. The image processor may be a system on a chip with multicore processor architecture.

[0297] The communication connection(s) refers to the hardware/software employed to connect the network interface to the bus. While the communication connection is shown for illustrative clarity inside the computer system, it can also be external to the computer system **210**. The hardware/software necessary for connection to the network interface includes, for illustrative purposes only, internal and external technologies such as modems, including regular telephone-grade modems, cable modems, and DSL modems, ISDN adapters, and Ethernet cards.

[0298] FIG. **11** illustrates a functional block diagram of one aspect of a USB network hub **300** device, in accordance with at least one aspect of the present disclosure. In the illustrated aspect, the USB network hub device **300** employs a TUSB2036 integrated circuit hub by Texas Instruments. The USB network hub **300** is a CMOS device that provides an upstream USB transceiver port **302** and up to three downstream USB transceiver ports **304**, **306**, **308** in compliance with the USB 2.0 specification. The upstream USB transceiver port **302** is a differential root data port comprising a differential data minus (DM0) input paired with a differential data plus (DP0) input. The three downstream USB transceiver ports **304**, **306**, **308** are differential data ports where each port includes differential data plus (DP1-DP3) outputs paired with differential data minus (DM1-DM3) outputs.

[0299] The USB network hub **300** device is implemented with a digital state machine instead of a microcontroller, and no firmware programming is required. Fully compliant USB transceivers are integrated into the circuit for the upstream USB transceiver port **302** and all downstream USB transceiver ports **304**, **306**, **308**. The downstream USB transceiver ports **304**, **306**, **308** support both full-speed and low-speed devices by automatically setting the slew rate according to the speed of the device attached to the ports. The USB network hub **300** device may be configured either in bus-powered or self-powered mode and includes a hub power logic **312** to manage power.

[0300] The USB network hub **300** device includes a serial interface engine **310** (SIE). The SIE **310** is the front end of the USB network hub **300** hardware and handles most of the protocol described in chapter 8 of the USB specification. The SIE **310** typically comprehends signaling up to the transaction level. The functions that it handles could include: packet recognition, transaction

sequencing, SOP, EOP, RESET, and RESUME signal detection/generation, clock/data separation, non-return-to-zero invert (NRZI) data encoding/decoding and bit-stuffing, CRC generation and checking (token and data), packet ID (PID) generation and checking/decoding, and/or serial-parallel/parallel-serial conversion. The **310** receives a clock input **314** and is coupled to a suspend/resume logic and frame timer **316** circuit and a hub repeater circuit **318** to control communication between the upstream USB transceiver port **302** and the downstream USB transceiver ports **304**, **306**, **308** through port logic circuits **320**, **322**, **324**. The SIE **310** is coupled to a command decoder **326** via interface logic **328** to control commands from a serial EEPROM via a serial EEPROM interface **330**.

[0301] In various aspects, the USB network hub **300** can connect **127** functions configured in up to six logical layers (tiers) to a single computer. Further, the USB network hub **300** can connect to all peripherals using a standardized four-wire cable that provides both communication and power distribution. The power configurations are bus-powered and self-powered modes. The USB network hub **300** may be configured to support four modes of power management: a bus-powered hub, with either individual-port power management or ganged-port power management, and the self-powered hub, with either individual-port power management or ganged-port power management. In one aspect, using a USB cable, the USB network hub **300**, the upstream USB transceiver port **302** is plugged into a USB host controller, and the downstream USB transceiver ports **304**, **306**, **308** are exposed for connecting USB compatible devices, and so forth.

[0302] Additional details regarding the structure and function of the surgical hub and/or surgical hub networks can be found in U.S. Provisional Patent Application No. 62/659,900, titled METHOD OF HUB COMMUNICATION, filed Apr. 19, 2018, which is hereby incorporated by reference herein in its entirety.

Cloud System Hardware and Functional Modules

[0303] FIG. **12** is a block diagram of the computer-implemented interactive surgical system, in accordance with at least one aspect of the present disclosure. In one aspect, the computer-implemented interactive surgical system is configured to monitor and analyze data related to the operation of various surgical systems that include surgical hubs, surgical instruments, robotic devices and operating theaters or healthcare facilities. The computer-implemented interactive surgical system comprises a cloud-based analytics system. Although the cloud-based analytics system is described as a surgical system, it is not necessarily limited as such and could be a cloud-based medical system generally. As illustrated in FIG. **12**, the cloud-based analytics system comprises a plurality of surgical instruments **7012** (may be the same or similar to instruments **112**), a plurality of surgical hubs **7006** (may be the same or similar to hubs **106**), and a surgical data network **7001** (may be the same or similar to network **201**) to couple the surgical hubs **7006** to the cloud **7004** (may be the same or similar to cloud **204**). Each of the plurality of surgical hubs **7006** is communicatively coupled to one or more surgical instruments **7012**. The hubs **7006** are also communicatively coupled to the cloud **7004** of the computer-implemented interactive surgical system via the network **7001**. The cloud **7004** is a remote centralized source of hardware and software for storing, manipulating, and communicating data generated based on the operation of various surgical systems. As shown in FIG. **12**, access to the cloud **7004** is achieved via the network **7001**, which may be the Internet or some other suitable computer network. Surgical hubs **7006** that are coupled to the cloud **7004** can be considered the client side of the cloud computing system (i.e., cloud-based analytics system). Surgical instruments **7012** are paired with the surgical hubs **7006** for control and implementation of various surgical procedures or operations as described herein.

[0304] In addition, surgical instruments **7012** may comprise transceivers for data transmission to and from their corresponding surgical hubs **7006** (which may also comprise transceivers). Combinations of surgical instruments **7012** and corresponding hubs **7006** may indicate particular locations, such as operating theaters in healthcare facilities (e.g., hospitals), for providing medical

operations. For example, the memory of a surgical hub **7006** may store location data. As shown in FIG. **12**, the cloud **7004** comprises central servers **7013** (which may be same or similar to remote server **113** in FIG. **1** and/or remote server **213** in FIG. **9**), hub application servers **7002**, data analytics modules **7034**, and an input/output (“I/O”) interface **7007**. The central servers **7013** of the cloud **7004** collectively administer the cloud computing system, which includes monitoring requests by client surgical hubs **7006** and managing the processing capacity of the cloud **7004** for executing the requests. Each of the central servers **7013** comprises one or more processors **7008** coupled to suitable memory devices **7010** which can include volatile memory such as random-access memory (RAM) and non-volatile memory such as magnetic storage devices. The memory devices **7010** may comprise machine executable instructions that when executed cause the processors **7008** to execute the data analytics modules **7034** for the cloud-based data analysis, operations, recommendations and other operations described below. Moreover, the processors **7008** can execute the data analytics modules **7034** independently or in conjunction with hub applications independently executed by the hubs **7006**. The central servers **7013** also comprise aggregated medical data databases **2212**, which can reside in the memory **2210**.

[0305] Based on connections to various surgical hubs **7006** via the network **7001**, the cloud **7004** can aggregate data from specific data generated by various surgical instruments **7012** and their corresponding hubs **7006**. Such aggregated data may be stored within the aggregated medical databases **7011** of the cloud **7004**. In particular, the cloud **7004** may advantageously perform data analysis and operations on the aggregated data to yield insights and/or perform functions that individual hubs **7006** could not achieve on their own. To this end, as shown in FIG. **12**, the cloud **7004** and the surgical hubs **7006** are communicatively coupled to transmit and receive information. The I/O interface **7007** is connected to the plurality of surgical hubs **7006** via the network **7001**. In this way, the I/O interface **7007** can be configured to transfer information between the surgical hubs **7006** and the aggregated medical data databases **7011**. Accordingly, the I/O interface **7007** may facilitate read/write operations of the cloud-based analytics system. Such read/write operations may be executed in response to requests from hubs **7006**. These requests could be transmitted to the hubs **7006** through the hub applications. The I/O interface **7007** may include one or more high speed data ports, which may include universal serial bus (USB) ports, IEEE 1394 ports, as well as Wi-Fi and Bluetooth I/O interfaces for connecting the cloud **7004** to hubs **7006**. The hub application servers **7002** of the cloud **7004** are configured to host and supply shared capabilities to software applications (e.g. hub applications) executed by surgical hubs **7006**. For example, the hub application servers **7002** may manage requests made by the hub applications through the hubs **7006**, control access to the aggregated medical data databases **7011**, and perform load balancing. The data analytics modules **7034** are described in further detail with reference to FIG. **13**.

[0306] The particular cloud computing system configuration described in the present disclosure is specifically designed to address various issues arising in the context of medical operations and procedures performed using medical devices, such as the surgical instruments **7012**, **112**. In particular, the surgical instruments **7012** may be digital surgical devices configured to interact with the cloud **7004** for implementing techniques to improve the performance of surgical operations. Various surgical instruments **7012** and/or surgical hubs **7006** may comprise touch controlled user interfaces such that clinicians may control aspects of interaction between the surgical instruments **7012** and the cloud **7004**. Other suitable user interfaces for control such as auditory controlled user interfaces can also be used.

[0307] FIG. **13** is a block diagram which illustrates the functional architecture of the computer-implemented interactive surgical system, in accordance with at least one aspect of the present disclosure. The cloud-based analytics system includes a plurality of data analytics modules **7034** that may be executed by the processors **7008** of the cloud **7004** for providing data analytic solutions to problems specifically arising in the medical field. As shown in FIG. **13**, the functions of the cloud-based data analytics modules **7034** may be assisted via hub applications **7014** hosted

by the hub application servers **7002** that may be accessed on surgical hubs **7006**. The cloud processors **7008** and hub applications **7014** may operate in conjunction to execute the data analytics modules **7034**. Application program interfaces (APIs) **7016** define the set of protocols and routines corresponding to the hub applications **7014**. Additionally, the APIs **7016** manage the storing and retrieval of data into and from the aggregated medical data databases **7011** for the operations of the applications **7014**. The caches **7018** also store data (e.g., temporarily) and are coupled to the APIs **7016** for more efficient retrieval of data used by the applications **7014**. The data analytics modules **7034** in FIG. 13 include modules for resource optimization **7020**, data collection and aggregation **7022**, authorization and security **7024**, control program updating **7026**, patient outcome analysis **7028**, recommendations **7030**, and data sorting and prioritization **7032**. Other suitable data analytics modules could also be implemented by the cloud **7004**, according to some aspects. In one aspect, the data analytics modules are used for specific recommendations based on analyzing trends, outcomes, and other data.

[0308] For example, the data collection and aggregation module **7022** could be used to generate self-describing data (e.g., metadata) including identification of notable features or configuration (e.g., trends), management of redundant data sets, and storage of the data in paired data sets which can be grouped by surgery but not necessarily keyed to actual surgical dates and surgeons. In particular, pair data sets generated from operations of surgical instruments **7012** can comprise applying a binary classification, e.g., a bleeding or a non-bleeding event. More generally, the binary classification may be characterized as either a desirable event (e.g., a successful surgical procedure) or an undesirable event (e.g., a misfired or misused surgical instrument **7012**). The aggregated self-describing data may correspond to individual data received from various groups or subgroups of surgical hubs **7006**. Accordingly, the data collection and aggregation module **7022** can generate aggregated metadata or other organized data based on raw data received from the surgical hubs **7006**. To this end, the processors **7008** can be operationally coupled to the hub applications **7014** and aggregated medical data databases **7011** for executing the data analytics modules **7034**. The data collection and aggregation module **7022** may store the aggregated organized data into the aggregated medical data databases **2212**.

[0309] The resource optimization module **7020** can be configured to analyze this aggregated data to determine an optimal usage of resources for a particular or group of healthcare facilities. For example, the resource optimization module **7020** may determine an optimal order point of surgical stapling instruments **7012** for a group of healthcare facilities based on corresponding predicted demand of such instruments **7012**. The resource optimization module **7020** might also assess the resource usage or other operational configurations of various healthcare facilities to determine whether resource usage could be improved. Similarly, the recommendations module **7030** can be configured to analyze aggregated organized data from the data collection and aggregation module **7022** to provide recommendations. For example, the recommendations module **7030** could recommend to healthcare facilities (e.g., medical service providers such as hospitals) that a particular surgical instrument **7012** should be upgraded to an improved version based on a higher than expected error rate, for example. Additionally, the recommendations module **7030** and/or resource optimization module **7020** could recommend better supply chain parameters such as product reorder points and provide suggestions of different surgical instrument **7012**, uses thereof, or procedure steps to improve surgical outcomes. The healthcare facilities can receive such recommendations via corresponding surgical hubs **7006**. More specific recommendations regarding parameters or configurations of various surgical instruments **7012** can also be provided. Hubs **7006** and/or surgical instruments **7012** each could also have display screens that display data or recommendations provided by the cloud **7004**.

[0310] The patient outcome analysis module **7028** can analyze surgical outcomes associated with currently used operational parameters of surgical instruments **7012**. The patient outcome analysis module **7028** may also analyze and assess other potential operational parameters. In this

connection, the recommendations module **7030** could recommend using these other potential operational parameters based on yielding better surgical outcomes, such as better sealing or less bleeding. For example, the recommendations module **7030** could transmit recommendations to a surgical hub **7006** regarding when to use a particular cartridge for a corresponding stapling surgical instrument **7012**. Thus, the cloud-based analytics system, while controlling for common variables, may be configured to analyze the large collection of raw data and to provide centralized recommendations over multiple healthcare facilities (advantageously determined based on aggregated data). For example, the cloud-based analytics system could analyze, evaluate, and/or aggregate data based on type of medical practice, type of patient, number of patients, geographic similarity between medical providers, which medical providers/facilities use similar types of instruments, etc., in a way that no single healthcare facility alone would be able to analyze independently.

[0311] The control program updating module **7026** could be configured to implement various surgical instrument **7012** recommendations when corresponding control programs are updated. For example, the patient outcome analysis module **7028** could identify correlations linking specific control parameters with successful (or unsuccessful) results. Such correlations may be addressed when updated control programs are transmitted to surgical instruments **7012** via the control program updating module **7026**. Updates to instruments **7012** that are transmitted via a corresponding hub **7006** may incorporate aggregated performance data that was gathered and analyzed by the data collection and aggregation module **7022** of the cloud **7004**. Additionally, the patient outcome analysis module **7028** and recommendations module **7030** could identify improved methods of using instruments **7012** based on aggregated performance data.

[0312] The cloud-based analytics system may include security features implemented by the cloud **7004**. These security features may be managed by the authorization and security module **7024**. Each surgical hub **7006** can have associated unique credentials such as username, password, and other suitable security credentials. These credentials could be stored in the memory **7010** and be associated with a permitted cloud access level. For example, based on providing accurate credentials, a surgical hub **7006** may be granted access to communicate with the cloud to a predetermined extent (e.g., may only engage in transmitting or receiving certain defined types of information). To this end, the aggregated medical data databases **7011** of the cloud **7004** may comprise a database of authorized credentials for verifying the accuracy of provided credentials. Different credentials may be associated with varying levels of permission for interaction with the cloud **7004**, such as a predetermined access level for receiving the data analytics generated by the cloud **7004**.

[0313] Furthermore, for security purposes, the cloud could maintain a database of hubs **7006**, instruments **7012**, and other devices that may comprise a “black list” of prohibited devices. In particular, a surgical hub **7006** listed on the black list may not be permitted to interact with the cloud, while surgical instruments **7012** listed on the black list may not have functional access to a corresponding hub **7006** and/or may be prevented from fully functioning when paired to its corresponding hub **7006**. Additionally or alternatively, the cloud **7004** may flag instruments **7012** based on incompatibility or other specified criteria. In this manner, counterfeit medical devices and improper reuse of such devices throughout the cloud-based analytics system can be identified and addressed.

[0314] The surgical instruments **7012** may use wireless transceivers to transmit wireless signals that may represent, for example, authorization credentials for access to corresponding hubs **7006** and the cloud **7004**. Wired transceivers may also be used to transmit signals. Such authorization credentials can be stored in the respective memory devices of the surgical instruments **7012**. The authorization and security module **7024** can determine whether the authorization credentials are accurate or counterfeit. The authorization and security module **7024** may also dynamically generate authorization credentials for enhanced security. The credentials could also be encrypted, such as by

using hash based encryption. Upon transmitting proper authorization, the surgical instruments **7012** may transmit a signal to the corresponding hubs **7006** and ultimately the cloud **7004** to indicate that the instruments **7012** are ready to obtain and transmit medical data. In response, the cloud **7004** may transition into a state enabled for receiving medical data for storage into the aggregated medical data databases **7011**. This data transmission readiness could be indicated by a light indicator on the instruments **7012**, for example. The cloud **7004** can also transmit signals to surgical instruments **7012** for updating their associated control programs. The cloud **7004** can transmit signals that are directed to a particular class of surgical instruments **7012** (e.g., electrosurgical instruments) so that software updates to control programs are only transmitted to the appropriate surgical instruments **7012**. Moreover, the cloud **7004** could be used to implement system wide solutions to address local or global problems based on selective data transmission and authorization credentials. For example, if a group of surgical instruments **7012** are identified as having a common manufacturing defect, the cloud **7004** may change the authorization credentials corresponding to this group to implement an operational lockout of the group.

[0315] The cloud-based analytics system may allow for monitoring multiple healthcare facilities (e.g., medical facilities like hospitals) to determine improved practices and recommend changes (via the recommendations module **2030**, for example) accordingly. Thus, the processors **7008** of the cloud **7004** can analyze data associated with an individual healthcare facility to identify the facility and aggregate the data with other data associated with other healthcare facilities in a group. Groups could be defined based on similar operating practices or geographical location, for example. In this way, the cloud **7004** may provide healthcare facility group wide analysis and recommendations.

The cloud-based analytics system could also be used for enhanced situational awareness. For example, the processors **7008** may predictively model the effects of recommendations on the cost and effectiveness for a particular facility (relative to overall operations and/or various medical procedures). The cost and effectiveness associated with that particular facility can also be compared to a corresponding local region of other facilities or any other comparable facilities.

[0316] The data sorting and prioritization module **7032** may prioritize and sort data based on criticality (e.g., the severity of a medical event associated with the data, unexpectedness, suspiciousness). This sorting and prioritization may be used in conjunction with the functions of the other data analytics modules **7034** described above to improve the cloud-based analytics and operations described herein. For example, the data sorting and prioritization module **7032** can assign a priority to the data analysis performed by the data collection and aggregation module **7022** and patient outcome analysis modules **7028**. Different prioritization levels can result in particular responses from the cloud **7004** (corresponding to a level of urgency) such as escalation for an expedited response, special processing, exclusion from the aggregated medical data databases **7011**, or other suitable responses. Moreover, if necessary, the cloud **7004** can transmit a request (e.g. a push message) through the hub application servers for additional data from corresponding surgical instruments **7012**. The push message can result in a notification displayed on the corresponding hubs **7006** for requesting supporting or additional data. This push message may be required in situations in which the cloud detects a significant irregularity or outlier and the cloud cannot determine the cause of the irregularity. The central servers **7013** may be programmed to trigger this push message in certain significant circumstances, such as when data is determined to be different from an expected value beyond a predetermined threshold or when it appears security has been comprised, for example.

[0317] Additional details regarding the cloud analysis system can be found in U.S. Provisional Patent Application No. 62/659,900, titled METHOD OF HUB COMMUNICATION, filed Apr. 19, 2018, which is hereby incorporated by reference herein in its entirety.

Situational Awareness

[0318] Although an “intelligent” device including control algorithms that respond to sensed data can be an improvement over a “dumb” device that operates without accounting for sensed data,

some sensed data can be incomplete or inconclusive when considered in isolation, i.e., without the context of the type of surgical procedure being performed or the type of tissue that is being operated on. Without knowing the procedural context (e.g., knowing the type of tissue being operated on or the type of procedure being performed), the control algorithm may control the modular device incorrectly or suboptimally given the particular context-free sensed data. For example, the optimal manner for a control algorithm to control a surgical instrument in response to a particular sensed parameter can vary according to the particular tissue type being operated on. This is due to the fact that different tissue types have different properties (e.g., resistance to tearing) and thus respond differently to actions taken by surgical instruments. Therefore, it may be desirable for a surgical instrument to take different actions even when the same measurement for a particular parameter is sensed. As one specific example, the optimal manner in which to control a surgical stapling and cutting instrument in response to the instrument sensing an unexpectedly high force to close its end effector will vary depending upon whether the tissue type is susceptible or resistant to tearing. For tissues that are susceptible to tearing, such as lung tissue, the instrument's control algorithm would optimally ramp down the motor in response to an unexpectedly high force to close to avoid tearing the tissue. For tissues that are resistant to tearing, such as stomach tissue, the instrument's control algorithm would optimally ramp up the motor in response to an unexpectedly high force to close to ensure that the end effector is clamped properly on the tissue. Without knowing whether lung or stomach tissue has been clamped, the control algorithm may make a suboptimal decision.

[0319] One solution utilizes a surgical hub including a system that is configured to derive information about the surgical procedure being performed based on data received from various data sources and then control the paired modular devices accordingly. In other words, the surgical hub is configured to infer information about the surgical procedure from received data and then control the modular devices paired to the surgical hub based upon the inferred context of the surgical procedure. FIG. 14 illustrates a diagram of a situationally aware surgical system **5100**, in accordance with at least one aspect of the present disclosure. In some exemplifications, the data sources **5126** include, for example, the modular devices **5102** (which can include sensors configured to detect parameters associated with the patient and/or the modular device itself), databases **5122** (e.g., an EMR database containing patient records), and patient monitoring devices **5124** (e.g., a blood pressure (BP) monitor and an electrocardiography (EKG) monitor).

[0320] A surgical hub **5104**, which may be similar to the hub **106** in many respects, can be configured to derive the contextual information pertaining to the surgical procedure from the data based upon, for example, the particular combination(s) of received data or the particular order in which the data is received from the data sources **5126**. The contextual information inferred from the received data can include, for example, the type of surgical procedure being performed, the particular step of the surgical procedure that the surgeon is performing, the type of tissue being operated on, or the body cavity that is the subject of the procedure. This ability by some aspects of the surgical hub **5104** to derive or infer information related to the surgical procedure from received data can be referred to as “situational awareness.” In one exemplification, the surgical hub **5104** can incorporate a situational awareness system, which is the hardware and/or programming associated with the surgical hub **5104** that derives contextual information pertaining to the surgical procedure from the received data.

[0321] The situational awareness system of the surgical hub **5104** can be configured to derive the contextual information from the data received from the data sources **5126** in a variety of different ways. In one exemplification, the situational awareness system includes a pattern recognition system, or machine learning system (e.g., an artificial neural network), that has been trained on training data to correlate various inputs (e.g., data from databases **5122**, patient monitoring devices **5124**, and/or modular devices **5102**) to corresponding contextual information regarding a surgical procedure. In other words, a machine learning system can be trained to accurately derive contextual

information regarding a surgical procedure from the provided inputs. In another exemplification, the situational awareness system can include a lookup table storing pre-characterized contextual information regarding a surgical procedure in association with one or more inputs (or ranges of inputs) corresponding to the contextual information. In response to a query with one or more inputs, the lookup table can return the corresponding contextual information for the situational awareness system for controlling the modular devices **5102**. In one exemplification, the contextual information received by the situational awareness system of the surgical hub **5104** is associated with a particular control adjustment or set of control adjustments for one or more modular devices **5102**. In another exemplification, the situational awareness system includes a further machine learning system, lookup table, or other such system, which generates or retrieves one or more control adjustments for one or more modular devices **5102** when provided the contextual information as input.

[0322] A surgical hub **5104** incorporating a situational awareness system provides a number of benefits for the surgical system **5100**. One benefit includes improving the interpretation of sensed and collected data, which would in turn improve the processing accuracy and/or the usage of the data during the course of a surgical procedure. To return to a previous example, a situationally aware surgical hub **5104** could determine what type of tissue was being operated on; therefore, when an unexpectedly high force to close the surgical instrument's end effector is detected, the situationally aware surgical hub **5104** could correctly ramp up or ramp down the motor of the surgical instrument for the type of tissue.

[0323] As another example, the type of tissue being operated can affect the adjustments that are made to the compression rate and load thresholds of a surgical stapling and cutting instrument for a particular tissue gap measurement. A situationally aware surgical hub **5104** could infer whether a surgical procedure being performed is a thoracic or an abdominal procedure, allowing the surgical hub **5104** to determine whether the tissue clamped by an end effector of the surgical stapling and cutting instrument is lung (for a thoracic procedure) or stomach (for an abdominal procedure) tissue. The surgical hub **5104** could then adjust the compression rate and load thresholds of the surgical stapling and cutting instrument appropriately for the type of tissue.

[0324] As yet another example, the type of body cavity being operated in during an insufflation procedure can affect the function of a smoke evacuator. A situationally aware surgical hub **5104** could determine whether the surgical site is under pressure (by determining that the surgical procedure is utilizing insufflation) and determine the procedure type. As a procedure type is generally performed in a specific body cavity, the surgical hub **5104** could then control the motor rate of the smoke evacuator appropriately for the body cavity being operated in. Thus, a situationally aware surgical hub **5104** could provide a consistent amount of smoke evacuation for both thoracic and abdominal procedures.

[0325] As yet another example, the type of procedure being performed can affect the optimal energy level for an ultrasonic surgical instrument or radio frequency (RF) electrosurgical instrument to operate at. Arthroscopic procedures, for example, require higher energy levels because the end effector of the ultrasonic surgical instrument or RF electrosurgical instrument is immersed in fluid. A situationally aware surgical hub **5104** could determine whether the surgical procedure is an arthroscopic procedure. The surgical hub **5104** could then adjust the RF power level or the ultrasonic amplitude of the generator (i.e., "energy level") to compensate for the fluid filled environment. Relatedly, the type of tissue being operated on can affect the optimal energy level for an ultrasonic surgical instrument or RF electrosurgical instrument to operate at. A situationally aware surgical hub **5104** could determine what type of surgical procedure is being performed and then customize the energy level for the ultrasonic surgical instrument or RF electrosurgical instrument, respectively, according to the expected tissue profile for the surgical procedure. Furthermore, a situationally aware surgical hub **5104** can be configured to adjust the energy level for the ultrasonic surgical instrument or RF electrosurgical instrument throughout the

course of a surgical procedure, rather than just on a procedure-by-procedure basis. A situationally aware surgical hub **5104** could determine what step of the surgical procedure is being performed or will subsequently be performed and then update the control algorithms for the generator and/or ultrasonic surgical instrument or RF electrosurgical instrument to set the energy level at a value appropriate for the expected tissue type according to the surgical procedure step.

[0326] As yet another example, data can be drawn from additional data sources **5126** to improve the conclusions that the surgical hub **5104** draws from one data source **5126**. A situationally aware surgical hub **5104** could augment data that it receives from the modular devices **5102** with contextual information that it has built up regarding the surgical procedure from other data sources **5126**. For example, a situationally aware surgical hub **5104** can be configured to determine whether hemostasis has occurred (i.e., whether bleeding at a surgical site has stopped) according to video or image data received from a medical imaging device. However, in some cases the video or image data can be inconclusive. Therefore, in one exemplification, the surgical hub **5104** can be further configured to compare a physiologic measurement (e.g., blood pressure sensed by a BP monitor communicably connected to the surgical hub **5104**) with the visual or image data of hemostasis (e.g., from a medical imaging device **124** (FIG. 2) communicably coupled to the surgical hub **5104**) to make a determination on the integrity of the staple line or tissue weld. In other words, the situational awareness system of the surgical hub **5104** can consider the physiological measurement data to provide additional context in analyzing the visualization data. The additional context can be useful when the visualization data may be inconclusive or incomplete on its own.

[0327] Another benefit includes proactively and automatically controlling the paired modular devices **5102** according to the particular step of the surgical procedure that is being performed to reduce the number of times that medical personnel are required to interact with or control the surgical system **5100** during the course of a surgical procedure. For example, a situationally aware surgical hub **5104** could proactively activate the generator to which an RF electrosurgical instrument is connected if it determines that a subsequent step of the procedure requires the use of the instrument. Proactively activating the energy source allows the instrument to be ready for use as soon as the preceding step of the procedure is completed.

[0328] As another example, a situationally aware surgical hub **5104** could determine whether the current or subsequent step of the surgical procedure requires a different view or degree of magnification on the display according to the feature(s) at the surgical site that the surgeon is expected to need to view. The surgical hub **5104** could then proactively change the displayed view (supplied by, e.g., a medical imaging device for the visualization system **108**) accordingly so that the display automatically adjusts throughout the surgical procedure.

[0329] As yet another example, a situationally aware surgical hub **5104** could determine which step of the surgical procedure is being performed or will subsequently be performed and whether particular data or comparisons between data will be required for that step of the surgical procedure. The surgical hub **5104** can be configured to automatically call up data screens based upon the step of the surgical procedure being performed, without waiting for the surgeon to ask for the particular information.

[0330] Another benefit includes checking for errors during the setup of the surgical procedure or during the course of the surgical procedure. For example, a situationally aware surgical hub **5104** could determine whether the operating theater is setup properly or optimally for the surgical procedure to be performed. The surgical hub **5104** can be configured to determine the type of surgical procedure being performed, retrieve the corresponding checklists, product location, or setup needs (e.g., from a memory), and then compare the current operating theater layout to the standard layout for the type of surgical procedure that the surgical hub **5104** determines is being performed. In one exemplification, the surgical hub **5104** can be configured to compare the list of items for the procedure scanned by a suitable scanner for example, and/or a list of devices paired with the surgical hub **5104** to a recommended or anticipated manifest of items and/or devices for

the given surgical procedure. If there are any discontinuities between the lists, the surgical hub **5104** can be configured to provide an alert indicating that a particular modular device **5102**, patient monitoring device **5124**, and/or other surgical item is missing. In one exemplification, the surgical hub **5104** can be configured to determine the relative distance or position of the modular devices **5102** and patient monitoring devices **5124** via proximity sensors, for example. The surgical hub **5104** can compare the relative positions of the devices to a recommended or anticipated layout for the particular surgical procedure. If there are any discontinuities between the layouts, the surgical hub **5104** can be configured to provide an alert indicating that the current layout for the surgical procedure deviates from the recommended layout.

[0331] As another example, a situationally aware surgical hub **5104** could determine whether the surgeon (or other medical personnel) was making an error or otherwise deviating from the expected course of action during the course of a surgical procedure. For example, the surgical hub **5104** can be configured to determine the type of surgical procedure being performed, retrieve the corresponding list of steps or order of equipment usage (e.g., from a memory), and then compare the steps being performed or the equipment being used during the course of the surgical procedure to the expected steps or equipment for the type of surgical procedure that the surgical hub **5104** determined is being performed. In one exemplification, the surgical hub **5104** can be configured to provide an alert indicating that an unexpected action is being performed or an unexpected device is being utilized at the particular step in the surgical procedure.

[0332] Overall, the situational awareness system for the surgical hub **5104** improves surgical procedure outcomes by adjusting the surgical instruments (and other modular devices **5102**) for the particular context of each surgical procedure (such as adjusting to different tissue types) and validating actions during a surgical procedure. The situational awareness system also improves surgeons' efficiency in performing surgical procedures by automatically suggesting next steps, providing data, and adjusting displays and other modular devices **5102** in the surgical theater according to the specific context of the procedure.

[0333] Referring now to FIG. 15, a timeline **5200** depicting situational awareness of a hub, such as the surgical hub **106** or **206** (FIGS. 1-11), for example, is depicted. The timeline **5200** is an illustrative surgical procedure and the contextual information that the surgical hub **106**, **206** can derive from the data received from the data sources at each step in the surgical procedure. The timeline **5200** depicts the typical steps that would be taken by the nurses, surgeons, and other medical personnel during the course of a lung segmentectomy procedure, beginning with setting up the operating theater and ending with transferring the patient to a post-operative recovery room.

[0334] The situationally aware surgical hub **106**, **206** receives data from the data sources throughout the course of the surgical procedure, including data generated each time medical personnel utilize a modular device that is paired with the surgical hub **106**, **206**. The surgical hub **106**, **206** can receive this data from the paired modular devices and other data sources and continually derive inferences (i.e., contextual information) about the ongoing procedure as new data is received, such as which step of the procedure is being performed at any given time. The situational awareness system of the surgical hub **106**, **206** is able to, for example, record data pertaining to the procedure for generating reports, verify the steps being taken by the medical personnel, provide data or prompts (e.g., via a display screen) that may be pertinent for the particular procedural step, adjust modular devices based on the context (e.g., activate monitors, adjust the field of view (FOV) of the medical imaging device, or change the energy level of an ultrasonic surgical instrument or RF electrosurgical instrument), and take any other such action described above.

[0335] As the first step **5202** in this illustrative procedure, the hospital staff members retrieve the patient's EMR from the hospital's EMR database. Based on select patient data in the EMR, the surgical hub **106**, **206** determines that the procedure to be performed is a thoracic procedure.

[0336] Second step **5204**, the staff members scan the incoming medical supplies for the procedure.

The surgical hub **106, 206** cross-references the scanned supplies with a list of supplies that are utilized in various types of procedures and confirms that the mix of supplies corresponds to a thoracic procedure. Further, the surgical hub **106, 206** is also able to determine that the procedure is not a wedge procedure (because the incoming supplies either lack certain supplies that are necessary for a thoracic wedge procedure or do not otherwise correspond to a thoracic wedge procedure).

[0337] Third step **5206**, the medical personnel scan the patient band via a scanner that is communicably connected to the surgical hub **106, 206**. The surgical hub **106, 206** can then confirm the patient's identity based on the scanned data.

[0338] Fourth step **5208**, the medical staff turns on the auxiliary equipment. The auxiliary equipment being utilized can vary according to the type of surgical procedure and the techniques to be used by the surgeon, but in this illustrative case they include a smoke evacuator, insufflator, and medical imaging device. When activated, the auxiliary equipment that are modular devices can automatically pair with the surgical hub **106, 206** that is located within a particular vicinity of the modular devices as part of their initialization process. The surgical hub **106, 206** can then derive contextual information about the surgical procedure by detecting the types of modular devices that pair with it during this pre-operative or initialization phase. In this particular example, the surgical hub **106, 206** determines that the surgical procedure is a VATS procedure based on this particular combination of paired modular devices. Based on the combination of the data from the patient's EMR, the list of medical supplies to be used in the procedure, and the type of modular devices that connect to the hub, the surgical hub **106, 206** can generally infer the specific procedure that the surgical team will be performing. Once the surgical hub **106, 206** knows what specific procedure is being performed, the surgical hub **106, 206** can then retrieve the steps of that procedure from a memory or from the cloud and then cross-reference the data it subsequently receives from the connected data sources (e.g., modular devices and patient monitoring devices) to infer what step of the surgical procedure the surgical team is performing.

[0339] Fifth step **5210**, the staff members attach the EKG electrodes and other patient monitoring devices to the patient. The EKG electrodes and other patient monitoring devices are able to pair with the surgical hub **106, 206**. As the surgical hub **106, 206** begins receiving data from the patient monitoring devices, the surgical hub **106, 206** thus confirms that the patient is in the operating theater.

[0340] Sixth step **5212**, the medical personnel induce anesthesia in the patient. The surgical hub **106, 206** can infer that the patient is under anesthesia based on data from the modular devices and/or patient monitoring devices, including EKG data, blood pressure data, ventilator data, or combinations thereof, for example. Upon completion of the sixth step **5212**, the pre-operative portion of the lung segmentectomy procedure is completed and the operative portion begins.

[0341] Seventh step **5214**, the patient's lung that is being operated on is collapsed (while ventilation is switched to the contralateral lung). The surgical hub **106, 206** can infer from the ventilator data that the patient's lung has been collapsed, for example. The surgical hub **106, 206** can infer that the operative portion of the procedure has commenced as it can compare the detection of the patient's lung collapsing to the expected steps of the procedure (which can be accessed or retrieved previously) and thereby determine that collapsing the lung is the first operative step in this particular procedure.

[0342] Eighth step **5216**, the medical imaging device (e.g., a scope) is inserted and video from the medical imaging device is initiated. The surgical hub **106, 206** receives the medical imaging device data (i.e., video or image data) through its connection to the medical imaging device. Upon receipt of the medical imaging device data, the surgical hub **106, 206** can determine that the laparoscopic portion of the surgical procedure has commenced. Further, the surgical hub **106, 206** can determine that the particular procedure being performed is a segmentectomy, as opposed to a lobectomy (note that a wedge procedure has already been discounted by the surgical hub **106, 206** based on data

received at the second step 5204 of the procedure). The data from the medical imaging device 124 (FIG. 2) can be utilized to determine contextual information regarding the type of procedure being performed in a number of different ways, including by determining the angle at which the medical imaging device is oriented with respect to the visualization of the patient's anatomy, monitoring the number or medical imaging devices being utilized (i.e., that are activated and paired with the surgical hub 106, 206), and monitoring the types of visualization devices utilized. For example, one technique for performing a VATS lobectomy places the camera in the lower anterior corner of the patient's chest cavity above the diaphragm, whereas one technique for performing a VATS segmentectomy places the camera in an anterior intercostal position relative to the segmental fissure. Using pattern recognition or machine learning techniques, for example, the situational awareness system can be trained to recognize the positioning of the medical imaging device according to the visualization of the patient's anatomy. As another example, one technique for performing a VATS lobectomy utilizes a single medical imaging device, whereas another technique for performing a VATS segmentectomy utilizes multiple cameras. As yet another example, one technique for performing a VATS segmentectomy utilizes an infrared light source (which can be communicably coupled to the surgical hub as part of the visualization system) to visualize the segmental fissure, which is not utilized in a VATS lobectomy. By tracking any or all of this data from the medical imaging device, the surgical hub 106, 206 can thereby determine the specific type of surgical procedure being performed and/or the technique being used for a particular type of surgical procedure.

[0343] Ninth step 5218, the surgical team begins the dissection step of the procedure. The surgical hub 106, 206 can infer that the surgeon is in the process of dissecting to mobilize the patient's lung because it receives data from the RF or ultrasonic generator indicating that an energy instrument is being fired. The surgical hub 106, 206 can cross-reference the received data with the retrieved steps of the surgical procedure to determine that an energy instrument being fired at this point in the process (i.e., after the completion of the previously discussed steps of the procedure) corresponds to the dissection step. In certain instances, the energy instrument can be an energy tool mounted to a robotic arm of a robotic surgical system.

[0344] Tenth step 5220, the surgical team proceeds to the ligation step of the procedure. The surgical hub 106, 206 can infer that the surgeon is ligating arteries and veins because it receives data from the surgical stapling and cutting instrument indicating that the instrument is being fired. Similarly to the prior step, the surgical hub 106, 206 can derive this inference by cross-referencing the receipt of data from the surgical stapling and cutting instrument with the retrieved steps in the process. In certain instances, the surgical instrument can be a surgical tool mounted to a robotic arm of a robotic surgical system.

[0345] Eleventh step 5222, the segmentectomy portion of the procedure is performed. The surgical hub 106, 206 can infer that the surgeon is transecting the parenchyma based on data from the surgical stapling and cutting instrument, including data from its cartridge. The cartridge data can correspond to the size or type of staple being fired by the instrument, for example. As different types of staples are utilized for different types of tissues, the cartridge data can thus indicate the type of tissue being stapled and/or transected. In this case, the type of staple being fired is utilized for parenchyma (or other similar tissue types), which allows the surgical hub 106, 206 to infer that the segmentectomy portion of the procedure is being performed.

[0346] Twelfth step 5224, the node dissection step is then performed. The surgical hub 106, 206 can infer that the surgical team is dissecting the node and performing a leak test based on data received from the generator indicating that an RF or ultrasonic instrument is being fired. For this particular procedure, an RF or ultrasonic instrument being utilized after parenchyma was transected corresponds to the node dissection step, which allows the surgical hub 106, 206 to make this inference. It should be noted that surgeons regularly switch back and forth between surgical stapling/cutting instruments and surgical energy (i.e., RF or ultrasonic) instruments depending upon

the particular step in the procedure because different instruments are better adapted for particular tasks. Therefore, the particular sequence in which the stapling/cutting instruments and surgical energy instruments are used can indicate what step of the procedure the surgeon is performing. Moreover, in certain instances, robotic tools can be utilized for one or more steps in a surgical procedure and/or handheld surgical instruments can be utilized for one or more steps in the surgical procedure. The surgeon(s) can alternate between robotic tools and handheld surgical instruments and/or can use the devices concurrently, for example. Upon completion of the twelfth step **5224**, the incisions are closed up and the post-operative portion of the procedure begins.

[0347] Thirteenth step **5226**, the patient's anesthesia is reversed. The surgical hub **106, 206** can infer that the patient is emerging from the anesthesia based on the ventilator data (i.e., the patient's breathing rate begins increasing), for example.

[0348] Lastly, the fourteenth step **5228** is that the medical personnel remove the various patient monitoring devices from the patient. The surgical hub **106, 206** can thus infer that the patient is being transferred to a recovery room when the hub loses EKG, BP, and other data from the patient monitoring devices. As can be seen from the description of this illustrative procedure, the surgical hub **106, 206** can determine or infer when each step of a given surgical procedure is taking place according to data received from the various data sources that are communicably coupled to the surgical hub **106, 206**.

[0349] Situational awareness is further described in U.S. Provisional Patent Application Ser. No. 62/659,900, titled METHOD OF HUB COMMUNICATION, filed Apr. 19, 2018, which is herein incorporated by reference in its entirety. In certain instances, operation of a robotic surgical system, including the various robotic surgical systems disclosed herein, for example, can be controlled by the hub **106, 206** based on its situational awareness and/or feedback from the components thereof and/or based on information from the cloud **104**.

Data Manipulation, Analysis, and Storage

[0350] In various aspects, a surgical hub system can be configured to collect rich contextual data pertaining to the use of surgical devices that are connected to the system, providing a hierarchy of awareness for the surgical hub system.

Data Indexing and Storage

[0351] Various techniques are described herein for data transformation, validation, organization, and fusion.

[0352] One issue that arises in the surgical hub system context is how to fuse data from diverse and different sources into a common data set that is useful. For example, what solutions are available to fuse two types of data that are recorded at different sampling rates? Can systems be made to be sampled at a similar rate? Can one insert a timing signal into all data to assist in synchronizing the data sets? The solution selected for various applications can depend on the particular types of surgical devices that are collecting the data sets being fused and other such factors.

[0353] In one aspect, the surgical hub can be configured to perform automated data scaling, alignment, and organizing of the collected data based on predefined parameters within the surgical hub prior to the data being transmitted. In one aspect, the predefined parameters could be established or altered by interaction between the surgical hub and the cloud system when configured for use. The cloud system may provide solutions from other hubs in the network that addressed a similar problem, and may also provide offline processing using various learning mechanisms to determine how to align the data. This enables the data collected by the surgical hub to be directly integrated into a larger cloud based database for analysis. In another aspect, the surgical hub and/or cloud system can be configured to change the sampling rate of measured systems. For example, the datasets could be organized into a functional database and/or analyzed via functional data analysis. For example, the system can be configured to include computational tabulation of multiple measures from the same or different devices into a single measure. For example, the system can be configured to include a hash function to encrypt or authenticate the

source or sources of the data.

Data Wrangling

[0354] In one aspect, the surgical hub system can be configured to perform “data wrangling” or “munging,” i.e., the reorganization of raw data into a usable form. For example, the surgical hub system can be configured to organize data from the surgical hub and other equipment into a unified dataset.

Data Warehousing and Fusing

[0355] One challenge with integrating and fusing datasets are differing data rates, data configurations, file formats, and organizational methods of the data sources. Further, in moving the data into a single storage framework, it is useful to include the context of what were the conditions under which the data was recorded, any modification of the data into the expected format, and any calibration alteration of the data to make it what is directly comparable, as some examples.

[0356] FIG. **16** is a block diagram of a system for automated scaling, organizing, fusing and aligning of disparate data sets, in accordance with at least one aspect of the present disclosure. Illustration **204000** provides an example of how the data may be organized through several different stages. The automated system may be implemented in a medical hub that takes in data from multiple sources, such as one or more medical devices, enterprising software, and administrative software. In one aspect of the automated system, data can be extracted from a variety of data sources **204005** and placed in a staging module **204010**. The sources may come from typical vendors and other proprietary data types from medical devices and other inputs that may be received at a medical hub. The staging module **204010** may include one or more programs to house the various types of data in its various native formats, including a staging database or universal file transfer protocols and repositories. This data then gets loaded into a data warehouse **204015**, which may be configured to perform the analysis necessary to fuse or combine the data into common data types useful for people to analyze. The data warehouse module **204015** shows a few examples of the types of programs that may be used to perform these functions. Once the data is properly aligned, additional analysis can be performed on the composite data, which can then be utilized for providing business metadata (via the metadata layer **204020**), reporting, and analysis (via the reporting and analysis layer **204025**). The data sources from which data is extracted can include, for example, electronic medical records (EMR) databases and supply databases of product moving into the operating room (OR).

[0357] In one aspect, data can be loaded into the data warehouse **204015** via a number of different techniques. Known techniques for transferring data may be used, preserving the formats of the data. The speed at which the data may be transferred may be based more so on the means of transferring the data, such as through what physical means or if the data is transferred wirelessly. Utilizing hardware may be much faster than relying on software, as another example.

[0358] In one aspect, organized data can be loaded into a functional database for analysis. The process of data loading can depend on the structure of the data warehouse **204015**. For example, metadata tends to be dramatically larger in size and significantly more ancillary to the primary data itself. Therefore, the data might be pooled/stored in one location so that it can be referenced faster/on the fly and the metadata could be stored in another location (e.g., offsite) and/or in a storage medium more appropriate for long term storage so that it is referenced when necessary or when directly asked for. Accordingly, the system can be configured to parse out the data and send it to the appropriate repositories. In such aspects, different datasets or data types could be manipulated in the absence of each other or the metadata could be used as a means to modify the primary data and then be put back in storage or otherwise removed from the combined dataset to limit the size of the dataset.

[0359] In another aspect, all the parts of the data (i.e., primary data and/or metadata) can be stored in predefined locations and a reference database can be configured to retrieve each of the pieces of data that are required by the current analysis, rather than having all the data stored in one cohesive

database.

[0360] In one aspect, the data warehousing system **204015** can be configured to fuse dissimilar data, such as high and low-volume data. In one aspect, data that is received which is in a different format or structure than another dataset could use the metadata linked to the data point to allow the data to be fused into a format that is compatible with the other dataset. For example, data that is recorded at a vastly different data rate could be duplicated and placed in empty cells of a data storage structure. This technique can be used if, for example, the data source is a non-critical or supplementary data element or metadata to another critical data point. As another example, if the data rate is very high and it is being merged into a lower, more critical data form, the average of the data points or dropped data point methods could be used to provide a mean homogenous data flow. To illustrate, if kHz harmonic transducer data (i.e., transducer data that is sampled at a kHz rate) is being combined with or into outcomes-based 30 Hz data, the average of each 1,000 data points for blade impedance could be used with the lower sampling jaw clamp force to create a uniform time-based data stream. As another illustration, 3D imaging data could be transformed into a 2D planer version in the plane being measured by the adjacent mechanical device. In some aspects, the cloud system may help facilitate the determination of which data sets are more important than others, so as to determine how to effectively combine and align the data. Using situational awareness, the cloud system may recall from other datasets or various medical procedures which types of data are relied upon and most commonly adhered to. These datasets used by surgeons, analysts or others may provide probabilistic indications of what types of data are most useful, and then determine how to fuse the data for these purposes.

[0361] In various aspects, the data coming from one or more sources connected to a medical hub may be sent to the data warehouse and organized, scaled, and/or aligned using predefined parameters within the medical hub. That is, before integration or aggregation into the cloud system, the data may already be processed to fit a predefined format, scale, or other alignment when it is collected at the medical hub. In some cases, these predefined parameters may be adjusted after interaction with the cloud system. For example, the cloud system may determine that some data needs to be revised after including new medical devices into its system. As another example, the cloud system may utilize situational awareness or other machine learning to determine a more efficient scale of certain types of data that is more useful to an end user. These kinds of changes may be propagated to each of the medical hubs such that the automated data scaling, alignment, and organizing at the medical hub can provide more relevant data before being uploaded to the cloud system.

Data Cleansing

[0362] Data cleaning, also called data pre-processing, refers to the removal of duplicates, re-orientation of columns or rows, and linking of interrelated data.

[0363] In one aspect, the data warehousing system **204015** can be configured to remove duplicate data. Data duplication can result from the fact that data could be coming into the data warehouse system **204015** from multiple sources (see block **204005**), several of which might be being used together during the course of a surgical procedure. For example, a robotic hub, an energy/visualization hub, and a monitor tower hub could all be interfacing within the same procedure, and each of these hubs may generate at least some data that overlaps, but is ultimately useful to have combined and aligned. Further, the hubs could be moved in and out of the OR for portions of a surgical procedure and even moved into other procedures. Being able to look for duplicate data sets coming from different sources and then being able to remove the duplicate data would keep specific users, usages, or regions from overly influencing conclusions drawn from the overall dataset resulting merely from duplication of the data. As another example, data might be duplicated due to an interruption or loss of data in transit, initiating a second transmission of the same data. As yet another example, data could be intentionally uploaded multiple times. All of these duplicates would affect the weighting of certain conclusions drawn from the datasets, which

could interfere with trends and analysis.

[0364] In one aspect, the data warehousing system **204015** can be configured to merge separate streams of data. An alternative to multiple duplicates of data might occur when each of a series of devices or hubs that were used in the same procedure all send their data separately to the data warehouse **204015** assembling them. This presents an unrelated problem in that each device will require some fashion of synchronizing some continuous measure that allows the devices to be related with respect to one another. In aspects where the patient data is anonymized, then synchronization of the data from the different data sources can be even more challenging because a single synchronized real-time clock may not be an acceptable synchronizer (as storing real times associated with data could be used to ascertain confidential patient data). Further, if a randomized date and time are generated, then the randomizer would need to communicate that starting point to all devices to allow them to use the same time measure.

[0365] In one aspect, the surgical devices are configured to use the time of their internal clocks, rather than real-time, and communicate a synchronizer signal between the devices within the same procedure. Accordingly, each device records and time-stamps the data from their individualized points of view and then once all of the data is transmitted to a data warehouse (e.g., data warehouse **204015**), the data warehouse could synchronize the signals and use that to interrelate or fuse the different data feeds into a signal unified dataset. This addresses the patient privacy issue while still successfully synchronizing the data.

Calculating Values from Independent Imported Data Elements

[0366] In one aspect, a portion of the metadata can be utilized to transform primary data points into related aspect data. For example, the data warehousing system **204015** could be configured to use tissue type to calculate a constant that is then multiplied with the tissue impedance to balance collagen level and conductivity with the impedance to create a comparable impedance value to evaluate seal strength that is comparable between tissue types. As another example, the data warehousing system **204015** could be configured to use tissue thickness and cartridge color for a surgical stapling instrument to calculate a constant that is then multiplied by force-to-fire (FTF) to create a tissue-independent value of device firing performance.

[0367] In one aspect, the generation of a particular surgical instrument or its serial number can be utilized to transform the instrument's behavior into a cascade that allows all devices to be compared across multiple design changes. The cloud system may propagate the change from one medical hub that is connected to the particular surgical instrument to all of the other medical hubs to the extent relevant. The new changes may also be incorporated into updating situational awareness for the medical device, noting that a new or updated version of the surgical instrument leads to a modification that should be taken into account.

Chronological Interrelated Data

[0368] In one aspect, chronological interrelated data can be stored as part of the patient's electronic health record (EHR) within HIPAA-controlled and protected privacy limits. The patient may then have access to a combined set of data derived from multiple different data sources. If for example multiple medical hubs and/or multiple medical devices were used in a surgery, the patient may be able to see how all of the instruments may have interacted in a chronological fashion, based on the fused and aligned data according to the processes described herein.

Randomized Data Pairs

[0369] In one aspect, non-trackable, seemingly unrelated data pairs or clusters can be integrated with outcomes. In one such aspect, the data wrangling process can include randomized data pairs and allowing the metadata resulting from the data to continue to be correlated to the outcomes which exist as part of the data pair or bundle.

Data Fit and Form Transformation

[0370] In various aspects, the fit and form of the data can be transformed so that the data is in an expected format (e.g., a format expected by the data warehousing system **204015**). In one aspect,

raw data can be mapped into particular functional forms at the data staging module **204010**. For example, numeric data elements can be substituted for alphabetic data elements. In another aspect, data can be transformed into a predefined configuration, such as a particular arrangement of rows, columns, fields, cells, and so on.

[0371] Illustration **204100** in FIG. **17** shows a set of graphs including a first graph **204105** depicting measured blood pressure verse time, a second graph **204110** depicting fused blood pressure verse time, and a third graph **204115** depicting blood pressure verse time for different sample rates, in accordance with at least one aspect of the present disclosure. The graphs in illustration **204100** show examples of how some of the data coming into the data warehouse **204015** may contain different scales, sampling rates, and different measurements over time, and how the system of FIG. **16** may properly fuse and align the data to create a usable format. Here, graphs **204105** and **204110** are different data sets but shown on the same time scale. Graph **204105** represents a small set of measured blood pressure over time (over the period t.sub.1). In this example, the line plot **204125** represents the actual blood pressure, while there are sampled data points shown mostly along the smooth, continuous line. The sampled data points actually contain a couple of error data points **204120**. The data warehouse **204015**, via processing by one or more medical hubs and/or the cloud system, may utilize techniques to account for the error points in order to form the correct blood pressure curve.

[0372] This measured blood pressure curve in graph **204105** may then may fused with other sampling data to create a fused blood pressure graph **204110**. The time period is aligned as shown, along with additional data that may be gathered from other data sets. For example, the line plot **204140** may be a set of data from a slower sampling rate but that was recorded over the same time period. The blood pressure plot **204135** may be generated in part by the sampled data points in plot **204105**, but also additional data. In the fused plot **204110**, because the data warehouse **204015** would have processed the data to integrate it, error data points like points **204120** may be smoothed out. They may be removed and the revised plot **204135** may have in their places an average of the last data points before and after, in some cases. In other cases, simply the last data point may replace the error points if the rate of change of the data over unit time is greater than a predefined threshold.

[0373] The plot **204115** shows an example of low frequency data scaling. A downward sloping plot **204155** sampled at 100 Hz may be overlaid with data sampled at a lower frequency but then upsampled to be aligned with the higher sampling rate. Shown around the 100 Hz sampled plot **204155** are some data plots **204145** and **204150** that are sampled at 10 Hz but scaled to 100 Hz to match the higher sampled plot. Plot **204145** is an example of an error in the lower sampled plot but which is filled in and/or replacing the error points. As shown, the lower sampled points are scaled simply in a horizontal fashion at the higher frequency rate, in this example. In other cases, if enough data points are shown to establish a non-horizontal slope, the data warehouse **204015** may extrapolate the lower frequency sampling to create a more smooth fit, for example following the downward slope of plot **204155**.

[0374] Illustration **204200** in FIG. **18** shows a graph **204205** depicting blood pressure relative to high and low thresholds, in accordance with at least one aspect of the present disclosure. The graph **204205** may be one example of a data analysis report that may utilize metadata resulting from the fusion and alignment of data from the data warehouse **204200**. Graph **204205** may be a more finished product utilizing the data from FIG. **17**. Thus, in this example, the plot of blood pressure **204210** may be the result of one or more sets of sampled data of a patient that may have been fused together, similar to one or more of the processes described above. Multiple data sets that were sampled at different data rates, e.g., between 2 Hz and 10 Hz, may have been used to create the plot **204210**. Also, the system according to the present disclosure may add graphical overlays, such as the high limit line **204215** and low limit line **204220** to illustrate the range of blood pressure based on the data. Using the processes and systems described herein, a patient or analyst may be able to

obtain more comprehensive data that has more beneficial use than if the multiple data sets were examined independently.

[0375] FIG. **19** is a graph depicting ultrasonic system frequency verse time, in accordance with at least one aspect of the present disclosure. Illustration **204300** may be another example of an end product report that is the result of data fusion and alignment by the data warehouse **204015** and subsequent analysis using metadata layer **204020** and/or reporting and analysis layer **204025**. Here, graph **204305** shows a plot of ultrasonic frequency over time of a medical device. The plot **204315** may be the combined result of multiple data sets monitoring the device, with each data set having a sampling rate varying from 100 Hz to 2 kHz. The data sets may have been combined to produce this final plot **204315**. As shown, the various sampling rates may have revealed that the frequency rate changed. The data sampled within the circle **204310** time period shows that the frequency of the device was decreasing somewhat gradually, while the data sampled within the circle **204320** time period shows that the frequency of the device decreased even more dramatically.

[0376] FIG. **20** is a graph depicting expected blood pressure for different vessel types, in accordance with at least one aspect of the present disclosure. Illustration **204400** may be another example of an end product report that is the result of data fusion and alignment by the data warehouse **204015** and subsequent analysis using metadata layer **204020** and/or reporting and analysis layer **204025**. Here, graph **204405** shows how different types of blood vessels are either expected or reported to have different blood pressures. Each single data point in the graph **204405** may be the result of aggregate data compilation from multiple data samples. The data warehouse **204015** may have combined the data into a common scale and then organized it to be viewed like the plot as shown. Additional overlays may have been included, such as the labels and the vertical lines, to help better visualize the data.

[0377] As discussed above and illustrated in FIGS. **17-20**, in various aspects a computer system can be configured to smooth or fuse data based on, for example, expected variations in the source data. For example, blood pressure as measured in the large arteries is not necessarily equivalent to the blood pressure measured in the smaller vessels (e.g., smaller arteries, arterioles, etc.) where an energy surgical instrument (i.e., an electrosurgical instrument or an ultrasonic surgical instrument) may be operating. Accordingly, the computer system can apply a scaling factor to the pressure measurement (i.e., the measured pressure in larger arteries) to put it in line with the pressure being experienced by the end effector of the device (i.e., the actual pressure in the smaller arteries being dissected, sealed, or otherwise manipulated by the surgical instrument end effector). Further, there can also be a lag in the blood pulse (pressure) measured in the artery and the correlated pressure to tissue property measured local to the end effector. Accordingly, the computer system can apply a time delay factor to shift the pressure measurement. Still further, different types of sampled data can have vastly different sampling rates and bit sizes. For example, ultrasonic feedback/control data may be sampled at, e.g., 100 Hz to 2 kHz (see FIG. **19**), whereas blood pressure may be sampled at, e.g., 2 Hz to 0.25 Hz. Accordingly, the computer system can be configured to pair or fuse the data having the different sample rates using techniques discussed above in order to perform deep analysis on the data.

Dataset Validation

[0378] In various aspects, the computer system can be configured to validate the datasets themselves and/or the sources of the datasets, including the hubs, the individual instruments, or sensing systems. Further, a computer system (e.g., the surgical hub and/or cloud system) can be configured to validate received data and provide reactions to invalid data.

[0379] In one aspect, hub, instrument, and/or cloud can be configured to provide particular responses based on validation of a received dataset and authentication of its source and integrity. Further, the response(s) provided by the hub, instrument, and/or cloud could be selected from a set of reactions corresponding to the data and/or metadata. In one aspect, the cloud can be configured to isolate data from the primary data group in response to poor data integrity, a lack of data

authenticity (i.e., the inability to authenticate the data or the ability to determine that the data is inauthentic), or user behavior. In another aspect, a computer system can be configured to provide a variety of responses, including identifying the user or the facility, isolating the data from other datasets, compiling the effects of the data to determine warehouse reactions, and/or providing warnings of inviolate instruments for procedures and their implications. In one aspect, the hub can be configured to provide a variety of responses, including flagging the data for later analysis, varying control algorithm changes of linked instruments, or preventing of usage of the hub or the instruments based on the validation or authenticity of the data, instruments, user behaviors, or linked data sources. In one aspect, a local user could have the ability to override the local Hub's reaction.

Data Trend Verification

[0380] In one aspect, the computer system can be configured to verify trends within the data to confirm that its behavior and therefore its data are unaltered. There could be several sources of error or invalid information that might move into large datasets. If all data is treated as valid, it could impact the statistical significance of other data points as it would create data that could move an average/correlation or increase the error term of the analysis to make insignificant something that was a significantly different event. Still further, data could be maliciously altered with the intent to hinder the ability to improve or detect something or cause the computer system to modify the devices' behaviors (e.g., control algorithms for the devices) in the wrong direction. Malicious intent could come from hiding the use when a device is being used off label, too many times (i.e., more times than recommended by the device manufacturer), or even in abusive manner. The intent could also be to damage the ability to effectively determine trends in the data or even to misdirect the data analysis.

[0381] There are sequential trends and repetitive data points that could be used throughout any normal surgical procedures that could not be fooled if the device was used for the jobs and when it was said to have been used. In one aspect, these comparison points could be used to verify the integrity of the data. This analysis of sequential trends and repetitive data points could not only be a check to verify a validation or encryption term, but it would also be as an on-the-fly means to assure that the data itself has not been affected in some way.

[0382] In one aspect, a validation term and/or private key encrypted checksum can be utilized to verify the data received is truly from the instrument it says it is from. For example, a validation term could be used as opposed to encrypting all data and metadata, which could be costly from a bandwidth, storage size, and processor speed point of view. Using a validation term could allow the data to be scanned in a less onerous manner via an encryption algorithm and key to allow the cloud and surgical hub units to verify the data is real and came from the purported specific source.

[0383] In one aspect, the cloud system may utilize data from other sources, such as one or more other medical hubs, to determine whether a dataset from a different medical hub is valid. The cloud system may be configured to draw from patterns of known valid datasets in multiple other medical hubs, and/or known invalid datasets from these multiple sources. In other cases, the cloud system may cross check data to determine whether the dataset is unique and whether that dataset should in fact be unique. For example, if data from a medical instrument that has a certain serial number happens to match a serial number from another known medical instrument, the dataset could be flagged.

[0384] In one aspect, if new malicious actors are discovered, the cloud system may utilize situational awareness to propagate the known instance of fraud or malicious activity to other hubs in the network. In general, situational awareness may be used to determine patterns of valid or invalid data and may apply those patterns to new situations or new nodes (e.g., hubs) in the network when determining the validity of any dataset.

[0385] In one aspect, if the data is found to be altered, the computer system can be configured to determine if the data is entirely contrived or has been modified.

[0386] If the data is determined to be entirely contrived or artificial, then the computer system can respond by notifying a security agent of the intrusion and to initiate an investigation into the data source and behavior; quarantining all data and data requests from the affected hubs, regions, or system users; and/or preventing erroneous data from being added to any of the databases (e.g., a data warehousing database) or from affecting or being considered as part of any analyses.

[0387] If the data is determined to be altered (e.g., in order to affect data correlation analyses), but is determined to be from a valid source, then the computer system can respond by flagging the data and identifying the data source as a source of contaminated data. An example of this would be for a knock-off device that knows it is being monitored to generate slightly off data with the intent to hide the fact that it is not as effective as the original devices. Another example would be for a repossessed device to contain a mathematical constant that is used to offset the aging calibration of the original device that has been affected by its uses (i.e., overuse) or re-sterilization. These datasets could be verified by the instrument being instructed to operate in a given way during a controlled situation. For example, an instrument can be programmed to close the jaws at first start up, activate the transducer with a known power level, and then review the blade harmonics. As another example, a powered stapler can be programmed to retract the firing member when the knife is in its fully retracted state and monitor the force measured by the system. By being programmed to operate in a particular known manner in a controlled situation, the instrument can thus determine whether its operation is being altered or otherwise affected.

[0388] If the data is determined to be invalid and it has characteristics of invalid data that the system has seen before, the data could be used to determine the source and purpose of the invalidation. The data could then be relayed back to the hub from which it came to inform the users that they are being affected by products or individuals that are interfering with the proper operation and best outcomes of their devices.

Layered Contextual Information

[0389] In some aspects, contextual information can be layered onto data to enable contextual transformation, rather than merely aggregation, of datasets. In other words, contextual metadata can be linked to the outcome and device data to enable contextual transformation of the datasets.

[0390] In one aspect, a system (e.g., a surgical hub system, a cloud analytics system, and so on) can be programmed to adjust devices' control programs based on stratified contextual data, in addition to the data. The contextual data represents the circumstances around the collection of the data or related patient, procedure, surgeon, or facility information. The stratified analysis for determining interrelationships of influencing factors can be utilized to create an improved causational response for the surgical hub and instrument control program updates. In one aspect, the stratification of context includes a hierarchy of influencing factors, where some can be more important or functionally interrelated at a higher priority than other interrelations. In one aspect, the data pairs include the outcome of the instrument operation and its functional parameters. In one aspect, the contextual parameters are derived from the patient complications, other treatments, co-morbidities, procedure complications, previous instrument functional parameters, and so on. In one aspect, the adjustments based on these contextual limitations or influencing factors can proportionately affect the adjustments.

Identification of Relevant Contextual Cues

[0391] FIG. **21** is a block diagram **204500** depicting layered contextual information, in accordance with at least one aspect of the present disclosure. In this illustration, there are four examples of types of layered contextual information that may be accounted for by the systems of the present disclosure. In general, the cloud system, in connection with one or more medical hubs, may be configured to adjust connected devices based on one or more sets of contextual data. FIG. **21** provides some examples of how to determine what adjustments to make when there are multiple contextual datasets, and the datasets come into conflict with one another or otherwise need to be reconciled with one another. In the example A **204505**, the present disclosure allows for a

hierarchical conflict resolution system of tiered contextual information for when at least some of the contextual information conflicts. For example, a first set of primary contextual information relevant to a medical device may include managing the speed of a medical device that leads to an instruction to not exceed an initial activation rate of 10 mm/sec. However, a second set of primary contextual information may be entered that relates to diseased tissue that is being operated on, and leads to an instruction to not exceed an initial activation rate of 8 mm/sec. The hierarchical conflict resolution scheme **24505** may include logic to create a combined set of instructions that satisfies all combined constraints. In this case, the resolved primary set of contextual information therefore leads to an instruction to have the initial activation rate of the medical instrument not exceed 8 mm/sec. In other cases where instructions directly contradict one another, contextual information at a higher tier in the hierarchy may take precedence. In cases where there are contradicting instructions in the same tier, a flag may be presented that highlights the irresolvable conflict. In other cases, situational awareness may be utilized to refer back to past instances of such conflict to help determine how to resolve.

[0392] In the example B **204510**, adjustments to device control parameters may be made after resolving conflicts between different tiers in a non-standard manner. For example, a primary dataset of contextual information may contain a max force to fire of 400 lbs, while a secondary dataset of contextual information related to what type of medication a patient is taking may indicate a max force of only 150 lbs. A tertiary dataset of contextual information may have additional instructions for max force to fire based on patient parameters. In this case, the secondary contextual information may override the parameters of the primary contextual information because the patient possesses a high BMI, or there is some other overriding constraint. In some cases, the primary contextual dataset may include one or more exceptions to defer to different parameters, if they exist in other lower tiered datasets. In this case, the primary dataset may provide an exception to use different max force to fire if patient medication requires it, and thus the secondary contextual dataset will override this condition for this case.

[0393] In the example C **204515**, adjustments of control parameters may be determined by combining multiple pieces of contextual information, rather than simply overriding one over another. As an example, a primary set of contextual information may lead to an instruction to set the initial activation rate of the instrument to 8 mm/sec, while a tertiary set of contextual information about patient parameters may lead to an instruction to decrease speed by an additional 20% due to a diseased tissue state. In this case, the effects do not directly contradict, but rather they may be combined to create a revised instruction. Here, the speed set to 8 mm/sec is a decrease from the default 10 mm/sec, which is an initial decrease of 2 mm/sec. The tertiary instruction leads to an additional decrease of 20%, or $2.0 \times 0.2 = 0.4$ mm/sec. Therefore, the final reduced speed is $8 - 0.4 = 7.6$ mm/sec. The cloud system may be configured to interpret the logic of the instructions and generate the adjusted device parameters based on the combination of contextual information.

[0394] In the example D **204520**, secondary or tertiary effects can still be used to override any predefined control parameters that a primary contextual dataset does not speak to. Generally, the secondary and tertiary contextual datasets may be based on patient specific parameters, and therefore lead to changes made at the time of surgery or “on the fly.” In some cases, new contextual information may be provided in real time, which may then cause additional adjustments to the device(s).

[0395] FIG. 22 is a block diagram **204600** depicting instrument functional settings, in accordance with at least one aspect of the present disclosure. Illustration **204600** of FIG. 22 provides an example of how multiple sets of contextual information may apply to the same device simultaneously, but are applied conditionally at different times or in different functions occurring at the same time. For example, different settings may apply to the firing system of an instrument and separately to the clamping system of the instrument. How the control settings are applied according to any of the examples of FIG. 21 may be applied in different instrument contexts as shown in FIG.

22. Therefore, multiple sets of contextual information may simultaneously apply to a single instrument. In some cases, this may result in some lower tiered effects being applied to the instrument during a particular function, while those same lower tiered effects would not be applied to the instrument during a different function.

[0396] FIG. 23 is a graph 204700 depicting force to fire (FTF) and firing velocity for patients having different complication risks. Illustration 204700 shows four plots, with two plots 204705 and 204720 corresponding to the vertical axis on the left pertaining to firing velocity, and two plots 204710 and 204715 corresponding to the vertical axis on the right pertaining to force to fire. The two plots corresponding to one type of axis correspond to two different patients and how an instrument's settings might vary between two patients with different health conditions. In various aspects, the contextual information adjustments to a single instrument may be queued into the same instrument, to account for different patients that the instrument may be applied to over the course of a day. In various aspects, the instrument may be configured to load in a particular file to disease contextual information, instrument contextual information, treatment contextual information, patient contextual information, and so on, where the combined contextual information from the multiple different datasets may form a particular combination that provides the optimal adjustments to the instrument for a particular surgery on a particular patient. For example, for the first patient, contextual information about emphysema stage 3 may be loaded into the instrument for the disease state. The force to fire information for the particular instrument may be loaded for the instrument state. Radiation treatment contextual information may be loaded for the treatment state, and steroid dosage contextual information may be loaded for the patient state. Any conflicts or combinations may be resolved using the example processes described in FIGS. 21 and 22, which then provides the instrument with a particular set of adjustments for precisely handling this first patient.

[0397] Next, a second set of contextual information for dealing with a second patient may also be loaded into the instrument but remained queued up before being implemented. For example, for the second patient, contextual information about high blood pressure (e.g., 165/110) may be loaded into the instrument for the disease state. The force to close information for the particular instrument may be loaded for the instrument state. Chemotherapy contextual information may be loaded for the treatment state, and blood thinner dosage contextual information may be loaded for the patient state. Any conflicts or combinations may be resolved using the example processes described in FIGS. 21 and 22, which then provides the instrument with a particular set of adjustments for precisely handling this second patient.

[0398] The resulting combinations of contextual information for the first patient may result in the two graphs 204705 and 204710, for example, for the firing velocity and the force to fire settings over time, respectively. Similarly, the resulting combinations of contextual information for the second patient may result in the two graphs 204720 and 204715, for example, for the firing velocity and the force to fire settings over time, respectively.

[0399] In one aspect, the adjustments to the instrument for a single setting may be weighted by the hierarchical tier in which the proposed adjustment derives from. For example, if adjustments to FTF are found in all of the primary, secondary, and tertiary tiers, then the adjustment to FTF may be made according to the following example weighting structure:

$$FTF = FTF(\text{default}) + 1.5 * FTF(\text{primary}) + 1.0 * FTF(\text{secondary}) + 0.75 * FTF(\text{tertiary}),$$

wherein FTF is force required to fire a surgical stapler.

[0400] Other weighting mechanisms may be used as well and are non-limiting.

[0401] Discussed below are some non-limiting examples of contextual information that may be included to cause adjustments to an instrument. The type and number of factors described may be used in the processes described in FIGS. 21, 22, and 23 to create a combined or comprehensive instrument adjustment. The contextual cues can include non-device-specific cues, device-specific cues, medical cues, patient-specific cues, procedure-specific cues, and surgeon-specific cues.

Non-Device-Specific Contextual Cues

[0402] Non-device-specific cues are contextual cues related to the operation of a device, but that are not specific to any particular type of device. Non-device-specific contextual cues can include, for example, device tissue clamping, tissue information, and instrument usage history.

[0403] Tissue clamping contextual cues can include, for example, implications of clamp force or pressure on tissue (i.e., the primary effect(s) of the tissue clamping), which can in turn include desired and adverse impacts on the tissue. Clamping of tissue can have multiple different desired effects on the clamped tissue. For example, clamping the tissue can drive the fluids out of the tissue, collapse the tissue layers, and collapse any interior opening(s). This allows the tissue layers to be in close proximity and prevents leaks from any hollow structures in the area (e.g., capillaries, bronchi, gastro-intestinal). Another desired effect of tissue clamping is that because body tissues are viscoelastic, the compressibility of the tissue is dependent on the type of tissue, its fluid content, the pressure level, and the rate of compression. Accordingly, for the same amount of compression, the faster the compression, the higher the applied force. For a constant pressure, the tissue will continue to move thinner and thinner until a stable state of full compression is achieved. This continued thinning is defined as tissue creep and is a function of the viscoelasticity of tissue. This is important in the discussion of pre-compression cycles, wait times, and overall instantaneous compression. Overall lower compression levels over a longer period of time are less detrimental to the treatment tissue and the pressure differential (shear) on the adjacent tissue.

[0404] Clamping the tissue can also have adverse impacts on the tissue due to compression. For example, as the tissue is clamped and tissue structures are collapsed, there may be structures in the tissue that are not intended to be collapsed. This can create a micro tissue tension or pressure differential between the adjacent unclamped tissue and the compressed tissue. Some tissues (e.g., parenchyma, solid organ parenchyma) are not particularly tolerant to such tension or pressure without causing ruptures of the tissue layers adjacent to the clamped tissue, which in turn causes inadvertent collateral tissue damage and, potentially, additional leaks. The total amount of pressure, the geometry of the clamping bodies, and the rate of clamping all have primary effects on the likelihood of collateral damage. Furthermore, the tissue composition, strength, and internal parameters also have an implication of the likelihood of damage. Many of these internal parameters of the tissue are influenced by other medical treatments, disease states, or physiologic states of the tissue. All of these can complicate the likelihood of collateral or primary site secondary damage. As another example, the maximum compression levels for different tissues and organ are at different levels. Most tissues in the body are a series of layers or structures enclosed within other structures. Once maximum compression occurs when the outer enclosing layer has too much pressure or pressure differential applied, it tears, allowing the internal constrained tissue out. The lung is a good example of this. The lung parenchyma is made up of alveoli, veins, arteries, and bronchi with an exterior visceral and parietal pleura covering the surface. When stapling lung parenchyma, it is desirable to staple the pleura to itself to promote good healing. But a tear in the pleura can expose the more fragile alveoli and without an outer constraining element, the alveoli easily rupture, creating air leaks. Another form of maximum tissue compression occurs when the fiber bundles of the tissue itself are ruptured or separated. This occurs at a much higher level and this tissue destruction is typically accompanied by wide spread cellular death and necrosis.

[0405] A number of different device control parameters influence clamping, such as the clamping force, the clamping rate, and the number of repetitive clamps. The clamping force characteristics can be illustrated by a clamp force vs. time curve, which can indicate the time rate of change of force, peak clamping force, time to clamp force stabilization, steady state clamping force, and difference between peak clamp force and stabilized clamping force.

[0406] Clamping force can be measured either directly or indirectly via a proxy. A number of different proxies can be utilized to measure the clamping force. For powered closure, the proxies can include the current through the motor and the difference between the target motor speed and

the actual motor speed. Strain gauges on components that are loaded during the act of clamping, such as the anvil, the closure member, and/or the support frame can also be utilized to measure the clamping force.

[0407] Clamping rate can be determined by comparing the actual clamping rate against the targeted clamping rate for powered closure. Clamping rate can also be determined according to the duration of the clamping process from start to finish.

[0408] The number of repetitive clamps can be important because heavy tissue manipulation prior to transection of tissue treatment can have a cumulative effect on the tissue due to the repetitive exposure of pressure to the tissue. Some devices have a maximum pressure that can be applied and also a minimum closure level for the next mechanism to begin its operation. In these instances, the jaws are opened and closed repeatedly to get the tissue to the minimum closed state while compressing over and over until that state is achieved. In one aspect, a robotic surgical system can signal if clamping parameters do not fall within a threshold to ensure the jaws are properly clamped.

[0409] Tissue information contextual cues can include, for example, placement in jaw (which can be considered, e.g., a secondary effect from the surgical procedure), tissue quality knowledge from other sources (e.g., imaging or EMR, which can indicate prior interventions, current/prior diseases, and so on), type, thickness or density, and impedance (which can be considered, e.g., a primary effect from the surgical procedure). The placement of the tissue within the jaws can correspond to the percentage of the jaws covered by the tissue, the region or locations of jaws covered by tissue (e.g., vessel, etc.), and the degree of bunching or degree of uniformity of the tissue along the length of the jaws. Any device that compresses tissue is applying a known measurable force to the tissue within the jaws. The amount of tissue in the jaws, the placement of the tissue (i.e., relative distal to proximal position), and its thickness variability impact the pressure on the tissue. Knowing the force applied to the tissue without knowing how much of the jaws are covered by the tissue or the placement of the tissue makes it challenging to determine the pressure on the tissue. Many devices are also technique sensitive. For example, often only the distal tip of an ultrasonic device is used for dissection, welding, and cutting, leaving the bulk of the jaw empty for many of the firings. As another example, surgical stapling and cutting instruments often have the tissue crammed into the proximal crotch of the jaws, leaving a differential of tissue from the proximal to distal end of the end effector. Unless the only meaningful information that is sought is the trends of the parameters (which can be the case in certain situations), adjustments to the device control parameters are functionally dependent on knowing how much of the jaw is loaded and where the jaws is loaded because those factors have implications of the forces measured.

[0410] Instrument usage history contextual cues can include, for example, the number of uses and the number of resterilizations of the devices.

Device-Specific Contextual Cues

[0411] There are a wide variety of device-specific contextual cues for staplers, ultrasonic instruments, laparoscopic or endoscopic suturing devices, dual bipolar instruments, monopolar instruments, and clip applicators.

[0412] Contextual cues for stapling devices can include device and reload identification cues and firing speed cues (which can be considered, e.g., a primary effect from the surgical procedure).

[0413] Device and reload identification cues can include, for example, the stapler type and reload (i.e., cartridge) information. Stapler type cues can include the brand, powered verse manual, shaft length, general purpose verse specialty, handheld verse robotic, single verse multiple use, usage history, whether the device has been reprocessed (e.g., authentic reprocessing or resterilized, off-label usage), and stapler configuration (e.g., linear, curved linear, or circular). Reload cues can include the color, length, uniform verse variable staple height, authentication (i.e., whether the reload is compatible and of the same brand; whether the reload is compatible, but of a different or unknown brand device or an illegal knock-off reload used with the manufacturer's stapler; not

compatible; or of the correct technology generation, such as a), specialty (e.g., curved tip, reinforced, radial, absorbable staples, medicament-coated staples, or tissue thickness compensation), use and type of buttress or other staple line adjunct, or provides medicament delivery via staple line adjunct.

[0414] Firing speed cues can include, for example, actual speed verse time throughout the firing cycle, the difference between the target speed and the actual speed, or adaptive firing control of firing speed (e.g., starting speed, the number of changes in target speed, and the maximum and minimum actual speeds recorded). Firing speed has multiple direct and/or indirect implications on device function. For example, firing speed can have implications of staple formation for multiple reasons.

[0415] As one reason, the rate at which an I-beam or bladed actuator is cycled causes tissue to move while the staples are being deployed. In a circular stapler, the knife advancement is often coupled to the staple driver advancement. If the knife begins to sever the tissue before the staples are fully formed (as may be the case), the tissue begins to move radially outward. This tissue flow can have implications on the staple formation. In sequentially linear deploying staplers, the knife/actuator progresses through the cartridge (typically proximal to distal, although sometimes it can be in the opposite direction) and severs tissue while progressively forming staples, which creates tissue flow in the direction of the movement. Pre-compression and tissue stabilization features can reduce this effect, while lower tissue compressions and I-beam local running compressions typically increase this effect. The tissue movement effect can create a wave in advance of the cutting member, which occurs in a related area to where the unformed staples are being advanced towards the anvil. Accordingly, this tissue flow has an effect on staple formation and cut line length.

[0416] As another reason, the rate at which the staple drivers are advanced has an effect on their advancement and the likelihood of them rotating or bending, causing the crown of the staple to move out of plane. For surgical stapling and cutting instruments where the diameter of the device is constrained to the trocar diameter, size, or type, the staple drivers are often short and have an aspect ratio that allows for driver roll, in addition to linear advancement perpendicular to the tissue contacting surface of the cartridge. This driver roll can result in a bind in the advancement of the driver so that as the sled continues to advance, the drivers are rotated, rather than advancing outwardly, resulting in the destruction of the cartridge and the staple line. This roll of the staple drivers is a function of sliding friction, lubrication, and cartridge geometry. The binding and therefore the driver rolling is amplified by the rate at which the sled is advanced and, therefore, the rate of the firing actuator of the device. Furthermore, many staplers overdrive the drivers above the tissue contacting surface of the cartridge. This overdrive exposes the driver to tissue flows occurring between the anvil and the cartridge. The drivers have a moment of inertia and are being driven up into contact with the tissue, as well as experiencing the loads from the staples being formed. The rate at which these drivers are advanced into the tissue and the extent of the overdrive both influence the likelihood of the driver remaining directly under the anvil forming pocket.

[0417] As another example, firing speed can have implications on local tissue compression. In the case of I-beam coupled surgical stapling and cutting instruments, the advancement of the firing member causes local tissue compression around the I-beam location, in addition to cutting tissue and deploying the staples against the anvil pockets. This local running compression wave moves proximal to distal with the I-beam location. The viscoelastic aspect of the tissue causes the rate of this advancement to be directly related to the local compression force applied, as well as the size of the rolling compression wave. This local rolling compression is capable of causing local tissue tension damage within the treatment area, as well as collateral damage because the rate of compression is likely more than the rate of pre-compression.

[0418] As yet another example, firing speed can have implications regarding forces within the device. The loads experienced within the instrument itself are a cumulative effect from the pre-

compression, as well as the local rolling compression. The faster the firing member is advanced, the higher the required force to advance the firing member. This is due to the dynamics of the moving structures within the device as well as the I-beam forces. The higher these forces are, the more stretch there is in all the components in the elongated tube and frame, which in turn results in some forces being impacted (e.g., pre-compression deteriorate as the system stretches). These losses then add more force to the firing member balance, resulting in even higher impacts on the firing speed to load relationship.

[0419] These various examples of stapler contextual cues can be further illustrated with regards to a specific example. In this example, through the procedure plan, EMR, and other hospital records, several things are known: (i) this is a right upper lobectomy procedure; (ii) the patient has had prior radiation to treat the tumor in this area; and (iii) the patient has interstitial lung disease. These pieces of information suggest that the lung will be stiffer than normal, healthy tissue. Based on this inference, the conservative approach would be to slow down the maximum rate of closure and adjust the thresholds. However, further layered contextual information can be further utilized in determining how the instrument should be controlled. On this same patient, during closure, the force to close is higher than anticipated, exceeding the new thresholds (i.e., the thresholds that we set according to the procedure and patient information noted above). As a result, the wait time prior to starting the firing sequence is increased and the initial firing speed is slowed. Firing algorithms will take over once firing has been initiated. Note that the contextual cues can influence thresholds within the firing algorithms.

[0420] Contextual cues for ultrasonic devices can include the activation time, the coherence tomography evaluation of the collagen content of the tissue, the current blend of energy modalities (e.g., whether the instrument utilizes an ultrasonic/bipolar blend, an ultrasonic/monopolar blend, or ultrasonic only), blade temperature, and pad condition. Blade temperature increases with the duration of contact with either tissue or the clamp arm pad and the power into the transducer. This temperature changes the natural frequency of the blade and has the ability to add heat into the welding of tissue that is not intended. It can also cause inadvertent damage to tissue that it comes into contact with, even in-between actuations of the instrument. Hot blades also cause local tissue charring (which can then stick to the blade). The blade temperature has a long-term effect on cut/coagulation performance, but also creates a shear or tearing force on the tissue weld as the jaws are unclamped and removed (as charred tissue sticks to the blade). The pad condition can depend upon the duration of time active without tissue in the jaws and/or the temperature history of the jaws. As the pad is degraded, the underlying metallic strength of the clamp arm is exposed to the blade, eventually impacting its performance.

[0421] These various examples of ultrasonic contextual cues can be further illustrated with regards to a specific example. In this example, through the procedure plan, EMR, and other hospital records, several things are known: (i) this is a vertical sleeve gastrectomy procedure; (ii) the patient's BMI **40**; and (iii) the patient's body composition suggest that they have a high level of visceral fat. These pieces of information suggest that the greater curvature takedown will have a higher than normal volume of fatty mesentery. Based on this inference, the blended algorithm leans more heavily toward cutting than sealing given the high expected fat content. Algorithm parameters can be concurrently adjusted to ensure a robust seal.

[0422] Contextual cues for laparoscopic or endoscopic suturing devices can include stitch tension (tension monitoring can be utilized to inform the suturing technique, for example), stitch type (e.g., mattress verse running, etc.), or suture type (e.g., braided verse monofilament, absorbable verse non-absorbable, suture diameter/size, or needle size/type). Pattern recognition models can be utilized to recognize the pattern of the stitches placed verse tension applied in applying the stitches (e.g., stitch/tension/stitch/tension verse stitch/stitch/tension/stitch/stitch/tension). The pattern recognition system can be configured to provide technique advisements, e.g., three stitches without a tension step with a braided suture may be difficult to cinch up without tissue damage, whereas

two stitches may be suitable.’

[0423] Contextual cues for dual jaw bipolar RF instruments can include the coating (which can include coatings disclosed in U.S. Pat. Nos. 5,693,052 and 5,843,080, which are each hereby incorporated by reference in their entirety), design (which can include the design disclosed in U.S. Pat. No. D399,561, which is hereby incorporated by reference in its entirety), bipolar coagulation, algorithms and load curves, smoke generated, conductivity contact of electrodes (e.g., the amount of charring present on electrodes or whether there is a detected short), jaw compression (e.g., the compressive force, pressure, or in the special case of bipolar shears the localized electrode cross-section/elevated geometry or higher max force, as is disclosed in U.S. Pat. No. 9,084,606, which is hereby incorporated by reference in its entirety), tissue gap (i.e., whether the jaw is open/feathering or closed/spacing between jaws), electrode configuration (e.g., opposed electrodes, offset electrodes, or electrodes/insulation, as disclosed in U.S. Pat. Nos. 5,100,402, 5,496,315, 5,531,743, 5,846,237, and/or 6,090,107, each of which is hereby incorporated by reference in its entirety).

[0424] Contextual cues for monopolar instruments can include power (e.g., constant voltage or current and variable control of the other given a tissue impedance), tissue impedance (e.g., rate of change of impedance, overall measured impedance, or time at a given impedance), algorithms and load curves, return path capacity, blade technology (e.g., the coating of blade, such as insulation breakdown or various coatings described in U.S. Pat. Nos. 5,197,962, 5,697,926, 5,893,849, 6,685,704, 6,783,525, 6,951,559, each of which is hereby incorporated by reference herein in its entirety; geometry of exposed conductive surfaces; conductivity of structural underling materials; or blade configurations, such as the configurations described in U.S. Pat. Nos. 6,039,735, 6,066,137, 8,439,910, each of which is hereby incorporated by reference in its entirety), heat dissipation, smoke generated, applied compression (e.g., compression force between the monopolar blade and the support arm or driving force of a monopolar probe pressed against tissue), or leakage current magnitude.

[0425] Contextual cues for clip applier devices can include clip size, first verse last clip from the applier, the timing of the detected forces (e.g., overload protection mechanisms if unexpectedly stiff structures are inadvertent within jaws, such as when clipping over another clip or closing jaws on another instrument), clip feeding monitoring (e.g., feeding loads or detection of the presence of clips in pre-defined locations or at pre-defined times), lateral end effector and bending loads, or jaw actuation load (e.g., clip closure, max load in displacement controlled actuation, max displacement in load controlled actuation, or tissue manipulation loads with jaws).

Medical Contextual Cues

[0426] Medical contextual cues can include contextual cues associated with medical complications, disease states, medications, and procedure complexity.

[0427] Contextual cues for medical complications can include functional constipation, functional diarrhea, sphincter control or strength insufficiency disorders, functional dyspepsia, and complications that effect tissue planes and tissue compositional makeup, as just some examples.

[0428] Functional constipation can result from colorectal surgery including a circular anastomosis, which would suggest a longer period before bowel motility after surgery to allow for more expansion. It could also suggest the use of an expandable staple configuration or counter indicate the use of buttress or compression ring technologies that would not tolerate larger more solid feces (which can be, e.g., a secondary effect from the surgical procedure as a tissue fragility complicating effect).

[0429] Functional diarrhea can indicate higher acid levels and more fluidics movements, which could dictate tighter staple forms, higher pre-compressions, and the potential need for abdomen-side applied secondary adjuncts to minimize the likelihood of fecal introduction into the abdomen cavity (which can be defined as a tertiary effect from the surgical procedure).

[0430] Poor sphincter control or other strength in sufficiency disorders can result in heartburn or acid reflux. These contextual cues can indicate, for example, that an esophageal anastomosis would

benefit from stronger staples, tighter forms, and/or longer precompression of the tissue to enable tighter staple forms and lower tissue tension thresholds (both micro and macro tension). Macro tissue tension is related to more extensive mobilization of the tissue from adjacent structures and could be measured by lateral forces on the stapling device. Micro tissue tension is due to the compression rate, max compression, and the gradient of tissue compression between the areas directly adjacent to the treatment area and the type of staple forms within the treatment area. 3D staples lower micro tension as does a lower max pre-compression level. Heartburn and/or acid reflux can be considered, e.g., a secondary effect of the surgical procedure. Device or surgery suggestions could include a reinforcement treatment for the sphincter or an adjunct therapy applied to the staple lines to improve robustness.

[0431] Functional dyspepsia can result from the sensation or inhibition of peristalsis, which could suggest rigidity of the anastomosis line (would further amplify the effect). Lower micro tissue tension would ease the effect (3D staples) or an expandable staple line. Compression of anastomosis lines or adjunct material used on the staple lines would cause more issues. Functional dyspepsia can be considered, e.g., a secondary effect as an adjunct therapy usage complication effect.

[0432] In the case of complications that effect tissue planes and tissue compositional makeup, healing from a first surgery and adhesions result in an increase in thickness of the tissue as well as disorganized remodeling of the tissue, resulting in increased toughness of the tissue. To adjust for these effects when stapling, it would be suggested to implement an increased gap, raised tissue load thresholds, slower actuation, and suggested larger or heavier staples. These complications can be considered, e.g., a secondary effect of the surgical procedure as a tissue thickness/toughness complication effect. Other such complications can include revisional surgery, adhesions, and altered tissue conditions from medical treatments (e.g., irradiated tissue or steroid induced changes).

[0433] Contextual cues for disease states can vary for colorectal, thoracic, metabolic, and cardiovascular diseases.

[0434] For colorectal diseases, inflammatory bowel disease can be a contextual cue. All of the repetitive inflammatory colorectal diseases cause an increase in thickness of the tissue, as well as disorganized remodeling of the tissue, resulting in increased toughness of the tissue. The stapling adjustments could include increased gap, raised tissue load thresholds, slower actuation, and suggested larger or heavier staples. These complications can be considered, e.g., a secondary effect of the surgical procedure as a stapling adjunct complication effect. Such colorectal diseases can include Crohn's disease and diverticulitis.

[0435] For thoracic diseases, bronchitis, emphysema, chronic obstructive pulmonary disease (COPD), asthma and interstitial lung disease can all be contextual cues. For example, emphysema results in thicker, stiffer lung tissue that would suggest load clamping rates, lower pre-compression levels, and slower firing actuation of the knife/I-beam to prevent adjacent collateral damage around the perimeter of the anvil and cartridge due to excessive pressure differential during treatment. These are primary effects. As another example, COPD results in artery walls that could be stiffer and less elastic, requiring a softer handling of the arteries prior to applying treatment. This could require the mechanical clamping elements to clamp at a slower rate and potentially a lower clamp pressure to minimize damage outside of the treatment area and premature damage to the treatment area. These adjustments could be for either energy or stapling. These are secondary effects. In stapling, the suggestion of an adjunct may be counter indicated to prevent additional uncontrolled remodeling and hardening. In advanced energy, the RF treatment modality is preferred and the balance of the RF to ultrasonic balance to weld more than cut.

[0436] For metabolic diseases, metabolic syndrome, obesity, and type 2 diabetes mellitus can all be contextual cues.

[0437] For cardiovascular diseases, arteriosclerosis, high cholesterol, and vascular disease can all

be contextual cues.

[0438] Contextual cues for medications can include blood thinning, blood clotting, steroids, radiation treatment, and chemotherapy.

[0439] For example, blood thinning can indicate that advanced energy devices would benefit from improved coagulation before cutting. This is a priority and primary effect as a blood pressure complication effect. In a hybrid energy device, the RF power could be increased or the ultrasonic application delayed in order to increase the coagulation before the cutting. In an ultrasonic-only device, the algorithm could be adjusted to apply a lower powered level for a longer period of time to in order to denature the collagen longer before cutting. Further, in an ultrasonic-only device the harmonic power could be adjusted lower as the temperature approaches a predefined optimum temperature and that temperature could be maintained for a longer period of time before elevating the power of the transducer to initiate cutting. Further, in a stapling instrument, the suggested cartridge color could be adjusted down, suggesting shorter formed staples or the closure clamping time increased or the pressure increased before firing. This is a secondary effect.

[0440] For example, blood clotting can indicate that precompression or the compression levels of either the advanced energy or stapling could be lowered prior to treatment to minimize inadvertent damage and therefore forming clotting outside of the treatment area. This is a secondary effect as an increased pressure complication effect.

[0441] For example, steroids cause physiologic effects that slow healing and raise blood pressure, which increases pre-existing complications. This is a tertiary effect as an amplification of disease complication effects, as well as longer term healing complications. Steroids raise blood pressure in many people who take it. One reason is that steroids and other corticosteroids cause the body to retain fluid. Extra fluid in the circulation can cause an increase in blood pressure. Further, anti-inflammatory corticosteroids significantly impair wound healing. Corticosteroids lower transforming growth factor- β (TGF- β) and insulin-like growth factor-I (IGF-I) levels and tissue deposition in wounds and that retinoids stimulate corticosteroid-impaired TGF- β and IGF-I release and collagen production.

[0442] For example, radiation treatment can result in inflammation of the organ and thickening of the tissue wall. This effect can increase the stiffness, thickness, and toughness of the tissue being treated. This increases the need for longer compression times and potentially higher compression thresholds, unless complications inhibit that. Radiation treatments can also have complication effects on blood oxygenation thru impacting the respiration system and can have a multiplicative effect on collagen vascular disease which could in turn require changes in any advanced energy welding energy blending or algorithms leaning towards more time to weld at a slower rate before cutting. This is a secondary effect as a tissue composition makeup complication effect.

[0443] For example, chemotherapy treatment can result in the tissue becoming thin and friable. These effects make collateral damage to the tissues much more likely and more difficult to treat. The implications for any mechanical device is lower manipulation forces and precompression levels, as well as lower rate thresholds and in general more gentle handling and tissue tensions needed. This is a primary effect as complication effects with higher tissue compressions.

[0444] Contextual cues associated with procedure complexity include the location of a tumor, remaining vascularization, the challenge of accessing the surgical site, the total time under anesthesia, the amount of work required to complete the procedure, and whether there were any prior procedures.

[0445] For example, the remaining vascularization can be a contextual cue because vascularization is directly related to the rate of healing and tissue viability. Further, it has longer-term implication on tissue strength and recovery. This is a primary physiologic impact on healing. This does not have any short-term instrument operation implications, but does have implications on recovery strength and reinforcement, needing additional time of the primary surgical treatment durability. This may impact the instrument's recommendations for post-surgery recovery, additional adjunct

therapies applied, and required monitoring. This is a secondary effect as an amplification of disease state, blood sugar level, and oxygenation impacts on tissue remodeling.

[0446] For example, the total time under anesthesia can be a contextual cue because the time under anesthesia is a complicating effect on recovery to pre-surgery levels relating to oxygenation levels and metabolic reactions. During the surgery, it has an amplification effect on lower blood oxygenation levels. This is a time-dependent effect that is not linear; the longer the time, the higher the impact of the effects become. This is a tertiary effect as a complication effect on lower blood oxygenation levels.

[0447] For example, the amount of work required to complete the procedure can be a contextual cue because it relates to the number of cycles of energy, the number of dissector or scissors moves, and/or the number of surgical stapling instrument firings.

[0448] For example, prior procedures can be a contextual cue because prior procedures increase the likelihood of adhesions and secondary remodeling of tissues. This typically creates more disorganized tissue planes and tougher tissues with more covering tissues. This is a tertiary effect as amplification of disease complication effects, as well as collagen level complication effects.

Patient-Specific Contextual Cues

[0449] Patient-specific contextual cues can include, for example, patient parameters and physiologic cues.

[0450] Patient parameters can include age, gender, whether the patient is a smoker, BMI, and body composition information.

[0451] For example, age results in friable tissues that would require a lower compression and lower rate of compression of the treatment devices, especially in the pre-treatment compressions. This is a secondary effect as a higher tissue compression complication effect.

[0452] For example, gender has threshold implication shifts for the ideal ranges of many physiologically related measures (e.g., BMI, body fat composition, and age impacts on physiology). This is a tertiary effect on other parameters.

[0453] For example, whether the patient is a smoker results in thicker, stiffer lung tissue that would suggest load clamping rates, lower pre-compression levels, and slower firing actuation of the knife/I-beam to prevent adjacent collateral damage around the perimeter of the anvil and cartridge due to excessive pressure differential during treatment. These are secondary effects as emphysema and oxygen saturation complication effects. Tissue oxygenation can be, as noted below, a metric available to quantify effect of smoking.

[0454] For example, BMI is a contextual cue because obesity tends to increase co-morbidities of many other medical complications. This is a tertiary effect as an amplification complication effect with blood sugar levels, congestive heart issues, oxygenation levels, and several other disease states.

[0455] For example, body composition information can be contextual cues because body fat percentage affects the collagen content of tissues, the compressive properties of the tissue types, and metabolic implications on healing and tissue remodeling. This has effects at both too high and too low of a percentage with differing effects at each extreme. Body fat percentages over a given level inhibit metabolic operation and add to complications around organ function. These complications will tend to amplify disease state complications on the mechanical device functions. Further, body fat percentages below a given level will tend to have impact on the tissue makeup itself. These tissue makeup changes can have impacts both on healing as well as advanced energy devices ability to weld consistently due to fluctuations in collagen levels, requiring more compression and longer weld times. These are tertiary effects as amplification of disease complication effects, as well as collagen level complication effects.

[0456] Physiologic cues can include the time since the patient last ate, fasting blood glucose level, blood pressure, macro tissue tension, tissue fluid levels, and tissue oxygenation.

[0457] For example, fasting blood glucose level can be a contextual cue because blood sugar level

is the main physiologic factor in healing. When blood sugar level is higher than normal, it prevents nutrients and oxygen from energizing cells and prevents your immune system from functioning efficiently. This is a secondary effect. Further, knowing the steady-state fasting state, as well as the post-meal changes and reactions, impact the implications of a measured value. In most humans this varies from about 82 mg/dl to 110 mg/dl (4.4 to 6.1 mmol/l). The blood sugar levels rises to nearly 140 mg/dl (7.8 mmol/l) or a bit more in normal humans after a full meal. In humans normal blood glucose levels are around 90 mg/dl, equivalent to 5 mM (mmol/l)

[0458] This measure is also time dependent. Consuming carbohydrate heavy food would cause a dramatic increase in blood sugar, but it would typically also begin to decrease after around 30 minutes.

[0459] For example, blood pressure can be utilized as a contextual cue because advanced energy devices benefit from improved coagulation before cutting. This is a priority and primary effect. In a hybrid energy device, the RF power could be increased or the ultrasonic application delayed in order to increase the coagulation before the cutting. In an ultrasonic-only device, the algorithm could be adjusted to apply a lower powered level for a longer period of time in order to denature the collagen longer before cutting. Further, in an ultrasonic-only device the harmonic power could be adjusted lower as the temperature approaches a predefined optimum temperature and that temperature could be maintained for a longer period of time before elevating the power of the transducer to initiate cutting. Blood pressure can be measured using different methods and at different locations. For example, the 10 minute resting pressure, i.e., resting blood pressure, can be very different than any blood pressure resulting from exertion. Knowing if this is a blood pressure measure based on an acute measure or is considered a systemic resting pressure will have implications on how to respond to the measure and its exceeding of upper or lower pressures. Further, differences exist for vascular levels of blood pressure. Typical blood pressure is taken in larger arteries within an arm or other extremity. A pressure in arteries in the arm of 129/80 could relate to a micro pressure of 70/40 in the capillaries and even lower 20/10 in veins where the actual tissue treatments are being preformed. Occlusions and variations in physiology can amplify or constrain the differences in pressure from one part of the system to another. Knowing where the pressure is being taken and any long-term measures could help adjust the effects needed due to changes in pressure.

[0460] For example, tissue fluid levels can be a contextual cue because dehydration reduces blood flow throughout the body, while also consequently lowering blood pressure, it can starve the wound bed of white blood cells that protect against infection, while also limiting oxygen reaching the wound site by way of blood flow, as do vitamins and nutrients. In general, lower fluid levels inhibit every aspect of wound healing. This is a tertiary effect as an amplification of healing complication effects. For surface tissue remodeling and potentially colorectal site healing, dehydration can delay healing in several ways. A warm, damp environment is ideal for the growth of new tissue, and a lack of moisture to the affected area can halt cellular development and migration. Without proper moisture, the epithelial cells that migrate across the wound bed to repair tissue along the way cannot properly navigate and cover the wound site. This interrupts the creation of new tissue and leaves the wound open and susceptible to harmful bacteria that can cause infection. Potential measures as related to dehydration can include, for example, electrolytes, blood urea nitrogen, creatine, urinalysis, complete blood count, and urine and/or blood osmolality.

[0461] For example, tissue oxygenation can be a contextual cue because tissue oxygenation is widely recognized to play a role in nearly every part of the wound healing stages. When healing, a surgical site develops an increased need for bacterial defense, cell proliferation, collagen synthesis and angiogenesis, among other reparative functions. Collagen accumulation is a direct function of oxygen tension and levels below 20 mmHg have been shown to impair accumulation. Collagen synthesis is dependent on functions of enzymes that are in turn a function of local oxygen levels. By contrast, hyperbaric oxygen therapy has been shown to increase healing rates by increasing the

oxygen concentrations above normal. While the tumor tissue is metabolically designed to thrive under conditions of hypoxia, hypoxia of the wound primarily caused by vascular limitations is intensified by coincident conditions (e.g., infection, pain, anxiety and hyperthermia) and leads to poor healing outcomes. This is a tertiary effect as an amplification of healing complication effects. Tissue oxygenation can be measured according to oxygen delivery (DO₂), oxygen uptake/consumption (VO₂), oxygen tension (PO₂), or hemoglobin oxygen saturation (SO₂). Further, several other techniques are available for measuring tissue oxygenation, such as near infrared spectroscopy (NIR).

Procedure-Specific Contextual Cues

[0462] Procedure-specific contextual cues can include the time of day the procedure occurred, whether it was an emergency versus a planned surgery, the length of the procedure, the type of procedure (e.g., laparoscopic, robotic, or open), and whether it was a reoperative or original procedure.

Surgeon-Specific Contextual Cues

[0463] Surgeon-specific contextual cues can include whether the surgeon was a specialist or a general practitioner (this comparison made against the procedure to be performed), the skill level of the surgeon (which can be indicated by, e.g., the total number of procedures performed, total number of times performed the current operation, and/or training level), and the focus or energy of the surgeon (which can be indicated by, e.g., the number of other procedures performed that day and the duration of current procedure).

Management of Metadata and Data

[0464] In various aspects, the metadata (e.g., the contextual cues described above) can be included with the general data generation.

[0465] In one aspect, the metadata can be stored by attaching the metadata to the primary data with the ability to filter the data out from the metadata. In another aspect, the metadata can be stored in a location other than the primary data, but can be linked to, allowing for reaching into the metadata for key metadata.

[0466] The accessibility of metadata can be controlled in a variety of different manners. In one aspect, the linked metadata can be transported with the original collected data. In another aspect, data can be extracted from the metadata by filtering data and relevant context.

Knowledge Hierarchy

[0467] In various aspects, contextual cues can be organized to provide data based on the needed context. Accordingly, the computer system can be configured to identify or determine the relevance of specific metadata to provide context. Further, a computer system can be programmed to provide navigation thru metadata.

Utilization Methodology

[0468] In one aspect, the metadata can be utilized for (i) identification and linking of isolated but interrelated data points or records, (ii) identification of linked occurrences, and/or (iii) algorithms can be programmed to automatically compare outcomes (and complaints) to any/all recorded data and compare regression trends and model capability of prediction to determine which factors can influence success. This data can be limited to a single device (e.g., speed of firing or energy verse leak) or can be combined between multiple devices to infer such things as time of firing relative to placement of trocars or start of anesthesia (beginning of surgery) or number of activations of scissors/dissectors/energy devices (e.g., degree of dissection/removal of fat).

Automated Interrelation/Integration of Data Streams

[0469] In various aspects, a computer system (e.g., a cloud analysis system, a surgical hub, and/or a data warehousing system) can be programmed to perform automated interrelation and/or integration of multiple data streams into a single interface which has focus areas or areas of additionally accessible or overlayable data. In one aspect, real-time interpreted information can be displayed on a device to the user, where the interpretation is based on data from at least one

function of the device and data from a second different source.

[0470] In one aspect, interpreted information can be displayed to the user based on at least one function of the device, including at least one data source not originating within the device. In another aspect, the at least one additional source can include a measurement device capable of determining a relevant parameter of the patient to the device's function and the ability of the instrument to automatically update aspects of the secondary information on the display of the device and update it in real-time. In one aspect, the real-time update is achieved by the surgical instrument being able to repeatedly calculate the new data in the select form the user has defined.

[0471] Referring to FIG. 24, a modular device in the form of a handheld surgical device **205001** includes a handle **205002** with a user interface **205004** in the form of a display or screen, for example. The surgical device **205001** includes an end effector **205006** configured to transect blood vessels. In the example, of FIG. 24, the end effector **205006** comprises a staple cartridge **205008**. Staples **205003** are deployed from the staple cartridge **205008** into tissue (T) grasped by the end effector **205006**, and a cutting member travels distally to sever the tissue during a firing stroke of the surgical device **205001**. In other instances, the surgical device **205001** can employ energy (e.g. RF energy and/or Ultrasonic energy) to seal, coagulate, and/or cut the tissue.

[0472] Referring to the top left corner of FIG. 24, a field of view **205007** of an imaging device is depicted. The field of view **205007** can be displayed on any suitable monitor or display within and/or outside the sterile field. The field of view **205007** shows the end effector **205006** about to staple and cut across a blood vessel. Another previously stapled and cut blood vessel also appears in the field of view **205007**. A close-up of the field of view **205007**, on the bottom left corner of FIG. 24, shows a leak (L) in the previously stapled and cut blood vessel. As described above, the leak (L) can be detected by one or more of the automated image interpretation techniques described herein.

[0473] In the example of FIG. 24, the surgical device **205001** is illustrated as a handheld surgical instrument that is similar in many respects to the handheld surgical instruments **112** (FIG. 2), **235** (FIG. 9). The user interface **205004** can be, for example, the device/instrument display **237**. In other examples, however, the surgical device **205001** can be adapted for use as a surgical tool **117** of a robotic system **110**. In such instances, the user interface **205004** and/or the controls of the surgical device **205001** can be located at a surgeon's console **118** (FIG. 2), for example. Alternatively, the user interface **205004** can be any suitable display (e.g. **107**, **109**, **119**, **135**, **210**, **217**).

[0474] Referring to FIG. 25, a schematic diagram of the surgical device **205001** is illustrated. A control circuit **205012** is in electrical communication with a motor driver **205013** which controls a motor **205014** that is configured to motivate a firing member **205010** to deploy the staples **205003** and advance the cutting member during the firing stroke of the surgical device **205001**.

[0475] As illustrated in the bottom right corner of FIG. 24, the user interface **205004** displays various parameters associated with the firing stroke of the surgical device **205001** including a speed setting **205005** of a firing member **205010** movable to deploy the staples **205003** and advance the cutting member. The user interface **205004** also displays a wait-time graph **205018** providing a recommended wait-time between two separate functions of the surgical device **205001**, the first function being grasping the tissue and the second function being deploying the staples **205003** and advancing the cutting member. In various examples, the user interface **205004** permits the user to select and/or adjust the speed of the firing member **205010** and/or the amount of wait-time.

[0476] The user interface **205004** is configured to display interpreted information based on at least one function of the surgical device **205001**. In the example of FIG. 24, the at least one function is the firing stroke, and the interpreted information relates to tissue hemostasis. Specifically, the interpreted information relates to hemostasis of tissue treated in previous firings of the surgical device **205001**. The user interface **205004** may display the interpreted information concurrently with at least one parameter setting of the firing stroke such as, for example, a firing speed or a wait-

time, as described below in greater detail. In addition, the tissue hemostasis can be monitored separately, and the tissue-hemostasis data and/or the interpreted information can be transmitted to the surgical device **205001** through a suitable communication link.

[0477] In another example, interpreted information associated with blood pressure of a blood vessel in the tissue being grasped, for example, can be concurrently displayed with the firing speed and/or wait-time parameters of the firing stroke. In various examples, the blood pressure can be monitored separately, and blood pressure data, or information interpreted from the blood pressure data, can be transmitted to the surgical device **205001** through a suitable communication link.

[0478] In another example, the at least one function can be grasping tissue. Tissue compression or pressure can be a parameter of tissue grasping, which can be displayed on the user interface **205004** and/or modified by user input through the user interface **205004**. Furthermore, interpreted information associated with blood pressure of a blood vessel in the tissue being grasped, for example, can be concurrently displayed with the tissue compression or pressure settings

[0479] In at least one example, the interpreted information is based on at least one data source not originating within the surgical device **205001**. In various aspects, the data source is a separate device that is different than the surgical device **205001**. The data from the data source could be interpreted into information relevant to the function performed by the surgical device **205001**.

[0480] The data interpretation can take place locally at the surgical device **205001**. Alternatively, the data interpretation could be performed locally at the data source, and the interpreted information can be routed to the surgical device **205001** either directly or indirectly through a surgical hub (e.g. **106**, **206**), for example. Alternatively, the data interpretation could be performed locally at a processing unit of the surgical hub (e.g. **106**, **206**), and the interpreted information can then be routed to the surgical device **205001**, for example. Alternatively, the data interpretation could take place at a cloud system **104**, which can be configured to route the interpreted information to the surgical device **205001** directly or indirectly through the surgical hub (e.g. **106**, **206**), for example.

[0481] Routing data and/or information between the surgical device **205001**, the data source, and/or the surgical hub (e.g. **106**, **206**) can be accomplished using any suitable wired or wireless communication link. For example, the modular communication hub **203** can be used to route the information and/or data.

[0482] In various examples, the data source could be a sensor on another modular device. In various examples, the data source is an imaging device. In at least one example, the data source could be one or more components of the visualization system **108** (FIG. **1**). Various components of the visualization system **108** are described under the heading “Advanced Imaging Acquisition Module” in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

[0483] In various examples, an imaging device can be configured to record and/or process imaging data that are relevant to a function performed by the surgical device **205001**. Automated image interpretation can then be performed locally at the imaging device, locally at the surgical device **205001**, locally at a surgical hub (e.g. **106**, **206**), and/or remotely at the cloud system **104**. A user interface **25004** can then be configured to display the image interpretations concurrently, or simultaneously, with one or more parameter settings associated with the function.

[0484] In various examples, the interpreted information is displayed by the user interface **205004** in real-time, or close to real-time. The real-time update is achieved by repeatedly interpreting new data in a selected form the user has defined, for example. The interpreted information can be updated at a predetermined refresh rate, which can be selected by the user of the surgical device **205001**.

[0485] Referring to FIGS. **25** and **26**, the control circuit **205012** is configured to perform a process **205020**. In various examples, the control circuit **205012** includes a processor and a memory that

stores program instructions, which when executed by the processor, cause the processor to perform the process **205020**. As illustrated in FIG. **26**, the process **205020** comprises receiving **205022** input representative of interpreted information relevant to the firing stroke from an external data source, as described above. The process **205020** further causes **205023** the interpreted information to be displayed with the at least one parameter setting associated with the firing stroke on the user interface **205004**. In one example, the interpreted information is concurrently, or simultaneously, displayed with the at least one parameter setting to help the user selection. In at least one example, as illustrated in FIG. **24**, the parameter setting is a speed setting **205005** of the firing member **205010**. In at least one example, the parameter setting is a wait-time setting before beginning the firing stroke.

[0486] Referring to FIGS. **25** and **27**, the control circuit **205012** is configured to perform a process **205030**. In various examples, the control circuit **205012** includes a processor and a memory that stores program instructions, which when executed by the processor, cause the processor to perform the process **205030**. As illustrated in FIG. **27**, the process **205030** comprises receiving **205032** input from an external data source representative of interpreted information relevant to a tissue-treatment function of the surgical device **205001**, as described above. The process **205030** further causes **205033** the interpreted information to be displayed with at least one parameter setting associated with the tissue-treatment function on the user interface **205004**. In one example, the interpreted information is concurrently, or simultaneously, displayed with the at least one parameter setting to help the user selection.

Automated Image Interpretation

[0487] In various aspects, a surgical hub (or surgical instrument or other system) can be configured to reduce captured images into a representation of outcomes of a transection, for example. In one aspect, a surgical hub (e.g. **106**, **206**) or an imaging module thereof (e.g. **138**, **238**) can include an algorithm to decompose pixels of an image (e.g., an image captured by a scope) and perform a calculation to determine the color differences between tissue and end effectors and/or leaks (e.g., air bubbles, dye, or blood) to determine the presence, amount, and locations of any leaks (L) or end effectors, as illustrated in the bottom left corner of FIG. **24**. For example, an algorithm can be programmed to compare the weight(s) of pixel and sub-pixels from the resulting image to the mathematical value of the pixel. Detected leaks in previously treated tissue can then be presented to a user of the surgical device **205001** along with one or more recommended settings for parameters of an upcoming firing stroke, for example.

[0488] In one aspect, a surgical hub (e.g. **106**, **206**) or an imaging module thereof (e.g. **138**, **238**) can include a classification algorithm for performing digital image processing on an image (e.g., an image captured by a scope) to identify (classify), an end effector, bleeding, bubbles, and other events from other classes of tissue within the image. For example, an algorithm can be programmed to perform comparative pixelation where the image is reduced to a constrained grid pattern and each element of the grid is reduced to, e.g., a 256 color designation for the pixel. A first scan can remove all pixels from the analysis that are not in the correct coloration of the class being sought (e.g., bleeding). Then the potential bleeding areas are compared with either adjacent areas or backward one frame in order to identify flowing blood within the image.

[0489] In one aspect, a surgical hub (e.g. **106**, **206**) or an imaging module thereof (e.g. **138**, **238**) can include an algorithm to perform feature extraction image processing to reduce an image from a near infinite variation of aspects to zones that are formed by reducing the number of random variables to groups of similar variables. For example, a user can select a type of tissue or a feature of anatomy and the imaging system can simplify the characteristic variation within the image to a unified average aspect of the selected feature. This would allow users to find boundaries of tissue planes, different organs, or limits of a tissue surface disrupted by infection or cancer, for example.

[0490] In one aspect, a surgical hub (e.g. **106**, **206**) or an imaging module thereof (e.g. **138**, **238**) can include a pattern recognition algorithm to identify target features. Various such techniques are

disclosed in Misawa, Masashi et al., "Artificial Intelligence-Assisted Polyp Detection for Colonoscopy: Initial Experience" Gastroenterology, Volume 154, Issue 8, 202-2029.e3, which is hereby incorporated by reference in its entirety, and can be accessed at www.gastrojournal.org. [0491] In various examples, the control circuit **205012** may receive interpreted information related to hemostasis of tissue previously treated by the surgical device **205001**. The interpreted information can be based on imaging data processed by on one or more of the above-described algorithms. As illustrated in the bottom right corner of FIG. **24**, the interpreted information is illustrated in a graph **205019** depicting a second-to-last firing and a last firing of the surgical device **205001**.

Image Manipulation

[0492] In various aspects, algorithms can be programmed to manipulate one image feed to fit another feed to allow for visualization of a static image on a dynamic image. In one aspect, an algorithm can use landmarks and the ability to define the elasticity of the overlay shapes on the primary feed to allow the image to be distorted and forced to fit the moving underlying anchors. This would allow, for example, a pre-surgery CT scan of the tumor or surgical site to be layered over the live feed from the scope during the surgical procedure. This could be used, for example, to extrapolate a pre-surgery image landscape or complex from a portion of the scan which is open to visualization, allowing a surgeon to see tissues or structures that are currently occluded from visible view on a user display.

User Selectable Datasets

[0493] In various aspects, a surgical hub (e.g. **106**, **206**) or an imaging module thereof (e.g. **138**, **238**) can be configured to receive user-selectable, highlightable, or flagable data sets that would display their varying data either numerically, graphically, or as highlightable areas on another image feed.

[0494] The user-selectable datasets can be utilized in various surgical contexts. For example, selection of a blood pressure monitor of a selected blood vessel or capillary could be selected because it is to be transected and the surgeon would like to watch the pressure calculated in that region continuously in order to monitor the proximity of dissection to the vessel or adjacent nerve or as a means to decide how long to coagulate a specific region before transecting. As another example, a surgeon could select a series of blood vessels for the surgical hub system to provide a continuous updated visualization feed of the amount of blood moving through the series of blood vessels while they are skeletonized, dissected, and then transected individually. In this example a laser Doppler visualization system can show the magnitude of blood flow measures in a wide area overlaid on the visual image and the fluctuation in blood flow during the interactive dissection steps with the blood flow areas. Such interpreted information can be displayed on the user interface **205004** along with one or more parameters of a function of the surgical device **205001** such as, for example, the firing stroke.

[0495] In various aspects, the user can interact with the display of information on the user interface **205004** and select specific sources of additional information derived from data measured, beyond the displayed information. The user then could select the form and frequency the data should be refreshed. The internal processor of the surgical hub (e.g. **106**, **206**) would then continually update that shading, digital data point, etc. and move it on the display as the selected areas moves on the display. This would allow the user to move and refocus a camera or imaging system and the selected and highlight data would still be measuring and displaying the desired information relative to the user selection.

[0496] FIG. **28** is a diagram illustrating the surgical device **205001** providing unprompted recommendations to a user based on contextual cues, in accordance with at least one aspect of the present disclosure. In various aspects, unprompted recommendations can be provided to the user based on prior actions and intraoperative outcome assessments.

[0497] In at least one example, highlighting based on hyperspectral imaging (i.e., processing an

image to visualize particular types of structures) could trigger a warning indicator if the processed image shows something the user should be made aware of, even if the user did not request the particular imaging associated with the warning. For example, if a critical structure is detected, but is not visible under direct visualization, the surgical hub (e.g. **106, 206**) and/or device **205001** can automatically trigger a warning so that the user can be made aware of the critical structure.

[0498] In at least one example, as illustrated in FIG. **28**, an unprompted adjustment **205009** to a parameter setting (firing speed setting **205005**) is recommended through the user interface **205004** based on information interpreted from external data received from an external data source.

[0499] Referring to FIGS. **25** and **29**, the control circuit **205012** is configured to perform a process **205040**. In various examples, the control circuit **205012** includes a processor and a memory that stores program instructions, which when executed by the processor, cause the processor to perform the process **205040**. As illustrated in FIG. **29**, the process **205020** comprises receiving **205042** input representative of interpreted information relevant to a firing stroke of the firing member **205010** from an external data source, as described above. The process **205040** further causes **205043** the interpreted information to be displayed with the at least one parameter setting associated with the firing stroke on the user interface **205004**. In one example, the interpreted information is concurrently, or simultaneously, displayed with the at least one parameter setting to help the user selection. In at least one example, as illustrated in FIG. **28**, the parameter setting is a speed setting **205005** of the firing member **205010**. In at least one example, the parameter setting is a wait-time setting before beginning the firing stroke.

[0500] Referring to FIGS. **25** and **30**, the control circuit **205012** is configured to perform a process **205050**. In various examples, the control circuit **205012** includes a processor and a memory that stores program instructions, which when executed by the processor, cause the processor to perform the process **205050**. As illustrated in FIG. **30**, the process **205050** comprises receiving **205052** input from an external data source representative of interpreted information relevant to a tissue-treatment function of the surgical device **205001**. The process **205050** further causes **205053** the interpreted information to be displayed with at least one parameter setting associated with the tissue-treatment function on the user interface **205004**. In one example, the interpreted information is concurrently, or simultaneously, displayed with the at least one parameter setting to help the user selection. Furthermore, the process **205050** further comprises recommending **205054** an adjustment of the parameter setting based on the interpreted information. In various aspects, the recommendations **205044** and **205054** can be unprompted recommendations.

[0501] FIG. **31** is a diagram illustrating a surgical device **205001** that includes a user interface **205004** receiving input from a user to automatically adjust a field of view of a medical imaging device based on contextual cues, in accordance with at least one aspect of the present disclosure. The field of view of the medical imaging device can be displayed on a monitor **205011**, which can be outside the sterile field, for example.

[0502] In various examples, the automatic adjustment of the field of view of the medical imaging device may include automatic focusing and/or centering based on the location of a critical structure such as, for example, an end effector of the surgical device **205001**. In other examples, the critical structure can be an anatomical structure or surgical site location. In at least one example, the center of the area visualized on a monitor **205011** could be automatically adjusted based on user actions or device locations.

[0503] FIG. **32** is a logic flow diagram of a process **205100** depicting a control program or a logic configuration for automatically adjusting a field of view of a medical imaging device with respect to a detected critical structure. The process **205100** includes detecting **205102** the critical structure and assessing **205104** its position with respect to the field of view of the imaging device.

[0504] Various suitable image interpretation techniques, as described above, can be employed by an imaging module (e.g. **138, 238**) to detect **205100** the critical structure and/or assess **205104** its position with respect to the field of view of the imaging device. In one example, a surgical hub (e.g.

106, 206) or an imaging module thereof (e.g. **138, 238**) can include an algorithm to decompose pixels of an image (e.g., an image captured by a scope) and perform a calculation to determine the color differences between the critical structure and the surrounding environment. The determined color differences are utilized to detect **205100** the critical structure and/or assess **205104** its position with respect to the field of view of the imaging device. In another aspect, a surgical hub (e.g. **106, 206**) or an imaging module thereof (e.g. **138, 238**) can include a classification algorithm for performing digital image processing on an image (e.g., an image captured by a scope) to detect **205100** (classify) a critical structure and/or assess **205104** its position with respect to the field of view of the imaging device.

[0505] In the event it is determined **205106** that the critical structure is at the edge of the current field of view of the imaging device, and the medical imaging device is capable of adjusting the field of view on the locus of the critical structure (e.g. end effector), a monitor **205011** or the surgical field input device (e.g. user interface **205004**) could provide feedback by prompting **205108** the user that the field of view could be automatically adjusted with respect to the critical structure, if desired. The automatically adjusted could be a one-time adjustment or a continuous adjustment.

[0506] In at least one example, the visualization system (e.g. visualization systems **108, 208**) could determine, for example through the user interface **205004**, if the user would prefer the system to track and adjust the center focus area of the field of view of the imaging device on a locus of the critical structure. If the user selects the auto-tracking option, as illustrated in FIG. **31**, the visualization system can then control the display(s) accordingly.

[0507] In the event the critical structure is an end effector of a surgical device, the end effector can be articulated to a new position at the center of the field of view per instructions from the surgical hub, for example. Alternatively, the imaging device can be moved to reposition the field of view with respect to the critical structure.

[0508] FIG. **33** is a logic flow diagram of a process **205200** depicting a control program or a logic configuration for automatically adjusting a field of view of a medical imaging device with respect to a detected critical structure. The process **205100** includes receiving an input from the surgical hub (e.g. **106, 206**) indicative of a position of a critical structure with respect to a current field of view of a medical imaging device as determined by the visualization or imaging module (e.g. **138, 238**). The process **205200** further includes causing **205204** the user interface **205004** to recommend an adjustment that changes the position of the critical structure with respect to the current field of view of the medical imaging based on the received input.

[0509] In various examples, the medical imaging device comprises a camera pointed at the end effector **205006** of the surgical device **205001** at a surgical site within a patient cavity. In certain instances, as illustrated at the top left corner of FIG. **31**, the end effector **205006** is off-center with respect to the field of view **205007** of the medical imaging device. As described above, an imaging module (e.g. **138, 238**) of the surgical hub (e.g. **106, 206**) may detect that the end effector **205006** is off-center with respect to the field of view **205007**. Consequently, the surgical hub (e.g. **106, 206**) may cause the user interface **205004** to prompt the user of the surgical device **205001** for permission to automatically center the end effector **205006** with respect to the field of view **205007**, as illustrated at the bottom left corner of FIG. **31**. For example, the communication module **130** may wirelessly transmit instructions to a wireless receiver of the surgical device **205001** to prompt the user permission to automatically center the end effector **205006**.

Supervised Learning

[0510] In various aspects, the machine learning systems of the present disclosure utilizes supervised learning methods to cluster data pairs into non-predefined categories based not only the outcomes, but also on the metadata context. In one aspect, supervised machine learning can be performed on cloud data from the surgical hubs (e.g. **102, 202**) to find unidentified data groups. The machine learning cloud data analysis can identify, for example, regional differences in surgical

and patient outcomes.

[0511] Methods of supervised learning can include, for example, parametric learning methods or non-parametric learning methods. Parametric learning methods can include, for example, regression (e.g., predicting continuous data, discrete data, or gradient descent, which is a sequential process that is used to determine the minimum of the model), classification, and vector clustering. Vector clustering is a process that, given a set of training examples, creates the capability of the system to take new examples and placing them within the grouping defined by the training examples. Vector clustering can also include margin defining, which is maximizing the margin between the groups to minimize the mis-grouping of marginal data points. Non-parametric learning methods can include, for example, decision trees and k-NN algorithms.

[0512] Data compilation methods can include, for example, singular value decomposition, normalization, and dimensionality reduction.

[0513] FIG. **34** is a graph depicting positive and negative outcomes for illustrative uses of a modular device (e.g. **1a-1n**, **2a-2m**) according to a calculated threshold, in accordance with at least one aspect of the present disclosure. FIG. **35** is a block diagram depicting surgical procedures having various characteristics grouped according to outcomes, in accordance with at least one aspect of the present disclosure. Machine learning can be utilized to identify interrelated causes and effects in the data, which can then be utilized to autonomously develop control program updates for the surgical instruments, surgical hubs, and/or other modular devices.

Metadata Context

[0514] In various aspects, the metadata context of the data can be utilized to widen the zone between at least two groups and therefore fit the data to their groups based on the outcomes and their contributing factors. Supervised machine learning techniques can be utilized to define data groups by their outcomes and contributing factors.

[0515] The supervision methodology can be adjusted according to validation of groupings and sensitivity.

[0516] FIG. **36** is a diagram depicting a validation analysis, in accordance with at least one aspect of the present disclosure. FIG. **37** is a block diagram depicting a validation analysis, in accordance with at least one aspect of the present disclosure. In various aspects, an analysis system can utilize an adjustable validation set for performing validation testing. Learning algorithms are trained by being provided a set of training examples, which the learning algorithms then use to apply the learned groupings to new data. A validation data set allows the algorithm to then check if the created data set contains the anticipated data and then remove any unanticipated data. As shown in FIGS. **36** and **37**, being able to adjust this validation set will allow for the fine tuning of the grouping process.

[0517] The sensitivity of the supervision methodology can be adjusted to control overlearning and threshold/probability cutoff of the machine learning model. Overlearning is the adjustability or the ratio of algorithm flexibility to bias. A highly flexible algorithm will fit all variable data into a data set, whereas a highly biased algorithm will fix a very limited set of data points into a data set. Threshold and probability cutoff is the machine learning model's tolerance of false positives versus false negatives.

Unsupervised Learning

[0518] In various aspects, unsupervised (or “untrained”) learning techniques can be used in order to determine relationships of data pairs from the surgical hub that contain linkages or result from a complication or morbidity, rather than the treatment step directly. In other words, machine learning can be utilized to identify non-device causes and adapt surgical hub responses accordingly.

[0519] Unsupervised learning is the process of training a machine learning model on unlabeled data. With time-based or sequential order-based learned outcomes, a program can determine the most likely to succeed next steps based on a predefined number of initial responses. These responses could be issues encountered, patient pre-operative conditions, or encountered issues in

the surgical procedure. A number of unsupervised learning techniques can be used, including, for example, clustering (e.g., hierarchical clustering and k-means clustering), which provides protection against potential threats that are outside “normal” learned behavior, and unsupervised simulation of one instance of the program attempting to beat another instance of the program. The unsupervised learning techniques can also include deep learning and neural networks, which are analysis techniques that analyze trends and connections without considering the meaning of the data. Artificial neural networks can be utilized for, e.g., image processing to improve the ability to identify underlying critical structures by identifying images with the areas previously called out on the image, while having no information of the parameters of the structures themselves.

[0520] These data points could be used to identify to the user that there is an underlying cause for future evaluation, or that there is the need for additional treatment reinforcement, not because the device is not capable of proofing in a predictive fashion, but because the tissue or patient requires special care.

[0521] The result could be the identification to the surgeon that an additional procedure step is warranted, and adjunct treatment considered, or an alternative more robust product is needed. For example, during a colorectal lower anterior resection procedure, an imaging system might detect the over-limitation of blood supply to the region after mobilization as a portion of the metadata generated from the procedure. Accordingly, a vessel regeneration product or drug might be warranted due to the implications of not being able to maintain an adequate blood supply to the transected tissue.

[0522] As another example, during a colorectal lower anterior resection procedure, the co-morbidities or chemo treatments might indicate in the metadata generated from the procedure that the tissue is too easily friable. Accordingly, a pressure distributing absorbable adjunct stapling product might be warranted in order to stress reduce the anastomosis, a diversion might need to be created to prevent bowel contents from placing undo stress on the new anastomosis, and/or an oversewing or fibrin/thrombin sealant might need to be applied to insure no undo leaks occurs in the first days of healing. A surgical system **200** can be programmed to recommend the appropriate product, drug, or treatment to the physician after completion of the procedure.

[0523] Accordingly, instrument, device, or machine settings can be set according to relationships determined via unsupervised learning techniques to provide better outcomes. In at least one example, a surgical stapling instrument is updated to run slower or faster depending on tissue type, resistance to push the knife, and so on. In another example, energy devices (e.g., ultrasonic surgical instruments and electrosurgical instruments) are updated to change the applied current over the transection (i.e., slowly ramping up or down or taking other such actions). Further, in various examples, one or more of the modular device (e.g. **1a-1n**, **2a-2m**) of the present disclosure are programmed to automatically update their settings based on the correlations learned by a machine learning performed by the cloud-based system (e.g. **104**, **204**).

[0524] In some aspects, the surgeon could have the ability to override the learned settings. In one aspect, the learned settings could be the default settings. In one aspect, a modular device (e.g. **1a-1n**, **2a-2m**) could suggest the new settings and allow the surgeon to choose between accepting the new settings and keeping the previous defaults.

[0525] A surgical system (e.g. **100**, **200**) is described herein that is configured to optimize outcome(s) of surgical procedures through machine learning. Patterns of treatment may be recognized over the course of numerous procedures that, when implemented, lead to a successful and/or safe result for a specific patient. Such patterns of treatment can include, for example, the type(s) of surgical instrument(s) to use during the procedure, additional procedure(s) to be performed, and/or any concerns that require further monitoring. Data collected during each particular procedure, the treatment performed, and/or the outcome of the particular procedure can be stored in a data bank (e.g. remote server **113**) for future analysis. The collected data can be used to bolster and/or update existing treatment recommendations. Analyzing various health statistics of

a patient, such as, for example a patient's medical record and/or current vital statistics, in light of the recognized patterns can further optimize the likelihood of success of the particular procedure. [0526] As described above in greater detail, a surgical system (e.g. **100, 200**) comprises an information hub, or a surgical hub (e.g. **106, 206**). The surgical hub is configured to facilitate communication between any surgical instruments used in the surgical procedure, the data source, and/or the clinician. An information hub, such as, for example, the surgical hub described herein, can store the data collected during each particular procedure. In various instances, the surgical hub can locally store data specific to the particular procedure and/or the specific patient. The surgical hub can store additional data relevant to different procedures in an external server (e.g. remove server **113**).

[0527] In various examples, referring to FIG. **10**, a surgical hub **206** comprises a control circuit including a processor **244** configured to receive an input from a data source. The data source is configured to supply the surgical hub **206** with data relevant to a particular surgical procedure and/or a particular patient. For example, the data source can provide the surgical hub **206** with the patient's medical history and/or current vital statistics, such as, for example, blood pressure. The data source can also provide the surgical hub **206** with data collected from the surgical site by a surgical instrument and/or a surgical visualization system.

[0528] Example surgical instruments that are suitable for use with the surgical hub **206** are described under the heading "Surgical Instrument Hardware" in U.S. Provisional Patent Application Ser. No. 62/611,341, entitled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety. Various components of the surgical visualization system, such as the visualization system **108** are described under the heading "Advanced Imaging Acquisition Module" in U.S. Provisional Patent Application Ser. No. 62/611,341, entitled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is hereby incorporated by reference in its entirety.

[0529] Upon receiving an input, or data, from the data source, the surgical hub **206** analyzes the received data against a stored set of data. In various examples, the stored data set is stored in a memory **249**. Such analysis is performed with the goal of optimizing an outcome of a surgical procedure. In various examples, the data storage and/or machine learning analysis can be performed locally at the surgical hub level. Additionally, or alternatively, the data storage and/or machine learning analysis can be performed at the cloud **204**, for example.

[0530] In various instances, the surgical hub **206** analyzes the received data against a stored set of data using one or more untrained machine learning techniques. Using an untrained, or unsupervised, machine learning technique can allow for the determination of relationships between data pairs and/or an identification of a complication resulting from the performed surgical procedure.

[0531] The surgical hub **206** identifies commonalities in the received data and the stored data and react based on the presence or absence of such commonalities in each analyzed piece of data. Primary techniques in untrained machine learning include, for example, clustering. The goal of clustering is to create groups of data points such that points in different clusters are dissimilar while points within a cluster are similar. In other words, clustering offers protection against potential threats that are outside "normal" learned behavior. In various instances, the control circuit is configured to utilize clustering in analyzing the received data against stored data. Clustering techniques include, for example, k-means clustering and hierarchical clustering. In k-means clustering, data points are clustered into k groups. A larger k creates smaller groups with more granularity, while a smaller k means larger groups and more granularity. Each group is defined by creating a centroid for each group, wherein the cluster captures the points closest to it and adds them to the cluster. In hierarchical clustering, similar data points are grouped into clusters. In order to decide which clusters should be combined, or where a cluster should be split, a measure of dissimilarity between sets of data is required. This is achieved by use of an appropriate metric, or a

measure of distance between pairs of data, and a linkage criterion which specifies the level of dissimilarity between data pairs.

[0532] Untrained machine learning techniques can develop recommendations based off of the creation of artificial neural networks. For example, a surgical imaging system, such as surgical visualization system **108**, can collect data representative of structures within the surgical site. After receiving the collected structural data, the surgical hub **206** analyzes the collected data against a stored set of structural data. Techniques in untrained machine learning analyze trends and connections without considering the meaning of the data. Such analysis allows for the processor **244** to recommend identities of underlying critical structures in a surgical site by utilizing images with previously-identified areas while having no information of the parameters of the critical structures themselves.

[0533] Based on the untrained machine learning analysis, the surgical hub **206** is configured to recommend an action. The recommended action(s) can be directed toward the clinician in the form of a prompt. The analyzed data can be used to identify to a clinician that an additional procedural step is warranted, an adjunct treatment should be considered, and/or an alternative treatment is needed. The alternative treatment can comprise using a different surgical instrument and/or implementing a more robust operating program on the surgical instrument.

[0534] For example, a clinician may be performing a colorectal lower anterior resection procedure on a patient. During the surgical procedure, a surgical visualization system, such as surgical visualization system **108**, may detect an over-limitation of blood supply to the region after mobilization. Once the surgical imaging system, or an alternative data source, communicates the detected over-limitation of blood supply to the surgical hub **206**, the processor **244** analyzes the detected over-limitation of blood supply against previously collected data. Using an untrained machine learning technique, the processor **244** assesses commonalities between the detected over-limitation of blood supply and previously collected data. Based on the assessed commonalities, the processor **244** is configured to develop a recommendation to optimize an outcome of the surgical procedure. A possible recommendation can be, for example, use a vessel regeneration product and/or drug due to the implications of not being able to maintain an adequate blood supply to the transected tissue.

[0535] In another example, a clinician may be performing the same colorectal lower anterior resection procedure as described above. Prior to and/or during the procedure, the patient's medical records may comprise information indicating that the patient has comorbidity, or a simultaneous presence of another chronic disease or condition, and/or has previously undergone chemotherapy. Once a data source communicates the data comprised in the patient's medical records to the surgical hub **206**, the processor **244** analyzes the current medical records against previously collected data.

[0536] Using an untrained machine learning technique, the processor **244** assesses commonalities between the current patient's medical records and previously collected data. Based on the assessed commonalities, the processor **244** develops a recommendation to optimize an outcome of the surgical procedure. Possible recommendations can include, for example: (1) Use a pressure-distributing absorbable adjunct stapling product in order to stress reduce the anastomosis; (2) Create a diversion to prevent bowel content from placing undue stress on the new anastomosis; and/or (3) Apply an over-sewing or fibrin/thrombin sealant to prevent leaks from occurring within the first few days of healing.

[0537] Using an untrained, or unsupervised, machine learning technique can allow for the determination of relationships between data pairs and/or the identification of a complication resulting from the performed surgical procedure. The processor **244** identifies commonalities in the received data and the stored data and reacts based on the presence or absence of such commonalities in each analyzed piece of data. Based on this analysis, the processor **244** recommends an action. The recommended action(s) can be directed toward a surgical instrument in

the form of an operating program and/or parameter. In other words the recommended action(s) can be in the form of modified machine and/or device (e.g. modular devices **1a-1n**, **2a-2m**) settings to provide improved outcomes. For example, various surgical instruments can receive the following alterations and/or updates to the instrument's operating program: (1) surgical staplers can run slower or faster depending on factors such as tissue type, resistance felt by the knife, etc.; and (2) surgical energy devices can change the current flowing over the transection by slowly ramping up or down, etc. In various instances, the surgical hub **206** is configured to automatically update a surgical instrument's settings based on the unsupervised machine learning analysis.

[0538] While a surgical instrument's operating program can be automatically updated and/or altered with the recommendations of the surgical hub **206**, the clinician can also have the ability to override the implemented recommendations. In various instances, the surgical hub **206** can prompt the clinician with the recommended alterations to the surgical instrument's operating program. The clinician is then able to accept the recommended alterations or maintain previous operating parameters. In other instances, the recommended alterations can automatically take effect, and the clinician is able to switch back to the previous operating parameters if the clinician desires.

[0539] As described above, upon receiving an input, or data, from the data source, the processor **244** of the surgical hub **206** analyzes the received data against a stored set of data. Such analysis is performed with the goal of optimizing an outcome of a surgical procedure. In various instances, the processor **244** analyzes the received data against a stored set of data using one or more trained machine learning techniques. In supervised learning, the processor **244** predicts a parameter as accurately as possible when given new examples where the inputs and outputs are unknown.

[0540] Primary techniques in trained machine learning include, for example, parametric learning and non-parametric learning. Regression, classification, and vector clustering are the two tasks involved in parametric learning. Regression predicts a continuous target variable, thereby allowing the processor **244** to estimate a value based on received input data. Continuous variables, such as, for example, a patient's height, weight, emphysema air leak rate, etc., means there are not discontinuities in the value that the predicted parameter can have. Discrete variables, on the other hand, can only take on a finite number of values—for example, the color of a staple cartridge within an end effector.

[0541] For example, a clinician may wish to predict the emphysema air leak rate based on factors such as, the stage of emphysema present in the patient, the firing speed of a surgical stapler, the force required to close the jaws of an end effector, the type and/or color of the staple cartridge within the end effector, etc. A surgical stapler is used to staple lung tissue in an emphysema patient. A suitable staple cartridge is selected based on the condition of the patient and/or the condition of the lung tissue to be stapled by the surgical stapler. Staple cartridge color may reflect the size of the surgical stapler, for example.

[0542] In various examples, the surgical hub **206** builds a model that approximates the relationship between the above factors and corresponding emphysema air leak rate. As shown in FIG. **36**, in supervised learning, the processor **244** attempts to learn the relationship between, for example, emphysema air leak rate and the previously-identified factors by running labeled training data **250810** through a learning algorithm. The learned function can be used by the processor **244** to estimate the emphysema air leak rate of a patient whose emphysema air leak rate is unknown. The estimation is based on various inputs including the stage of emphysema, the firing speed of the surgical stapler, the force required to close the jaws of the end effector, and/or the type and/or color of the staple cartridge within the end effector.

[0543] FIG. **34** represents exemplary data collected during previous surgical procedures. In FIG. **34**, the patients were diagnosed with a particular stage of emphysema. The calculated threshold **250700** represents the point at which the outcome(s) of the surgical procedure are acceptable. The positive (+) side represents a good, or beneficial, outcome of the surgical procedure, while the negative (−) side represents a bad, or detrimental, outcome of the surgical procedure. It has been

observed that radiation stiffens lung tissue, while emphysema softens lung tissue. Such an observation has an effect on the operating program of a surgical instrument, such as a surgical stapler. For example, advancing a firing member too quickly through a diseased tissue may have a detrimental impact on the outcome of the surgical procedure.

[0544] A first patient with stage 2 emphysema is represented by line **250710** and a second patient with stage 2 emphysema is represented by line **250715**. A firing member of a surgical stapler was advanced at a slow speed during the surgical procedure of the first patient **250710**, and a firing member was advanced at a fast speed during the surgical procedure of the second patient **250715**. As represented by the respective dashed lines **250710**, **250715**, the speed of the firing member appears to be irrelevant when the patient has stage 2 emphysema, as both patients had a good, or beneficial, outcome. Notably, the fast speed of the firing member brought the second patient **250715** closer to the threshold of acceptable outcomes of the surgical procedure.

[0545] A first patient with stage 3 emphysema is represented by line **250720** and a second patient with stage 3 emphysema is represented by line **250725**. A firing member of a surgical stapler was advanced at a slow speed during the surgical procedure of the first patient **250720**, and a firing member was advanced at a fast speed during the surgical procedure of the second patient **250725**. As represented by dotted line **250725**, the fast advancement of the firing member in a patient with stage 3 emphysema produces unacceptable and/or detrimental outcomes to the surgical procedure, while the dotted line **250720** shows that the slow advancement of the firing member in a patient with stage 3 emphysema results in an acceptable and/or beneficial outcome to the same surgical procedure. A first patient with stage 4 emphysema is represented by line **250730** and a second patient with stage 4 emphysema is represented by line **250735**. A firing member of a surgical stapler was advanced at a slow speed during the surgical procedure of the first patient **250730**, and a firing member was advanced at a fast speed during the surgical procedure of the second patient **250735**. As represented by the respective dashed lines **250730**, **250735**, the speed of the firing member appears to be irrelevant when the patient has stage 4 emphysema, as both patients had a bad, or detrimental, outcome.

[0546] As shown in FIG. 35, data similar to that collected in FIG. 34, is collected with respect to numerous parameters, including, for example: (1) the stage of emphysema **250750**; (2) if the patient has received radiation **250752**; (3) the force required for a surgical stapler to advance its firing member **250754**; (4) the patient's blood pressure **250756**; (5) advancing the firing member at a low speed **250758**; and/or (6) advancing the firing member at a high speed **250759**. As represented by the position of the measured parameters along the threshold line **250700**, each parameter has either a positive or negative effect on an outcome of a surgical procedure. As represented by the dashed line **250755**, the patient having received radiation **250752** and the force required for a surgical stapler to advance its firing member **250754** being high both have an inter-related cause and effect on the outcome of the surgical procedure. High blood pressure **250756** and a low speed of advancement of a firing member **250758** have positive effects on the outcome of the surgical procedure, while a high speed of advancement of a firing member **250759** and a greater stage of emphysema **250750** have negative effects on the outcome of the surgical procedure. Dashed line **250757** represents a surgical instrument control program change based on unlearned grouping.

[0547] Boxes **250760** represent measured parameters detected from individual patients whose combination resulted in a bad, or detrimental, outcome to the surgical procedure. Boxes **250770** represent measured parameters detected from individual patients whose combination resulted in a good, or beneficial, outcome to the surgical procedure. As described in greater detail with respect to FIG. 37, boxes **250765** represent measured parameters detected from individual patients whose combination resulted in unexpected results that are not representative of other patients. In various instances, extenuating circumstances affected the result of the surgical procedure for the patients represented by **250765**.

[0548] The previously collected data, as shown in FIG. 36, is split into a training data set **250810** and test data set(s) **250820, 250830**. The training data set **250810** has labels, so the surgical hub **206** can learn from these labeled examples. Such labels can include, for example, various determined emphysema air leak rates and their corresponding stage of emphysema, the firing speed of the surgical stapler, the force required to close the jaws of an end effector, and the type and/or color of the cartridge within the end effector. The test set(s) **250820, 250830** comprise data that does not have labels, i.e. the processor **244** does not yet know the value it is trying to predict. In other words, the processor **244** has to generalize to situations it hasn't encountered before so that it can provide the most accurate prediction for the test data **250820, 250830**.

[0549] The surgical hub **206** to create a mathematical model **250840** based off of the previously collected data marked as the training data set **250810**. The training data set is comprised of a large sample size, i.e. $n=1000$, to define a function. A test set **250820, 250830**, or a validation set, allows the surgical hub **206** to check if the test sets **250820, 250830** fit the determined mathematical model. The test sets **250820, 250830** comprise a smaller sample size, i.e. $n=50$. Analysis **250835** of the test sets **250820, 250830** results in a validation model **250850** for comparison to the generated mathematical model **250840**. In various instances, this gives the surgical hub **206** the opportunity to keep anticipated data and remove any unanticipated data. The ability to adjust the mathematical model **250840** allows for the fine tuning of the grouping process to populate **250855** a final mathematical model for use in data analysis. The developed model can be applied in analysis of collected data to estimate, for example, emphysema air leak rate.

[0550] As discussed above, some data points are outliers and must be removed from the data pool in order to fine tune the mathematical model. As shown in FIG. 37, and as discussed with respect to FIG. 35, boxes **250760** represent measured parameters detected from individual patients whose combination resulted in a bad, or detrimental, outcome to the surgical procedure. Boxes **250770** represent measured parameters detected from individual patients whose combination resulted in a good, or beneficial, outcome to the surgical procedure. Boxes **250765** represent measured parameters detected from individual patients whose combination resulted in unexpected results that are not representative of other patients. In various instances, extenuating circumstances affected the result of the surgical procedure for the patients represented by **250765**. The presence of the outlier boxes **250765** alters the median threshold as the mathematical model attempts to explain their results. The outlier data **250765** is excluded from the mathematical model, adjusting the actual threshold **250700** to its rightful position. The outlier boxers **250765** would have originally been grouped **250900** by the algorithm with the other detrimental outcome boxes **250760** as none of them produced an acceptable surgical procedure result. This would have resulted in inconsistent and inaccurate predictions of future outcomes. The actual grouping of parameters contributing to detrimental outcomes is represented by **250910**, and the actual grouping of parameters contributing to beneficial outcomes is represented by **250920**. A common problem in machine learning is overfitting. Overfitting is learning a function that perfectly explains the training data that the model learned from, but does not generalize well to unseen test data. Overfitting happens when a model overlearns from the training data to the point that it starts picking up characteristics that are not representative of patterns in collected data. Underfitting is a related issue where the generated model is not complex enough to capture an underlying trend in the data.

[0551] Referring now to FIG. 38, an algorithm **250950** for a surgical system **200** is depicted. The algorithm **250950** is intended to allow for adjustment of control programs of surgical instruments and/or other components of the surgical system **200** based on a remote analysis involving supervised and/or unsupervised learning techniques. Such techniques, as described in greater detail above, include repeated multivariate analysis of different variable combinations in search of correlations and an assessment of the predictive ability of outcome of the analysis against a separate group of data not used in the original analysis.

[0552] Various surgical systems disclosed herein such as, for example, the surgical systems **100**,

200 can utilize the algorithm **250950** of FIG. **38**. Moreover, the reader will readily appreciate that the algorithm **250950** of FIG. **38** can be combined with one or more additional algorithms described herein in certain instances. The decisions made by the surgical system **200** can be implemented, for example, by a control circuit that includes the processor **244**. The control circuit is configured to receive data **250952** from a data source. The received data can include, for example, data pertaining to the vital statistics of the patient, the medical history of the patient, the type of surgical procedure being performed, the type of surgical instrument being used, any data detected by a surgical instrument and/or surgical visualization system, etc.

[0553] The control circuit of the surgical system **200** is configured to analyze the received data **250954** against a stored set of data. Such analysis **250954** can be based on supervised or unsupervised learning. Such analysis **250954** can be performed within the local network of facility linked devices or could be exported to a remote location for compilation and returned to the network.

[0554] Following analysis **250954**, the control circuit is configured to communicate a recommendation **250956** based on the analysis. The recommendation can be based on the identification of device interrelated impacts, patient specific impacts, and/or interaction of used devices. The recommendation can be communicated to a clinician in the form of a prompt and/or can be communicated to a particular surgical instrument in the form of updated operating parameters. In one example, the recommendations are communicated by a cloud **204** to a modular device (e.g. **1a-1n**, **2a-2m**) either directly, or through the surgical hub **206**. In one example, the recommendations are communicated to a surgical hub **206**. In any event, the recommendation can include, for example, a prompt to update a control program of the modular device or adjusting the modular device's operating parameters. In various instances, the recommendations can comprise suggestions or other procedural modifications to be indicated to the clinician. In certain instances, the recommendations include, for example, adaptation of an actuation speed, wait time, and/or other operating parameters of a modular device.

Supervised Learning

[0555] In various aspects, the machine learning systems of the present disclosure utilizes supervised learning methods to cluster data pairs into non-predefined categories based not only the outcomes, but also on the metadata context. In one aspect, supervised machine learning can be performed on cloud data from the surgical hubs (e.g. **102**, **202**) to find unidentified data groups. The machine learning cloud data analysis can identify, for example, regional differences in surgical and patient outcomes.

[0556] Methods of supervised learning can include, for example, parametric learning methods or non-parametric learning methods. Parametric learning methods can include, for example, regression (e.g., predicting continuous data, discrete data, or gradient descent, which is a sequential process that is used to determine the minimum of the model), classification, and vector clustering. Vector clustering is a process that, given a set of training examples, creates the capability of the system to take new examples and placing them within the grouping defined by the training examples. Vector clustering can also include margin defining, which is maximizing the margin between the groups to minimize the mis-grouping of marginal data points. Non-parametric learning methods can include, for example, decision trees and k-NN algorithms.

[0557] Data compilation methods can include, for example, singular value decomposition, normalization, and dimensionality reduction.

[0558] FIG. **34** is a graph depicting positive and negative outcomes for illustrative uses of a modular device (e.g. **1a-1n**, **2a-2m**) according to a calculated threshold, in accordance with at least one aspect of the present disclosure. FIG. **35** is a block diagram depicting surgical procedures having various characteristics grouped according to outcomes, in accordance with at least one aspect of the present disclosure. Machine learning can be utilized to identify interrelated causes and effects in the data, which can then be utilized to autonomously develop control program updates for

the surgical instruments, surgical hubs, and/or other modular devices.

Metadata Context

[0559] In various aspects, the metadata context of the data can be utilized to widen the zone between at least two groups and therefore fit the data to their groups based on the outcomes and their contributing factors. Supervised machine learning techniques can be utilized to define data groups by their outcomes and contributing factors.

[0560] The supervision methodology can be adjusted according to validation of groupings and sensitivity.

[0561] FIG. 36 is a diagram depicting a validation analysis, in accordance with at least one aspect of the present disclosure. FIG. 37 is a block diagram depicting a validation analysis, in accordance with at least one aspect of the present disclosure. In various aspects, an analysis system can utilize an adjustable validation set for performing validation testing. Learning algorithms are trained by being provided a set of training examples, which the learning algorithms then use to apply the learned groupings to new data. A validation data set allows the algorithm to then check if the created data set contains the anticipated data and then remove any unanticipated data. As shown in FIGS. 41 and 42, being able to adjust this validation set will allow for the fine tuning of the grouping process.

[0562] The sensitivity of the supervision methodology can be adjusted to control overlearning and threshold/probability cutoff of the machine learning model. Overlearning is the adjustability or the ratio of algorithm flexibility to bias. A highly flexible algorithm will fit all variable data into a data set, whereas a highly biased algorithm will fix a very limited set of data points into a data set. Threshold and probability cutoff is the machine learning model's tolerance of false positives versus false negatives.

Unsupervised Learning

[0563] In various aspects, unsupervised (or “untrained”) learning techniques can be used in order to determine relationships of data pairs from the surgical hub that contain linkages or result from a complication or morbidity, rather than the treatment step directly. In other words, machine learning can be utilized to identify non-device causes and adapt surgical hub responses accordingly.

[0564] Unsupervised learning is the process of training a machine learning model on unlabeled data. With time-based or sequential order-based learned outcomes, a program can determine the most likely to succeed next steps based on a predefined number of initial responses. These responses could be issues encountered, patient pre-operative conditions, or encountered issues in the surgical procedure. A number of unsupervised learning techniques can be used, including, for example, clustering (e.g., hierarchical clustering and k-means clustering), which provides protection against potential threats that are outside “normal” learned behavior, and unsupervised simulation of one instance of the program attempting to beat another instance of the program. The unsupervised learning techniques can also include deep learning and neural networks, which are analysis techniques that analyze trends and connections without considering the meaning of the data. Artificial neural networks can be utilized for, e.g., image processing to improve the ability to identify underlying critical structures by identifying images with the areas previously called out on the image, while having no information of the parameters of the structures themselves.

[0565] These data points could be used to identify to the user that there is an underlying cause for future evaluation, or that there is the need for additional treatment reinforcement, not because the device is not capable of proofing in a predictive fashion, but because the tissue or patient requires special care.

[0566] The result could be the identification to the surgeon that an additional procedure step is warranted, and adjunct treatment considered, or an alternative more robust product is needed. For example, during a colorectal lower anterior resection procedure, an imaging system might detect the over-limitation of blood supply to the region after mobilization as a portion of the metadata generated from the procedure. Accordingly, a vessel regeneration product or drug might be

warranted due to the implications of not being able to maintain an adequate blood supply to the transected tissue.

[0567] As another example, during a colorectal lower anterior resection procedure, the co-morbidities or chemo treatments might indicate in the metadata generated from the procedure that the tissue is too easily friable. Accordingly, a pressure distributing absorbable adjunct stapling product might be warranted in order to stress reduce the anastomosis, a diversion might need to be created to prevent bowel contents from placing undo stress on the new anastomosis, and/or an oversewing or fibrin/thrombin sealant might need to be applied to insure no undo leaks occurs in the first days of healing. A surgical system **200** can be programmed to recommend the appropriate product, drug, or treatment to the physician after completion of the procedure.

[0568] Accordingly, instrument, device, or machine settings can be set according to relationships determined via unsupervised learning techniques to provide better outcomes. In at least one example, a surgical stapling instrument is updated to run slower or faster depending on tissue type, resistance to push the knife, and so on. In another example, energy devices (e.g., ultrasonic surgical instruments and electrosurgical instruments) are updated to change the applied current over the transection (i.e., slowly ramping up or down or taking other such actions). Further, in various examples, one or more of the modular device (e.g. **1a-1n**, **2a-2m**) of the present disclosure are programmed to automatically update their settings based on the correlations learned by a machine learning performed by the cloud-based system (e.g. **104**, **204**).

[0569] In some aspects, the surgeon could have the ability to override the learned settings. In one aspect, the learned settings could be the default settings. In one aspect, a modular device (e.g. **1a-1n**, **2a-2m**) could suggest the new settings and allow the surgeon to choose between accepting the new settings and keeping the previous defaults.

[0570] A surgical system (e.g. **100**, **200**) is described herein that is configured to optimize outcome(s) of surgical procedures through machine learning. Patterns of treatment may be recognized over the course of numerous procedures that, when implemented, lead to a successful and/or safe result for a specific patient. Such patterns of treatment can include, for example, the type(s) of surgical instrument(s) to use during the procedure, additional procedure(s) to be performed, and/or any concerns that require further monitoring. Data collected during each particular procedure, the treatment performed, and/or the outcome of the particular procedure can be stored in a data bank (e.g. remote server **113**) for future analysis. The collected data can be used to bolster and/or update existing treatment recommendations. Analyzing various health statistics of a patient, such as, for example a patient's medical record and/or current vital statistics, in light of the recognized patterns can further optimize the likelihood of success of the particular procedure.

[0571] As described above in greater detail, a surgical system (e.g. **100**, **200**) comprises an information hub, or a surgical hub (e.g. **106**, **206**). The surgical hub is configured to facilitate communication between any surgical instruments used in the surgical procedure, the data source, and/or the clinician. An information hub, such as, for example, the surgical hub described herein, can store the data collected during each particular procedure. In various instances, the surgical hub can locally store data specific to the particular procedure and/or the specific patient. The surgical hub can store additional data relevant to different procedures in an external server (e.g. remote server **113**).

[0572] In various examples, referring to FIG. **10**, a surgical hub **206** comprises a control circuit including a processor **244** configured to receive an input from a data source. The data source is configured to supply the surgical hub **206** with data relevant to a particular surgical procedure and/or a particular patient. For example, the data source can provide the surgical hub **206** with the patient's medical history and/or current vital statistics, such as, for example, blood pressure. The data source can also provide the surgical hub **206** with data collected from the surgical site by a surgical instrument and/or a surgical visualization system.

[0573] Example surgical instruments that are suitable for use with the surgical hub **206** are

described under the heading “Surgical Instrument Hardware” in U.S. Provisional Patent Application Ser. No. 62/611,341, entitled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety. Various components of the surgical visualization system, such as the visualization system **108** are described under the heading “Advanced Imaging Acquisition Module” in U.S. Provisional Patent Application Ser. No. 62/611,341, entitled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is hereby incorporated by reference in its entirety.

[0574] Upon receiving an input, or data, from the data source, the surgical hub **206** analyzes the received data against a stored set of data. In various examples, the stored data set is stored in a memory **249**. Such analysis is performed with the goal of optimizing an outcome of a surgical procedure. In various examples, the data storage and/or machine learning analysis can be performed locally at the surgical hub level. Additionally, or alternatively, the data storage and/or machine learning analysis can be performed at the cloud **204**, for example.

[0575] In various instances, the surgical hub **206** analyzes the received data against a stored set of data using one or more untrained machine learning techniques. Using an untrained, or unsupervised, machine learning technique can allow for the determination of relationships between data pairs and/or an identification of a complication resulting from the performed surgical procedure.

[0576] The surgical hub **206** identifies commonalities in the received data and the stored data and react based on the presence or absence of such commonalities in each analyzed piece of data. Primary techniques in untrained machine learning include, for example, clustering. The goal of clustering is to create groups of data points such that points in different clusters are dissimilar while points within a cluster are similar. In other words, clustering offers protection against potential threats that are outside “normal” learned behavior. In various instances, the control circuit is configured to utilize clustering in analyzing the received data against stored data. Clustering techniques include, for example, k-means clustering and hierarchical clustering. In k-means clustering, data points are clustered into k groups. A larger k creates smaller groups with more granularity, while a smaller k means larger groups and more granularity. Each group is defined by creating a centroid for each group, wherein the cluster captures the points closest to it and adds them to the cluster. In hierarchical clustering, similar data points are grouped into clusters. In order to decide which clusters should be combined, or where a cluster should be split, a measure of dissimilarity between sets of data is required. This is achieved by use of an appropriate metric, or a measure of distance between pairs of data, and a linkage criterion which specifies the level of dissimilarity between data pairs.

[0577] Untrained machine learning techniques can develop recommendations based off of the creation of artificial neural networks. For example, a surgical imaging system, such as surgical visualization system **108**, can collect data representative of structures within the surgical site. After receiving the collected structural data, the surgical hub **206** analyzes the collected data against a stored set of structural data. Techniques in untrained machine learning analyze trends and connections without considering the meaning of the data. Such analysis allows for the processor **244** to recommend identities of underlying critical structures in a surgical site by utilizing images with previously-identified areas while having no information of the parameters of the critical structures themselves.

[0578] Based on the untrained machine learning analysis, the surgical hub **206** is configured to recommend an action. The recommended action(s) can be directed toward the clinician in the form of a prompt. The analyzed data can be used to identify to a clinician that an additional procedural step is warranted, an adjunct treatment should be considered, and/or an alternative treatment is needed. The alternative treatment can comprise using a different surgical instrument and/or implementing a more robust operating program on the surgical instrument.

[0579] For example, a clinician may be performing a colorectal lower anterior resection procedure

on a patient. During the surgical procedure, a surgical visualization system, such as surgical visualization system **108**, may detect an over-limitation of blood supply to the region after mobilization. Once the surgical imaging system, or an alternative data source, communicates the detected over-limitation of blood supply to the surgical hub **206**, the processor **244** analyzes the detected over-limitation of blood supply against previously collected data. Using an untrained machine learning technique, the processor **244** assesses commonalities between the detected over-limitation of blood supply and previously collected data. Based on the assessed commonalities, the processor **244** is configured to develop a recommendation to optimize an outcome of the surgical procedure. A possible recommendation can be, for example, use a vessel regeneration product and/or drug due to the implications of not being able to maintain an adequate blood supply to the transected tissue.

[0580] In another example, a clinician may be performing the same colorectal lower anterior resection procedure as described above. Prior to and/or during the procedure, the patient's medical records may comprise information indicating that the patient has comorbidity, or a simultaneous presence of another chronic disease or condition, and/or has previously undergone chemotherapy. Once a data source communicates the data comprised in the patient's medical records to the surgical hub **206**, the processor **244** analyzes the current medical records against previously collected data.

[0581] Using an untrained machine learning technique, the processor **244** assesses commonalities between the current patient's medical records and previously collected data. Based on the assessed commonalities, the processor **244** develops a recommendation to optimize an outcome of the surgical procedure. Possible recommendations can include, for example: (1) Use a pressure-distributing absorbable adjunct stapling product in order to stress reduce the anastomosis; (2) Create a diversion to prevent bowel content from placing undo stress on the new anastomosis; and/or (3) Apply an over-sewing or fibrin/thrombin sealant to prevent leaks from occurring within the first few days of healing.

[0582] Using an untrained, or unsupervised, machine learning technique can allow for the determination of relationships between data pairs and/or the identification of a complication resulting from the performed surgical procedure. The processor **244** identifies commonalities in the received data and the stored data and reacts based on the presence or absence of such commonalities in each analyzed piece of data. Based on this analysis, the processor **244** recommends an action. The recommended action(s) can be directed toward a surgical instrument in the form of an operating program and/or parameter. In other words the recommended action(s) can be in the form of modified machine and/or device (e.g. modular devices **1a-1n**, **2a-2m**) settings to provide improved outcomes. For example, various surgical instruments can receive the following alterations and/or updates to the instrument's operating program: (1) surgical staplers can run slower or faster depending on factors such as tissue type, resistance felt by the knife, etc.; and (2) surgical energy devices can change the current flowing over the transection by slowly ramping up or down, etc. In various instances, the surgical hub **206** is configured to automatically update a surgical instrument's settings based on the unsupervised machine learning analysis.

[0583] While a surgical instrument's operating program can be automatically updated and/or altered with the recommendations of the surgical hub **206**, the clinician can also have the ability to override the implemented recommendations. In various instances, the surgical hub **206** can prompt the clinician with the recommended alterations to the surgical instrument's operating program. The clinician is then able to accept the recommended alterations or maintain previous operating parameters. In other instances, the recommended alterations can automatically take effect, and the clinician is able to switch back to the previous operating parameters if the clinician desires.

[0584] As described above, upon receiving an input, or data, from the data source, the processor **244** of the surgical hub **206** analyzes the received data against a stored set of data. Such analysis is performed with the goal of optimizing an outcome of a surgical procedure. In various instances, the

processor **244** analyzes the received data against a stored set of data using one or more trained machine learning techniques. In supervised learning, the processor **244** predicts a parameter as accurately as possible when given new examples where the inputs and outputs are unknown. [0585] Primary techniques in trained machine learning include, for example, parametric learning and non-parametric learning. Regression, classification, and vector clustering are the two tasks involved in parametric learning. Regression predicts a continuous target variable, thereby allowing the processor **244** to estimate a value based on received input data. Continuous variables, such as, for example, a patient's height, weight, emphysema air leak rate, etc., means there are not discontinuities in the value that the predicted parameter can have. Discrete variables, on the other hand, can only take on a finite number of values—for example, the color of a staple cartridge within an end effector.

[0586] For example, a clinician may wish to predict the emphysema air leak rate based on factors such as, the stage of emphysema present in the patient, the firing speed of a surgical stapler, the force required to close the jaws of an end effector, the type and/or color of the staple cartridge within the end effector, etc. A surgical stapler is used to staple lung tissue in an emphysema patient. A suitable staple cartridge is selected based on the condition of the patient and/or the condition of the lung tissue to be stapled by the surgical stapler. Staple cartridge color may reflect the size of the surgical stapler, for example.

[0587] In various examples, the surgical hub **206** builds a model that approximates the relationship *f* between the above factors and corresponding emphysema air leak rate. As shown in FIG. **41**, in supervised learning, the processor **244** attempts to learn the relationship between, for example, emphysema air leak rate and the previously-identified factors by running labeled training data **250810** through a learning algorithm. The learned function can be used by the processor **244** to estimate the emphysema air leak rate of a patient whose emphysema air leak rate is unknown. The estimation is based on various inputs including the stage of emphysema, the firing speed of the surgical stapler, the force required to close the jaws of the end effector, and/or the type and/or color of the staple cartridge within the end effector.

[0588] FIG. **34** represents exemplary data collected during previous surgical procedures. In FIG. **34**, the patients were diagnosed with a particular stage of emphysema. The calculated threshold **250700** represents the point at which the outcome(s) of the surgical procedure are acceptable. The positive (+) side represents a good, or beneficial, outcome of the surgical procedure, while the negative (−) side represents a bad, or detrimental, outcome of the surgical procedure. It has been observed that radiation stiffens lung tissue, while emphysema softens lung tissue. Such an observation has an effect on the operating program of a surgical instrument, such as a surgical stapler. For example, advancing a firing member too quickly through a diseased tissue may have a detrimental impact on the outcome of the surgical procedure.

[0589] A first patient with stage **2** emphysema is represented by line **250710** and a second patient with stage **2** emphysema is represented by line **250715**. A firing member of a surgical stapler was advanced at a slow speed during the surgical procedure of the first patient **250710**, and a firing member was advanced at a fast speed during the surgical procedure of the second patient **250715**. As represented by the respective dashed lines **250710**, **250715**, the speed of the firing member appears to be irrelevant when the patient has stage **2** emphysema, as both patients had a good, or beneficial, outcome. Notably, the fast speed of the firing member brought the second patient **250715** closer to the threshold of acceptable outcomes of the surgical procedure.

[0590] A first patient with stage **3** emphysema is represented by line **250720** and a second patient with stage **3** emphysema is represented by line **250725**. A firing member of a surgical stapler was advanced at a slow speed during the surgical procedure of the first patient **250720**, and a firing member was advanced at a fast speed during the surgical procedure of the second patient **250725**. As represented by dotted line **250725**, the fast advancement of the firing member in a patient with stage **3** emphysema produces unacceptable and/or detrimental outcomes to the surgical procedure,

while the dotted line **250720** shows that the slow advancement of the firing member in a patient with stage **3** emphysema results in an acceptable and/or beneficial outcome to the same surgical procedure. A first patient with stage **4** emphysema is represented by line **250730** and a second patient with stage **4** emphysema is represented by line **250735**. A firing member of a surgical stapler was advanced at a slow speed during the surgical procedure of the first patient **250730**, and a firing member was advanced at a fast speed during the surgical procedure of the second patient **250735**. As represented by the respective dashed lines **250730**, **250735**, the speed of the firing member appears to be irrelevant when the patient has stage **4** emphysema, as both patients had a bad, or detrimental, outcome.

[0591] As shown in FIG. **35**, data similar to that collected in FIG. **34**, is collected with respect to numerous parameters, including, for example: (1) the stage of emphysema **250750**; (2) if the patient has received radiation **250752**; (3) the force required for a surgical stapler to advance its firing member **250754**; (4) the patient's blood pressure **250756**; (5) advancing the firing member at a low speed **250758**; and/or (6) advancing the firing member at a high speed **250759**. As represented by the position of the measured parameters along the threshold line **250700**, each parameter has either a positive or negative effect on an outcome of a surgical procedure. As represented by the dashed line **250755**, the patient having received radiation **250752** and the force required for a surgical stapler to advance its firing member **250754** being high both have an inter-related cause and effect on the outcome of the surgical procedure. High blood pressure **250756** and a low speed of advancement of a firing member **250758** have positive effects on the outcome of the surgical procedure, while a high speed of advancement of a firing member **250759** and a greater stage of emphysema **250750** have negative effects on the outcome of the surgical procedure. Dashed line **250757** represents a surgical instrument control program change based on unlearned grouping.

[0592] Boxes **250760** represent measured parameters detected from individual patients whose combination resulted in a bad, or detrimental, outcome to the surgical procedure. Boxes **250770** represent measured parameters detected from individual patients whose combination resulted in a good, or beneficial, outcome to the surgical procedure. As described in greater detail with respect to FIG. **37**, boxes **250765** represent measured parameters detected from individual patients whose combination resulted in unexpected results that are not representative of other patients. In various instances, extenuating circumstances affected the result of the surgical procedure for the patients represented by **250765**.

[0593] The previously collected data, as shown in FIG. **34**, is split into a training data set **250810** and test data set(s) **250820**, **250830**. The training data set **250810** has labels, so the surgical hub **206** can learn from these labeled examples. Such labels can include, for example, various determined emphysema air leak rates and their corresponding stage of emphysema, the firing speed of the surgical stapler, the force required to close the jaws of an end effector, and the type and/or color of the cartridge within the end effector. The test set(s) **250820**, **250830** comprise data that does not have labels, i.e. the processor **244** does not yet know the value it is trying to predict. In other words, the processor **244** has to generalize to situations it hasn't encountered before so that it can provide the most accurate prediction for the test data **250820**, **250830**.

[0594] The surgical hub **206** to create a mathematical model **250840** based off of the previously collected data marked as the training data set **250810**. The training data set is comprised of a large sample size, i.e. $n=1000$, to define a function. A test set **250820**, **250830**, or a validation set, allows the surgical hub **206** to check if the test sets **250820**, **250830** fit the determined mathematical model. The test sets **250820**, **250830** comprise a smaller sample size, i.e. $n=50$. Analysis **250835** of the test sets **250820**, **250830** results in a validation model **250850** for comparison to the generated mathematical model **250840**. In various instances, this gives the surgical hub **206** the opportunity to keep anticipated data and remove any unanticipated data. The ability to adjust the mathematical model **250840** allows for the fine tuning of the grouping process to populate **250855** a final

mathematical model for use in data analysis. The developed model can be applied in analysis of collected data to estimate, for example, emphysema air leak rate.

[0595] As discussed above, some data points are outliers and must be removed from the data pool in order to fine tune the mathematical model. As shown in FIG. 37, and as discussed with respect to FIG. 35, boxes **250760** represent measured parameters detected from individual patients whose combination resulted in a bad, or detrimental, outcome to the surgical procedure. Boxes **250770** represent measured parameters detected from individual patients whose combination resulted in a good, or beneficial, outcome to the surgical procedure. Boxes **250765** represent measured parameters detected from individual patients whose combination resulted in unexpected results that are not representative of other patients. In various instances, extenuating circumstances affected the result of the surgical procedure for the patients represented by **250765**. The presence of the outlier boxes **250765** alters the median threshold as the mathematical model attempts to explain their results. The outlier data **250765** is excluded from the mathematical model, adjusting the actual threshold **250700** to its rightful position. The outlier boxers **250765** would have originally been grouped **250900** by the algorithm with the other detrimental outcome boxes **250760** as none of them produced an acceptable surgical procedure result. This would have resulted in inconsistent and inaccurate predictions of future outcomes. The actual grouping of parameters contributing to detrimental outcomes is represented by **250910**, and the actual grouping of parameters contributing to beneficial outcomes is represented by **250920**. A common problem in machine learning is overfitting. Overfitting is learning a function that perfectly explains the training data that the model learned from, but does not generalize well to unseen test data. Overfitting happens when a model overlearns from the training data to the point that it starts picking up characteristics that are not representative of patterns in collected data. Underfitting is a related issue where the generated model is not complex enough to capture an underlying trend in the data.

[0596] Referring now to FIG. 38, an algorithm **250950** for a surgical system **200** is depicted. The algorithm **250950** is intended to allow for adjustment of control programs of surgical instruments and/or other components of the surgical system **200** based on a remote analysis involving supervised and/or unsupervised learning techniques. Such techniques, as described in greater detail above, include repeated multivariate analysis of different variable combinations in search of correlations and an assessment of the predictive ability of outcome of the analysis against a separate group of data not used in the original analysis.

[0597] Various surgical systems disclosed herein such as, for example, the surgical systems **100**, **200** can utilize the algorithm **250950** of FIG. 38. Moreover, the reader will readily appreciate that the algorithm **250950** of FIG. 38 can be combined with one or more additional algorithms described herein in certain instances. The decisions made by the surgical system **200** can be implemented, for example, by a control circuit that includes the processor **244**. The control circuit is configured to receive data **250952** from a data source. The received data can include, for example, data pertaining to the vital statistics of the patient, the medical history of the patient, the type of surgical procedure being performed, the type of surgical instrument being used, any data detected by a surgical instrument and/or surgical visualization system, etc.

[0598] The control circuit of the surgical system **200** is configured to analyze the received data **250954** against a stored set of data. Such analysis **250954** can be based on supervised or unsupervised learning. Such analysis **250954** can be performed within the local network of facility linked devices or could be exported to a remote location for compilation and returned to the network.

[0599] Following analysis **250954**, the control circuit is configured to communicate a recommendation **250956** based on the analysis. The recommendation can be based on the identification of device interrelated impacts, patient specific impacts, and/or interaction of used devices. The recommendation can be communicated to a clinician in the form of a prompt and/or can be communicated to a particular surgical instrument in the form of updated operating

parameters. In one example, the recommendations are communicated by a cloud **204** to a modular device (e.g. **1a-1n**, **2a-2m**) either directly, or through the surgical hub **206**. In one example, the recommendations are communicated to a surgical hub **206**. In any event, the recommendation can include, for example, a prompt to update a control program of the modular device or adjusting the modular device's operating parameters. In various instances, the recommendations can comprise suggestions or other procedural modifications to be indicated to the clinician. In certain instances, the recommendations include, for example, adaptation of an actuation speed, wait time, and/or other operating parameters of a modular device.

[0600] FIG. **39** is a logic flow diagram of a process **206500** depicting a control program or a logic configuration for adjusting surgical hub responses. The process **206500** can be performed by any suitable control circuit such as, for example, a control circuit of a surgical hub (e.g. surgical hub **106**, **206**). The process **206500** includes receiving **206502** data from at least one data source communicably coupled to the surgical hub **5104**. The at least one data source can be, for example, a patient monitoring device, a surgical staff detection device, a module device detection device and/or hospital database is processed by the surgical hub **106**, **206** to determine a progress status of a surgical procedure. The received data can be determinative of a situational parameter of a surgical procedure that is being performed by the surgical hub. In various examples, the situational parameter represents a progress status of the surgical procedure, as illustrated in FIG. **40**.

[0601] FIG. **40** is a timeline **206501** of an illustrative surgical procedure and the corresponding information inferred by a surgical hub **5104** during the procedure, in accordance with at least one aspect of the present disclosure. The timeline **206501** is similar in many respects to the timeline **5200** of FIG. **15**. In addition, the timeline **206500** further depicts a progress status of the surgical procedure. The progress status may comprise a preoperative status **206503** that reflects that the surgical procedure is underway. In the preoperative status, various preoperative steps are performed to prepare the operating room for surgery. The progress status may also comprise an intraoperative status **206505** that reflects that the surgery has begun.

[0602] In various examples, activation of a surgical hub in an operating room signifies that a surgical procedure has started and, is underway, which causes the surgical hub to detect a preoperative status **206503**. Additional data sources could also be relied upon, alone or in combination, in detecting the preoperative status **206503**. For example, receiving **5206** a scan of a patient band via a scanner that is communicably connected to the surgical hub **5104** can indicate a preoperative status **206503**. Additionally, or alternatively, receiving **5210** data from a patient monitoring device such as, for example, an EKG can indicate a preoperative status **206503**.

[0603] As described above, the surgical hub **5104** can infer that the patient is under anesthesia based on data from the modular devices and/or patient monitoring devices, including EKG data, blood pressure data, ventilator data, or combinations thereof, for example. Upon completion of the sixth step **5212**, the pre-operative portion of the lung segmentectomy procedure is completed. Accordingly, receiving such data can indicate a transition from the preoperative status **206503** to the intraoperative status **206505**.

[0604] In one aspect, the surgical hub controls can be based on the awareness of whether or not a procedure is in process (i.e., whether a surgical procedure is currently being performed in connection with the particular surgical hub). Accordingly, the surgical hub controls can be based on the situation in which it senses itself.

[0605] The process **206500** further includes adjusting **206504** a response to a sensed parameter based on the determined situational parameter or progress status of the surgical procedure. In at least one example, as illustrated in FIG. **44**, the sensed parameter can be detecting security threat. In other examples, the sensed parameter can be detecting a surgeon. In other examples, the sensed parameter can be detecting an instrument fault such as, for example, a modular device.

[0606] FIG. **41** illustrates is a logic flow diagram of a process **206520** depicting a control program or a logic configuration for selecting operational modes of a surgical hub **5104**, in a surgical

procedure, depending on a determined progress status of the surgical procedure. The process **2065520** can be performed by any suitable control circuit such as, for example, a control circuit of a surgical hub **5104**. Data can be received **206522** from at least one data source, and may include patient data **206532** from a patient monitoring device, surgical staff data **206534** from a surgical staff detection device, modular device data **206536** from one or more modular devices and/or hospital data **206538** from a hospital database, as illustrated in FIG. **42**. The received **206522** data is processed by the surgical hub **5104** to determine a progress status of the surgical procedure. [0607] As illustrated in FIG. **41**, the received **206522** data can be utilized by the surgical hub **5104** to determine **206523** whether the surgical procedure is underway. If not, the surgical hub **5104** activates or selects a previous procedure/network interaction mode **206524**. If, however, the surgical hub **5104** determines **206523** that the surgical procedure is underway, it further determines **206525** whether surgery is in progress. If not, the surgical hub **5104** activates or selects an interactive/configurable control mode **206526**. If, however, the surgical hub **5104** determines **206525** that the surgery is in progress, the surgical hub **5104** activates or selects an instrument display control and procedural display mode **206528**.

[0608] The mode **206524** is more restrictive than the mode **206526**, and the mode **206526** is more restrictive than the mode **206528**. This arrangement is designed to take into consideration a user error in the form of inadvertent commands, for example. Before the surgical procedure starts, the mode **206524** only permits access to previous procedure data, and a limited interaction with a cloud-based system **104**, **204**, for example. During the preoperative steps, but before surgery is begun, the mode **206526** provides a less restrictive interface that permits a user to access and/or configure various parameters and/or controls without being able to use or activate such controls. In the least restrictive mode **206528**, which is only available during surgery, the user is allowed to use or activate controls of certain modular devices depending on the surgical step being performed, as illustrated in FIG. **46**.

[0609] A surgical hub can make inferences about events that are occurring during the course of a surgical procedure and then respond accordingly. FIG. **42** addresses determining **206523** whether a surgical procedure is underway, in accordance with the process **206520**, for example. A surgical hub can sense (or determine or infer) whether or not it is in a procedure based on attached devices and data feeds to, e.g., selectively show real-time data or the previous surgery's data.

[0610] As described above, data from various data sources can be analyzed, for example by a surgical hub **5104**, to determine **206523** whether a surgical procedure is underway. Patient data **206532** from one or more patient monitoring devices can be used to determine **206531** whether a patient is present in the operating room. Additionally, or alternatively, surgical staff data **206534** from a surgical staff detection device can be used to determine **206533** whether the necessary supporting staff for the surgical procedure is present in the operating room. Additionally, or alternatively, modular device data **206536** from one or more modular devices can be used to determine **206535** whether the equipment necessary for performing the surgical procedure is present in the operating room. Additionally, or alternatively, hospital data **206538** from one or more hospital databases can be used to determine **206537** the type of procedure being performed, for example.

[0611] The determinations at **206531**, **206533**, **206535**, **206537** can be used separately, or in any suitable combination, to determine **206523** whether a surgical procedure is underway. In various examples, each of the determinations at **206531**, **206533**, **206535**, **206537** can be assigned a predetermined value when achieved. The summation of all the values can then be compared to a predetermined threshold to determine **206523** whether the surgical procedure is underway. In certain examples, a surgical hub **5104** must achieve each of the determinations at **206531**, **206533**, **206535**, **206537** before determining **206523** that a surgical procedure is underway.

[0612] Referring still to FIG. **42**, the patient data **206532** may include patient blood pressure data from a blood pressure monitoring device, heart rate data from a heart rate monitoring device,

anesthesia data, and/or ventilator data from a ventilator and/or data from a patient identification tag. Other data sources from other devices that interact with the patient within the operating room are also contemplated by the present disclosure. For example, as described above in greater detail, a surgical hub **5104** can receive a unique identifier from, for example, a scanner for scanning the patient's wristband encoding the unique identifier associated with the patient when the patient enters the operating theater.

[0613] Furthermore, surgical staff data **206534** from a surgical staff detection device can be analyzed to determine the identity of the individuals in the operating room and/or where they are located with respect to the patient. For example, surgical staff data **206534** can be analyzed to assess whether a surgeon is standing in close proximity to the patient. For example, a surgical hub can receive a unique identifier from, for example, a scanner for scanning surgical staff wristbands encoding the unique identifier associated with each member of the surgical staff patient when they enters the operating theater. Other techniques for identifying a patient and/or a surgical staff are disclosed in U.S. Provisional Application No. 62/659,900, titled METHOD OF HUB COMMUNICATION, filed on Apr. 19, 2018, which is hereby incorporated by reference herein in its entirety.

[0614] Further to the above, the modular device data **206536** may include identification data for determining what devices are present in the operating room and/or what devices are active and/or paired with a surgical hub, for example. In one exemplification, a surgical hub can be configured to compare the list of items for a procedure (scanned by the scanner, for example) and/or a list of devices paired with the surgical hub to a recommended or anticipated manifest of items and/or devices for the given surgical procedure to determine **206535** whether the necessary equipment for the surgical are present in the operating room.

[0615] Further to the above, the hospital data **206538** may include EMR data that can be pulled from an EMR database containing patient records. Based on select patient data in the EMR, a surgical hub can determine the type of procedure to be performed, for example, which is helpful in assessing whether the surgical procedure is underway. In various examples, it is contemplated that other **206539** data can be received and analyzed by a surgical hub to determine **206523** whether a surgical procedure is underway, in accordance with the process **206520**, for example.

[0616] FIG. **43** addresses determining **206525** whether surgery is in progress, in accordance with the process **206520**, for example. Additional patient data **206542** such as, for example, active blood pressure data and/or sedation data can be analyzed by a surgical hub **5104** to determine **206525** whether surgery is in progress and/or assess the current surgical step being performed, as discussed below in connection with FIG. **46** Additionally, or alternatively, various device data **206544** such as, for example, generator data, device activation data, and/or live imaging data (with instrument identification and/or active moving images) can be analyzed by a surgical hub **5104** to determine **206525** whether surgery is in progress. In various examples, it is contemplated that other **206546** data can be received and analyzed by a surgical hub to determine **206525** whether surgery is in progress, in accordance with the process **206520**, for example.

[0617] As described above, the process **206500** includes adjusting **206504** a response to a sensed parameter based on a determined situational parameter or progress status of a surgical procedure. In at least one example, as illustrated in FIG. **44**, the sensed parameter can be detecting **206552** a security threat. In other examples, the sensed parameter can be detecting **206554** a surgeon. In other examples, the sensed parameter can be detecting **20559** an instrument fault such as, for example, a modular device.

[0618] Further to the above, responding to a detected **206552** security threat depends on whether surgery is progress, which can be determined **206525**, as described above in connection with FIG. **43**. If it is determined **206525** that surgery is in progress, an isolated operation mode **206553** can be activated. If surgery is not in progress, the current security level can escalated **206551** to a higher security level, and an appropriate reaction or response can be taken to address the detected **206552**

security threat.

[0619] In various examples, the isolated operation mode **206553** comprises interrupting communications with external systems such as, for example, the cloud-based system **104**, **204**. In certain examples, the communications interruption excludes local communications within an operating room such as, for example, instrument-to-instrument communications, instrument-to-surgical hub **106**, **206** communications, and/or remote controller-to-instrument communications.

[0620] Referring still to FIG. **44**, responding to a detected **206554** surgeon depends on whether the surgical procedure is underway, which can be determined **206523**, as described above in connection with FIG. **43**. If it is determined **206523** that a surgical procedure is underway, linked instruments can be set **206557** to pre-defined parameters based on previous use configurations for the detected **206554** surgeon, for example. If, however, a surgical procedure is not underway, previous captured data and/or previous surgeries data can be called up **206555**, for example.

[0621] Referring still to FIG. **44**, responding to a detected **206556** instrument fault depends on whether the surgical procedure is underway, and further depends on whether surgery is in progress which can be determined **206523**, **206525**, as described above in connection with FIG. **43**. An instrument can be, for example, a modular device. If it is determined **206523** that a surgical procedure is underway, and it is further determined **206525** that surgery is in progress, a limp mode can be activated **206565** for the instrument. If, however, a surgical procedure is not underway, a lockout of the surgical instrument can be engaged **206561** to prevent the surgical instrument from being used. Furthermore, if it is determined **206523** that a surgical procedure is underway, but surgery is not in progress, an alert or warning can be issued **206563** by the surgical hub **5104** to the surgical staff, for example, advising options.

[0622] Referring to FIG. **45**, an example is provided for adjusting operational parameters of a surgical stapler **206570** in the event of a detected **206572** security fault in accordance with one or more of the above-described processes. In the example of FIG. **45**, the stapler **206570** detected **206572** the security fault during communication with a surgical hub **5104** and/or with a cloud-based system **104**, **204**, for example. In response, the stapler **206570** severs **206573** communications with the surgical hub **5104** and/or with the cloud-based system **104**, **204**, and continues to allow autonomous operation unless a hack is detected.

[0623] FIG. **46** is a diagram depicting how a surgical hub **5104** can determine which step of a surgery is being performed according to various data feeds, in accordance with at least one aspect of the present disclosure. FIG. **46** also depicts various examples of responding to sensed parameters based on a determined situational parameter, in accordance with the process **206500** (FIG. **39**). In the examples of FIG. **46**, the determined situational parameter is the present step of an ongoing surgery.

[0624] In various instances, a surgical hub **5104** determines the present step of an ongoing surgery based on data received **206502** (FIG. **34**) from various data sources described in connection with FIG. **34**. In the example of FIG. **46**, the surgical hub **5104** identifies the surgical steps of the surgery as Access **206580**, Dissection **206582**, Transection **206584**, Anastomosis **206586**, and Closing **206588**, which is based on the received **206502** data. Furthermore, the surgical hub **5104** identifies the modular devices to be used in the surgery as a Trocar **206581**, a modular energy device **206583**, a linear surgical stapler **206585**, and a circular surgical stapler **206585**, also based on the received **206502** data.

[0625] In various examples, the surgical hub **5104** may store a database of various surgeries, the identity and order of the surgical steps pertaining to each of the surgeries, and/or the identity and/or usage or activation frequency of the modular surgical devices to be used in each of the surgical steps. In such examples, a user input selecting or identifying the surgery to be performed may be all that is needed for the surgical hub **5104** to identify the surgical steps and modular surgical devices associated with the surgery to be performed.

[0626] In other examples, the surgical hub **5104** deduces the type of surgery to be performed,

and/or its corresponding surgical steps, by detecting modular surgical devices that are in close proximity to the patient and/or are within the operating room. Additionally, or alternatively, the surgical hub **5104** may determine the type of surgery to be performed, and/or its corresponding surgical steps, from a received **206502** patient EMR, for example.

[0627] FIG. **46** depicts example information that is made available to and/or is deduced by the surgical hub **5104** in connection with an example surgery **206591** that is being coordinated by the surgical hub **5104**, which is based on the received **206502** data. In access **206580**, the trocar **206581** and the modular energy device **206583** are used. Usage of the modular energy device **206583** is depicted in activation instances **206590**, which can be repeated with or without interruptions, as depicted in FIG. **46**. Each activation instance **206590** may comprise a predefined time period of energy application by the modular energy device **206583**, for example. Similar activation instances **206592**, **206594**, **206596** are depicted for the trocar **206581**, the linear surgical stapler **206585**, and the circular surgical stapler **206585**, respectively.

[0628] FIG. **46** also depicts erroneous, inadvertent, or unexpected activation instances **206590'**, **206596'**, which constitute examples of sensed parameters, in accordance with the process **206500** (FIG. **39**). The example activation instances **206590'**, **206596'** are activation instances of the modular energy device **206583** and the circular surgical stapler **206585**, respectively, which are outside the expected, or normal, sequence of the surgery **206591** and, as such, are ignored. Alternatively, or additionally, an activation instance can be ignored if it is determined that the activated modular device is too far away from the surgical site. For example, an activation instance can be ignored if it is determined that the activated modular device is located on a sterile table, for example. Any suitable proximity sensors can be employed to determine the position of the activated modular device. Alternatively, or additionally, an activation instance can be ignored if it is determined that the activated modular device is not in contact with tissue. Any suitable continuity sensors can be employed to determine whether the activated modular device is in contact with tissue.

[0629] In various examples, the surgical hub **5104** could ignore hand piece activation based on depression of the buttons on the handle of a modular energy device if the surgical hub **5104** is aware the generator, surgical hub **5104**, and/or the modular energy device are not in an active surgery. This can reduce inadvertent activation of devices. Furthermore, this could operate on a finer control level as well: If the surgical hub **5104** determines that the modular energy device is not inside the patient or in contact with a patient's tissue by sensing continuity or the linking of the return path of the return pad, the activation of the modular energy device could be ignored. This could even be used relative to patient proximity in aspects where the system is capable of instrument tracking.

[0630] In one aspect, the surgical hub controls can be programmed such that fault detection during a surgery triggers a different response than fault detection does before or after a surgery. Accordingly, the severity of the surgical hub's response to faults can be based on its awareness of its use status.

[0631] FIG. **47** is a logic flow diagram of a process **206600** depicting a control program or a logic configuration for determining **206602** whether a modular device is in optimal condition and/or performing **206604** a severity assessment in the event it is determined that the modular device is in a suboptimal condition or is not functioning properly. A surgical hub **5104** may receive **206601** data from various sources indicative of device usage history, device status, recent performance, device authentication/identification data, device compatibility, and/or other data. The surgical hub **5104** may analyze such data to determine **206602** whether a modular device is in optimal condition. If so, the surgical hub **5104** permits the modular device to operate in a fully enabled device mode **206603**. If not, the surgical hub **5104** performs **206604** a severity assessment. Based on the severity assessment, the surgical hub **5104** may take steps to provide a warning **206605**, cause a reduction in available device functionality **206606**, force a device change **206607**, and/or take any other

suitable action **206608**.

[0632] In one aspect, a detected fault severity response can be based on whether the surgical hub **5104** believes the device is unsafe or could be put into a limp mode because it is currently in-use. For example, if a counterfeit cartridge reload is used in a procedure, the user is warned **206605**, but the product is allowed to be used with surgeon override if the modular device is already in-use. As opposed to a reload being installed at the beginning of a procedure, for example in the pre-operative stage, at which point the severity it senses might be a higher level and it may lock-out the use of that combination of products. As another example, the number of uses flag can be treated in one manner before and after the procedure, as opposed to it being triggered during a procedure. The surgical hub's inclination to lockout a device that has been reused too many times before a procedure can be overridden to allow the device to continue if the trigger is activated while in a procedure. The inclination to lockout the device can then be restored once the procedure is complete.

[0633] In one aspect, security responses in certain situations could be substantially more restrictive based on the situational awareness of the surgical hub **5104**. Accordingly, surgical hub's security response can be based on its perceived need for secure use. For example, the protective reaction to a system attack might be elevated if the system is aware that it is in-use. As another example, the protective response to an attack might be escalated if surgical hubs are already aware they are under attack in other locations. As yet another example, remote access for controlling aspects of the modules within the surgical hub **5104** could be limited when the surgical hub is in use in a procedure. In one implementation, the external control aspects of the modular devices could be disabled when local control is established. In another implementation, when remote access is requested and a procedure is in process or there is a sensed use by a local user, the surgical hub might request confirmation locally for permission before granting permission for remote control of the system or function.

[0634] In one aspect, the configuration and responses of the surgical hub **5104** could be altered based on the user the surgical hub senses is in-use. Accordingly, the surgical hubs can be programmed to automatically re-configure based on sensed users. For example, the default configuration of an attached device and the controls of the device could be adjusted based on the user the surgical hub senses is using the device. If a specific surgeon doing a specific procedure always tends to use a device or its control in a repeatable manner, the surgical hub could automatically configure the device for the user as it learns his or her behavior. As another example, the surgical hub **5104** can learn the behaviors of the users it works with. This could even be with a network of hubs, which could each communicate preferences, setups, and alterations in device setup for specific users. Accordingly, when a specific user is sensed, either by login or another technique, the surgical hub **5104** could then start configuring the systems based on previous uses by the user in question.

[0635] In various aspects, a modular device **5102** such as, for example, a surgical instrument may interact with other modular devices **5102** and/or surgical hubs **5104**. The interaction may occur before, during, and/or after a surgical procedure commences. For example, the modular device **5102** may receive a firmware update from a surgical hub **5104** before the surgical procedure. In another example, the modular device **5102** may receive commands from a remote controller during a surgical procedure. In yet another example, the modular device **5102** may transmit usage data to a surgical hub **5104** regarding a surgical procedure after completion of the surgical procedure. Unauthorized interactions between a modular device **5102** and other modular devices **5102** and/or surgical hubs **5104** can interfere with the proper operation of such devices and systems.

[0636] Ensuring a secure interaction between a modular device **5102** and other modular devices **5102** and/or surgical hubs **5104** can be achieved by generating an appropriate response to an unauthorized interaction. In various aspects, the response of a modular device **5102** to a potential security violation or unauthorized interaction can be adjusted based on situational awareness. In

some situations, the response of a modular device **5102** to a potential security violation or unauthorized interaction can be based on situational awareness that the modular device **5102** itself is attacked, instead of a surgical hub **5104**, for example.

[0637] In one aspect, a wirelessly pairable modular device **5102** can be programmed for detection and escalation of security responses in response to numerous or increasing severity threats. For example, the first response to a first violation results in a minor reaction and a second response to a second serial violation results in an escalated response. In another aspect, the escalated response could be termination of communication and/or autonomous usage only of the instrument.

[0638] In one aspect, a surgical instrument can be programmed to implement an escalation protocol to react according to the number or invasiveness of a security violation. Accordingly, a surgical instrument can be programmed to escalate security response in response to increasing threats. For example, a wireless device which pairs to another device or surgical hub and senses a first unauthorized or unauthentic interaction causes the device to execute a minor response (e.g., warn the user or raise the threat warning). When the device senses multiple additional issues or the severity of the additional issue is higher, the device's second response can be escalated much greater than the first response (e.g., end communication and only operate autonomously or only accept fully authenticated and encrypted requests). As another example, a device can execute a security response when communication interaction appears probing after already flagging an unauthentic handshake.

[0639] FIG. **48** is a logic flow diagram of a process **206610** depicting a control program or a logic configuration for generating suitable responses to unauthorized interactions with a modular device **5102**, in accordance with at least one aspect of the present disclosure. The process **206610** includes detecting **206612** a first security violation or unauthorized interaction, causing **206614** the modular device **5102** to generate a first response to the first security violation or unauthorized interaction, storing **206616** a record of the first security violation or unauthorized interaction; detecting **206618** a second security violation or unauthorized interaction, and causing **206620** the modular device **5102** to generate a second response to the second security violation or unauthorized interaction, wherein the second response is escalated from the first response.

[0640] Referring to FIG. **49**, in various examples, a modular device **5102** comprises a communication module **206622**, a control circuit **206624**, and a user interface **206626**. The control circuit **206624** is coupled to the communication module **206622** and the user interface **206626**, and is configured to execute the process of **206610**.

[0641] In various examples, any suitable wireless communication can be employed by the communication module **206622** including, for example, Bluetooth wireless technology standard for exchanging data over short distances (using short-wavelength UHF radio waves in the ISM band from 2.4 to 2.485 GHz) from fixed and mobile devices and building personal area networks (PANs). The communication module **206622** may employ anyone of a number of wireless or wired communication standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long-term evolution (LTE), and Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, and Ethernet derivatives thereof, as well as any other wireless and wired protocols that are designated as 3G, 4G, 5G, and beyond. In at least one example, the communication module **206622** may employ an Air Titan-Bluetooth.

[0642] In at least one example, a pairing attempt with the communication module **205522** by an unauthorized modular device or surgical hub is detected **206612**. In response, the control circuit **206624** may cause **206614** the user interface **206624** to issue a warning or an alert, which can a visual and/or an audible alert. Furthermore, the modular device **5102** may store **206616** a record of the first pairing attempt in a memory unit of the control circuit **206624**, for example.

[0643] If, however, a second pairing attempt by an unauthorized modular device or surgical hub is detected **206618**, the control circuit **206624** may cause **206620** a second response, escalated from

the first response, to be generated. For example, the control circuit **206624** may cause the communication module **206622** to terminate all external communications and/or may cause an autonomous operation mode to be activated.

[0644] In at least one example, the modular device **5102** is a surgical stapler, and the first security violation or unauthorized interaction involves loading a spent, or previously used, staple cartridge onto the surgical stapler. Detecting **206612** the spent staple cartridge can be achieved by one or more sensors and/or by interrogating a chip of the staple cartridge to assess whether its identification number is associated with a new or unspent staple cartridge. Once a first spent staple cartridge is detected **206612**, the control circuit **206624** causes **206614** the user interface **206626** to issue to issue a warning or an alert, which can a visual and/or an audible alert. The alert may include instructions to remove the spent staple cartridge.

[0645] Further to the above, the modular device **5102** may store **206616** a record of the spent staple cartridge in a memory unit of the control circuit **206624**, for example. If, however, the control circuit **206624** detects **206618** that the same spent staple cartridge, or another spent staple cartridge, has been loaded onto the surgical stapler, a more escalated response can be generated by the control circuit **206624**. For example, the control circuit **206624** may cause the surgical stapler to enter a permanent lockout, preventing the surgical stapler from further usage. The control circuit **206624** may also report the incident to a surgical hub, for example. Other suitable escalated responses are contemplated by the present disclosure.

[0646] FIG. **50** is a logic flow diagram of a process **206630** depicting a control program or a logic configuration for generating suitable responses to unauthorized interactions with a modular device **5102**, in accordance with at least one aspect of the present disclosure. A first unauthorized activation of the modular device **5102** is detected **206632** by the control circuit **206624**, for example. In at least one example, an activation of the modular device **5102** while the device is on a surgical tray, or while the device is separated from the patient beyond a predetermined distance, is considered an unauthorized activation. The control circuit **206624** causes the modular device **5102** to generate **206634** a first response to a first unauthorized activation, which can be a minor response (e.g., warn the user or raise the threat warning). In one example, the control circuit **206624** may cause the user interface **206626** to issue a warning or an alert, which can a visual and/or an audible alert. Furthermore, the modular device **5102** may store **206636** a record of the first unauthorized activation in a memory unit of the control circuit **206624**, for example. Furthermore, a second unauthorized activation of the modular device **5102** is detected **206638**, the control circuit **206624** cause the modular device **5102** to generate a second response to the second unauthorized activation of the modular device **5102**, wherein the second response is escalated from the first response.

[0647] In various aspects, a surgical hub can be configured to provide surgical hub **5104** feedback to a user. The feedback could be adjusted based on a connotation resulting from a different connotation. In one aspect, a surgical hub **5104** can be programmed to provide interactive feedback to the user that enables adjustment of a device or display based on presence of an actionable aspect of the task at hand for the user. In one aspect, the interactive feedback could include visualization improvements identified by the surgical hub/visualization module that could provide better or more complete imaging of the surgical site. In one aspect, the interactive feedback could include an alternate imaging overlay demonstrating how a device could be better articulated when a device is placed in an inopportune location.

[0648] In one aspect, the surgical hub or displays could be affected based on the sensing of an action context for the user. Accordingly, display adjustments could be based on the surgical hub's awareness of actionable context.

Improved Monopolar Return Pad

[0649] In various aspects, a monopolar return pad can be utilized in additional applications beyond simply a return path for monopolar energy. In one aspect, the radiative resistance measurement can

indicate patient position changes during surgery.

[0650] In one aspect, an interrogation circuit can constantly monitor the RF radiative resistance of the pad at high frequency. This allows for a determination of the base radiative resistance, which can then be utilized in combination with the situational awareness of the surgical hub to provide an indication of when the patient is on the pad. For example, a patient's shoulder placed on a portion of the pad may change a measure of radiative resistance in at least part of the pad. This may be compared against the base radiative resistance, and in some cases the comparison may be examined using situational awareness to determine more specifically if a particular body part or location of the body is placed on the pad. In some cases, the radiative resistance may be measurable in different places on the pad, and so different parts of the pad may provide different readings, based on if a body part is on that location of the pad. These different measurements may then be used to provide a more illustrative picture of where the body—and what parts of the body—is placed on the pad overall. Once this is determined, the quality of the patient to monopolar return pad coupling can be determined.

[0651] For additional clues or information, patients or other RF conductive elements will cause parasitic loading of the capacitively coupled pad when the pad is driven by a source with a frequency or frequencies that have wavelengths a significant fraction of the pad's transmitting antenna characteristics and resonances. Accordingly, in one aspect, this loading can be measured to track the changes during patient placement and the surgical procedure. The changes in the parasitic loading can indicate that something about the patient or the local environment has changed. For example, the patient may be maneuvered to a different spot on the pad, or may be turned to have different parts of the body on the pad, such as turned on the patient's side, or have an arm raised. In other cases, the patient may unintentionally be moved off the pad, the patient may react physiologically to the RF energy such that the capacitive load and/or radiative resistance changes, or the patient may experience a sudden change in response to the surgery that bears out in the capacitive coupling. This information can be used in conjunction with the situational awareness of the surgical hub to determine if any action or warning needs to take place. Further, a “trap” circuit may be employed to keep the electrosurgical generator out of the circuit that is being measured for radiative resistance.

[0652] In another aspect, vector network analysis can be utilized to measure antenna input impedance as a function of frequency.

[0653] In another aspect, the H-field (i.e., magnetic field intensity) can be measured in multiple (e.g., two) orthogonal directions to get a sense of what loading the patient is providing. These may provide additional pieces of information that can be factored into situational awareness. For example, the combination of radiative resistance measurements, capacitive coupling measurements, antenna input impedance and H-field measurements may create a particular signature for a type of surgery or patient condition, such that situational awareness as employed by the medical hub and/or the cloud system may incorporate the particular pattern to help determine if the course of the surgery is progressing as expected or if something wrong is occurring.

[0654] FIG. 51 is a circuit diagram **209000** of a circuit for calculating parameters of an antenna, in accordance with at least one aspect of the present disclosure. The RF generator **209005** is connected in a monopolar circuit through the body of the patient on top of a conductive pad **209010**. In one aspect, the surgical system can include a forward and reflected power meter (i.e., a directional coupler) **209015** coupled to the pad **209010** or other part of the system to measure the standing wave ratio (SWR), which is an indication of how in- or out-of-tune an antenna circuit is. This SWR is another proxy for changes in antenna loading. The example diagram **209025** shows a strip line F/R power measurement that may be expressed inside the F/R power meter **209015**. Accordingly, the antenna shape (e.g., rectangular) formed by the monopolar return pad **209010** can be considered, e.g., a patch antenna **209020**. Parameters for this patch antenna can be calculated using the formulae and assumptions disclosed in Dottorato, Pasquale, “Analysis and Design of the

Rectangular Microstrip Patch Antennas for TM0n0 Operating Mode,” Microwave Journal, October 2010, which is incorporated by reference herein in its entirety and is available at www.microwavejournal.com, and using the calculators and converters available at www.pasternack.com/t-calculator-microstrip-ant.aspx, for example. These SWR readings may be communicated to the medical hub through the circuitry attached to the pad. Therefore, the pad functioning as or being modeled as a patch antenna may be capable of transmitting signal information when the patient is on the pad that may contribute to forming a signature or pattern to be incorporated by situational awareness. Thus, when signal information from the pad acting as a patch antenna may match other known patterns based on a compilation of previous patients on a pad, the medical hub and/or the cloud system may utilize situational awareness to identify a state of the patient and may provide an indication that the surgery is on track or that something may be anomalous.

[0655] In one aspect, the radiative resistance measurement can be used in conjunction with nerve stimulation to detect patient movement that is in synchrony with the stimulation signal, as disclosed in Wikipedia contributors, “Radiation Resistance,” Wikipedia, the Free Encyclopedia, last revision Oct. 7, 2017, available at wikipedia.org/wiki/Radiation_resistance. Situational awareness may incorporate these readings as well, similar to the descriptions above.

Automatic Monopolar Return Pad Tuning

[0656] In various aspects, the surgical system can be configured for automatic tuning and compensation for variations in capacitance of the monopolar return pad-to-patient coupling. In various aspects, the system can compensate for variations in coupling capacitance by adjusting the power of the attached devices or adjusting the frequency.

[0657] In one aspect, the control system can include compensating networks capable of adjusting the power of the attached devices based on the detected connection. Accordingly, the networks could compensate for changes in capacitance of the monopolar return pad.

[0658] FIG. 52 is a diagram 209100 featuring a compensating circuit for adjusting the applied power, in accordance with at least one aspect of the present disclosure. The RF generator 209110 is coupled to an active probe that touches the patient 209105. The patient is on a conductive return pad 209115, which is connected to a return probe leading to a compensation relay 209120. A relay driver 209125 may be configured to control the compensation relay 209120. In one aspect, the surgical system could be configured to measure the power into the patient and capacitive pad and adjust, in real time, a compensating network so that the power is peaked. In this example, the compensation relay has five relays, A, B, C, D, and bypass. Each the relays A through D may be either on or off in a binary fashion, with each successive relay providing a compensation inductance equal to 2 times the previous relay. As an example, the inductance of D may be 2 times the inductance of C, and 4 times the inductance of B, and 8 times the inductance of A. The relay drive 209125 may be governed by a processor that is configured to activate one or more of the relays to provide compensation to the applied power to reach peak power. In an alternative aspect, a circuit can be configured to minimize the reflected power, rather than peaking the applied power. The relays may be utilized in a similar fashion to minimize reflected power instead.

[0659] FIG. 53 is a logic flow diagram 209200 of a process for peaking applied power, in accordance with at least one aspect of the present disclosure. In one aspect, switches in various networks (e.g., inductors) can be utilized to constantly “peak” the power output of the generator, such as those described in FIG. 52. This process is similar to an automatic antenna tuner in that it is always seeking to peak the power into the load when there are static and dynamic conditions that provide a mismatch between the source and the load. The algorithm may be implemented by a processor of a control circuit in the compensation network, for example. The process may start in default by engaging the bypass, as block 209205. The processor may then determine if the RF generator is active, at block 209210, in order to proceed with the process for peaking the applied power. Once the RF power is active, the processor may need to measure the power or access a

measurement of the power, at block **209215**. The bypass may then be disengaged, at block **209220**, to prepare for compensating. The processor may incrementally increase the inductance by changing the switches A-D, for example in FIG. **52**, one count at a time in a binary fashion, at block **209225**. At each increment, the power may be measured again, at block **209230**. This may be compared with the previous power measurement, at block **209235**. To reach peak power, if the current power measurement is higher than the previous measurement, then the process of block **209225** through **209235** may be repeated. At block **209240**, if the current measurement is found to be lower than the previous measurement, then the peak power has been reached, and the current power level may be maintained according to **209245** until the next RF activation, or in some cases the inductive switches may be decremented by one just to get back to the previous power measurement. In this way, the system can adjust networks connected to the output of the generator and the pad to provide a conjugate reactance so that the real power is peaked or the reactive power is minimized. This achieves the maximum power transfer from the generator to the load, as disclosed in. e.g., **7,837,680**, which is hereby incorporated herein by reference.

[0660] FIG. **54** is an illustration **209300** of a set of graphs **209305** and **209330** depicting generator power level and patient continuity over time, respectively, in accordance with at least one aspect of the present disclosure. The two graphs are synchronized to the same time axis. The illustration **209300** may show an example of the interrelations between generator power level and patient continuity to the return pad (expressed by ohms measured at the return pad), as one example of some kinds of information that may be incorporated into situational awareness when a patient is placed on a return pad.

[0661] In the first graph **209305**, some example readings may include a threshold line **209310** for maximum power of the generator and a threshold line **209315** for a user setting maximum power of the generator. The plot **204320** therefore reflects the actual applied power of the monopolar probe, and it can be seen that the applied power reaches a maximum at the user setting threshold **209315**. The extra plot **209325** reflects the theoretical or potential maximum power of the monopolar probe that could have been applied.

[0662] In the second graph **209330**, the plot **209335** represents a measure of resistance at the return pad, expressed in ohms. The resistance may be an expression of how connected the patient is to the return pad, where the higher resistance measured generally indicates that the patient is not as well connected to the return pad.

[0663] As shown, the markings at t.sub.1, t.sub.2, t.sub.3, t.sub.4, t.sub.5, and t.sub.6 show changes in the state of the generator and the patient that are reflected in both graphs but in different ways. For example, at t.sub.1, the resistance drops as the power of the monopolar probe is applied to the patient, closing the circuit and reflecting that the patient is aptly connected to the pad. When power is increased at t.sub.2 and ends at t.sub.3, the resistance drops even further, which is expected. However, at t.sub.4, something may have happened to the patient and the patient is somehow not sufficiently connected to the pad, reflecting a sudden increase in resistance. Correspondingly, the power from the generator may be turned off and/or the probe disconnected from the patient. This is represented by the period t.sub.w. During this time, steps may be taken to verify that the patient is fine and that any recovery steps are taken before resuming. From t.sub.5 to t.sub.6, the generator power may be incrementally increased and verified that the patient is duly connected to the return pad again. After that, the process resumes as intended as reflected in the rest of the graphs.

[0664] This kind of combined information may be used to make determinations about the position of the patient on the pad, and may be incorporated into kinds of patterns used by the situational awareness of the medical hub and/or the cloud system for future surgeries or operations of a similar nature. Thus, situational awareness may be applied to flag patterns of information that were consistent or even identical to past anomalous occurrences that may have looked like some of the problem situations described in FIG. **54**. Similarly, when the information looks consistent with times that did not have an anomaly occur, the medical hub and/or cloud system may utilize

situational awareness to permit the procedure to continue.

[0665] FIG. 55 is an illustration **209400** of a patient **209405** situated on a monopolar return pad **209410** during a surgical procedure, in accordance with at least one aspect of the present disclosure. The return pad **209410** is connected to a monopolar generator module **209420** with a continuity sensor capable of detecting continuity of the patient in accordance with the disclosures herein. The monopolar pen or probe **209415** may be configured to apply RF energy to the patient **209405** and complete the monopolar circuit. In some cases, the probe **209415** may also include functionality from a smoke evacuation module **209425** for performing evacuation functions of smoke and other debris. This setup is one example that may utilize the situational awareness and take the example measurements involving the pad **209410**, described herein.

[0666] FIG. 56 is a block diagram **209500** of a system for controlling a power level applied by a monopolar instrument, in accordance with at least one aspect of the present disclosure. The block diagram **209500** features example interrelationships of a monopolar generator system and the directionality of the power and inputs received from various components. Here, power from the RF generator **209530** is supplied to a monopolar pen **209505**. The energy flowing through the monopolar pen is applied to a patient **209510**, who is touching in some way to the capacitive pad **209515**. A measurement of continuity is determined via one or more of the example descriptions provided herein, expressed as block **209520**. That determination is ultimately fed back to the monopolar pen **209505** in the form of continuing to provide energy to the pen **209505** or not. This function may be provided through the control panel **209525** that controls the power generator **209530**. Additional information may be provided from the capacitive pad **209515**, such as the various other measurements described above. The monopolar circuit may be completed through the capacitive pad **209515** and back to the power generator **209530**. In general, various details of the control system described above are illustrated in FIGS. 54-56.

[0667] In another aspect, a control system can compensate for variations in capacitance of the monopolar return pad-to-patient coupling by changing frequency. Accordingly, the control system can change frequency to compensate for capacitance in the monopolar return pad.

[0668] In one aspect, a frequency agile generator can constantly sweep the output frequency through a predefined range, either in combination with a fixed inductor in series with the return pad or with a peaking compensating network to peak the power output, as described above. The reactance of the capacitor formed by the patient and the pad will change based on the contact the patient is making with the pad and this will compensate for those changes.

[0669] In another aspect, a controls system can adjust output frequency of the generator to seek the peak power output when using a reactive impedance (e.g., capacitive) pad.

Return Pad Feedback

[0670] In various aspects, a control system can provide feedback regarding the contact/return pad quality and indicate to the OR personnel of the current efficiency of the return pad. The medical hub may provide an indication, via a graphical user interface or one or more auditory signals, of the connectivity of the patient to the return pad or if there is some disconnect. Some example methods for making this determination are described above.

Nerve Stimulation Integration

[0671] In various aspects, the functionality of the generator and/or monopolar return pad can be integrated with nerve stimulation indications.

[0672] In one aspect, the surgical hub can be configured to modulate the energy (e.g., the energy applied by a generator to a monopolar electrosurgical instrument) based on nerve mapping and situational awareness. In addition to warning the surgeon when they're close to a nerve, the cutting energy (e.g., electrosurgical, ultrasonic, or the like) can be reduced as the surgeon cuts closer to a surgical hub-identified nerve structure. For example, energy can be stopped when the surgeon is about to damage a hub-identified nerve structure. In one aspect, the surgeon is required (e.g., by the surgical hub) to make a conscious decision to override a warning and a drop in power or inability to

deliver power based on the surgical hub's awareness of a nerve mapping. Accordingly, modulation of energy by the surgical hub can be based on nerve mapping and situational awareness. FIG. 57 shows an illustration 209600 of a probe 209605 approaching a nerve 209615. The probe 209605 may touch the surgical site at location 209610, which is a minor distance away from the nerve 209615. The determination of how close the probe 209605 is to the nerve 209615 may be based on a nerve mapping of the patient and augmented by aggregate mappings of this area from other past patient surgeries. In addition, in some cases, the resistance profile and other measurements through the use of the probe 209605 and the patient being on the monopolar return pad may provide an indication that the probe is touching an area that is close to the nerve 209615. The energy may respond differently to this area compared to when the probe 209605 is touching other parts of the patient, and based on situational awareness that may have aggregated information from past similar instances, the surgical hub may warn the surgeon.

[0673] Similarly, FIG. 58 shows an illustration 209700 of a probe 209705 directly touching a nerve 209715 at location 209710. The determination of directly touching the nerve may be based on similar methods described for FIG. 57, such as utilizing situational awareness from past sets of information in similarly situated surgeries, and on a nerve mapping of the patient and/or other patients. Accordingly, the hub warns the surgeon and terminates power to the probe, forcing the surgeon to make a conscious decision to continue cutting in an area where nerve damage is imminent.

Local Autonomous Adjustment of Functional Parameters

Limiting Adaptive Program Adjustment of a Powered Surgical Instrument

[0674] In various aspects, adjustable autonomous control programs can contain limits on surgical instrument algorithms. In one aspect, a powered surgical instrument 208100 (FIG. 62) with a predefined adjustable control algorithm for controlling at least one parameter of an end effector 208109 can further include a means for limiting the adjustment of the control algorithm to one or more predefined adjustability windows.

[0675] In one aspect, the adjustable control algorithm controls at least one function of the end effector 208109. In one aspect, the adjustability is dependent on at least one sensed parameter. In one aspect, the sensed parameter includes a historical dataset of previous uses of the surgical instrument 208100 by the surgeon, in the facility, in the region, or by the user base at large. In one aspect, the limit of the adjustment is predefined by the surgical instrument 208100 and/or a surgical hub (e.g. 102, 202). In one aspect, the limit is an overall maximum threshold. In one aspect, the limit is a per use adjustment. In one aspect, the limit is based on uses by a specific user, in a specific facility or in a specific region.

[0676] In one aspect, a control program can limit control-program learning adjustments. For example, in a qualified aggregation an event or behavior could have to pass a check to determine if it is going to be allowed to affect long term behavior of a particular surgical instrument 208100, or a class of surgical instrument 208100, for example. A control program executed by a surgical instrument 208100, or a surgical hub (e.g. 102, 202), may factor out individualized or one-time failures (e.g., a damaged or mis-inserted cartridge due to a non-repeatable error) that have a minimal effect on the behavior of the control program. In other words, the data associated with the individualized error may or may not be transmitted to a surgical hub (e.g. 102, 202) and/or main database depending on the nature of the individualized error. Even, however, if it is transferred, the individualized error could be excluded from the aggregated database used to affect long term behavior of the surgical instrument 208100 as a means to prevent or detect future flaws of the surgical instrument 208100.

[0677] As another example of qualified aggregation, the weighted effect of a behavior could be used to influence the amount of adjustment (e.g., a “class 0” defect resulting in a patient injury could have a greater influence as a single event on device performance than a number, e.g., 10×, of minor variations).

[0678] In one aspect, a control program can limit control program learning adjustments across a series of parameters. For example, learning adjustments can be limited to a maximum adjustment of the control algorithm over a given time interval (e.g., $\pm 10\%$ over a week, a month, or another interval). This would prevent different behaviors from a new user, rotation of OR staff, or other individuals, from dramatically shifting the instrument behavior for all other users (especially if, e.g., some other users are on vacation, not working over a weekend, or are otherwise not actively using the instruments for a period of time).

[0679] As another example, maximum and minimum total limits on a performance behavior can be applied for a given user. This could have a lifetime cumulative effect or a maximum adjustment for a given BIOS or control program version. Each time a control program is updated, the adjustment could be transferred over or it could be “reset” to a nominal target value and the system will have to re-learn the adjustment, for example. This would allow the system to benefit from improved control programs, without requiring that the control program re-learn the same adjustment if the program operates differently. As another example, users could be able to temporarily use other users' settings, if desired, while not having the ability to alter those settings.

[0680] In one aspect, a control program could set a cap or a maximum on the number of adjustments to the control program per procedural use. This would minimize what could appear as dramatic alterations in behavior from one use to the next. Further, this could also be factored as per use per user and therefore have different behaviors for different users and minimize the adjustments of the device performance from one user to the next.

[0681] In one aspect, a control program could be programmed to implement a predefined adjustability envelope. In this aspect, adaptive algorithms and techniques could be implemented to locally adjust (i.e., adjust a control program of a given surgical hub (e.g. **102**, **202**) or the control programs of a local network of surgical hubs of, e.g., a single facility) overall control schemes. The adjustment methods can be implemented by machine learning, e.g., as a neural network, for updating/controlling attached devices' algorithms.

[0682] In at least one instance, a GUI for controlling various device parameters, such as those parameters described above, is disclosed. The GUI can be displayed on, e.g., the device being controlled and/or a surgical hub (e.g. **102**, **202**) to which the device is connected. The GUI allows users to select settings for a particular surgeon (e.g., “Dr. Smith” or “Dr. Jones”) per device type (e.g., staplers, energy devices, scopes, and so on) per action type (e.g., clamping, firing, or articulating settings for staplers). Different settings for the devices can be learned over time as users are more experienced in using the devices.

[0683] In one aspect, the control programs can provide an overriding capability to allow the user to default the device to the nominal or manufacturer's suggested value of a device performance. For example, there could be an indication of the device's current learned parameters and allow the user to determine if they want to utilize this customized performance. As another example, the user could have the ability to select an override of an adjusted parameter. This could occur before a device is used, at the beginning or a procedure, or even during an actuation. As another example, the control programs could allow the user to reset the device to a non-adjusted state or even disable the ability for the parameter to be adjusted over time due to measured performance in the future.

[0684] In one aspect, a device could identify a user usage or behavior and determine a performance parameter adjustment to improve outcomes for that behavior. It could then in a later use detect the same behavior or usage, but because it is a different user, either limit the application of the adjustment or request the user confirm the use of the improvement before it was used. For example, if thicker than indicated tissue and an uneven distribution of the tissue with it skewing to the tissue stop end of the anvil is detected, the control program could adjust for these variables for by slowing the firing I-beam advancement in the beginning of the stroke and increasing the displayed stabilization wait period. Accordingly, if this same irregular tissue stuffing of the jaws is detected at a later time, but it appears to the instrument that the user is different than the first user, the

instrument could ask if the user wants to use the new performance program or the standard program rather than merely adjust the parameters automatically as it would for the first user using the device in a subsequent procedure.

[0685] In at least one embodiment, a surgical instrument system includes a surgical end effector, such as surgical end effector **208109**, for example, or surgical instrument such as those disclosed herein (e.g. **208100**), for example, configured to deliver at least one end effector function to a patient and a control circuit, such as the control circuit **208103**, for example, configured to operate the surgical end effector and/or the function of the surgical end effector **208109**. Function(s) of the end effector **208109** can be actuated by a surgical robot and/or by way of a handheld instrument handle, for example. The handheld instrument handles may be manually operated by a clinician. The end effectors attached to surgical robot may be manually operated by a clinician operating the surgical robot and/or automatically operated by a control circuit of the surgical robot, for example. Functions of an end effector may include firing staples, for example, which may include cutting tissue and/or deploying staples in a surgical stapling end effector. Another end effector function may include clamping tissue with a surgical stapling end effector. Yet another example of an end effector function may include energizing tissue with a surgical energy device. It should be appreciated that any suitable end effector functions can be used with the surgical systems described herein.

[0686] The control circuits of such surgical systems can include adaptive control programs configured to control the end effector function and adapt itself over time to better accommodate subsequent uses of the end effector function(s) and/or the surgical instrument systems. Such adaptive control programs can utilize various types of information to automatically adjust and/or adapt the control program of the end effector function. For example, the adaptive control programs can be directly based on inputs including parameters sensed within an end effector, such as end effector **208109**, for example, itself, within a patient, and/or within a surgical suite. The adaptive control programs can also be based on inputs from a surgical hub (e.g. **102**, **202**) for example. Machine learning can be used to analyze the inputs and make adjustments to the adaptive control program in an attempt to provide better end results of the end effector function for each subsequent use.

[0687] In at least one instance, the adaptiveness of the control program is based on a locally-sensed parameter within the end effector, such as end effector **208109**, for example. For example, the load on a tissue-cutting knife or firing member **208111** applied by tissue and/or other aspects of the system, in a surgical stapling end effector can be measured within the end effector **208109**. Information about the load on the tissue-cutting knife can be fed to the control circuit **208103** so that the control circuit **208103** can adjust the control program of the tissue-cutting knife automatically. For example, if the load is monitored and becomes increasingly high during a firing sequence, the adaptive control program may predict that the next firing sequence will include a similar load profile and, in at least one instance, the adaptive control program can automatically slow the firing speed of the tissue-cutting knife for the next firing sequence to prevent the tissue-cutting knife from becoming jammed.

[0688] In at least one instance, the adaptiveness of the control program is based on information collected over a period of time. Further to the above, the adaptiveness of the control program can be based on specific information collected over time. For example, the adaptiveness may only be based on data collected while a certain surgeon was using the device. In at least one instance, the adaptiveness may only be based on data collected during use on a specific patient, during use in a specific operating room, during use in a specific region of the country, and/or during use on specific types of procedures. Any suitable groupings of data can be used for control program adaptiveness. In at least one instance, multiple groupings of data are used cooperatively and the adaptiveness of the control program is based on the multiple groupings of data.

[0689] In systems utilizing adaptive control programs, it may be advantageous to restrict the

adaptiveness of the control program itself. Placing limits, automatically based on locally-sensed parameters, for example, and/or manually based on direct input from a surgeon, for example, on the adaptiveness of the control program can prevent undesirable adaptive adjustments to the control program. Further to the above, such restrictions and/or limitations placed on the adaptive control programs, whether applied automatically and/or applied manually can provide more information for machine learning aspects of the control circuit to better operate the end effector functions in subsequent uses. Such limitations may be put in place by an adaptive-limiting program, for example.

[0690] Referring again to the tissue-cutting knife example discussed above, a clinician may be aware that the adaptive control program is going to slow down the firing speed of the tissue-cutting knife for a subsequent firing sequence; however, in such an instance, the clinician may not want the firing speed of the tissue-cutting knife to slow down for the next firing sequence. The clinician may want a limit automatically placed on the adaptive control program controlling the firing function of the end effector, such as end effector **208109**, for example. In at least one instance, the clinician may want to manually place a limit on the adaptiveness of the control program controlling the firing function. In the discussed example, the clinician may want define a slowest-possible firing speed value that the adaptive control program is permitted to automatically slow to. In such an instance, after such a restriction and/or limitation is set in place, the adaptive control program may not be permitted to adjust the firing speed of the tissue-cutting knife to a speed that would fall below the defined slowest-possible firing speed. Restrictions and/or limitations may be set during a procedure, before a procedure, and/or after a procedure. In at least one instance, the clinician may be made aware by way of a display or audible alert of the adaptiveness of the control program in real time to allow the clinician to make real-time adjustments to the adaptiveness of the control program.

[0691] In at least one instance, the adaptive adjustments made by the control program can have bounds placed on them. For example, a control circuit, such as the control circuit **208103**, for example, could analyze behavior of an end effector function to determine whether or not that the behavior would affect the adaptiveness of the control program thereby affecting the long term behavior of the end effector, such as end effector **208109**, for example. In such an instance, one-time inadvertent and/or preventable failures of the end effector **208109** could be ruled out so that that the one-time failure is not factored into the adaptiveness of the control program of the end effector **208109**. For example, if a staple cartridge is improperly loaded into a surgical stapling end effector and firing is attempted, this irregular load sensed due to the improperly loaded staple cartridge can be treated as outlier and not factored into the adaptiveness of the control program of the end effector. In at least one instance, such a misfire could still be factored in to the adaptiveness of the control program but not with the same weight as a tissue-jam incident resembling a similar load level as a misfired end effector would. In other words, it may be desirable to not completely ignore an improperly loaded cartridge misfire event and, rather, to apply it to the adaptiveness of a control program in a manner that would be less aggressive than a tissue-jam incident where a cartridge was properly loaded. At any rate, outlier events or behavior can be excluded from the aggregated database of usage such that the outlier events do not affect long term behavior of the adaptive control program.

[0692] In at least one instance, certain events, such as the improperly-loaded cartridge misfire event discussed above could be given different weight values when determining the amount of influence such an event would have on the adaptiveness of the control program. For example, a misfire due to an improperly loaded staple cartridge may be given considerably less weight providing considerably less influence to the adaptiveness of the control program than a complete tissue jam incident resulting after a properly assembled cartridge firing. In such an instance, the type of tissue may have caused the complete tissue jam incident which may be much more desirable to have influence the adaptiveness of the control program in case the clinician and the end effector

encounters that type of tissue again. On the same hand, a clinician may not want the adaptive control program adjusting itself based on user error of an improperly loaded cartridge and/or misuse of the instrument.

[0693] During normal operation and assuming no misuse of the end effector, such as end effector **208109**, for example, an event that causes patient harm and/or injury could be given a much higher weight and thus influence the adaptiveness of the control program greater than an event that causes little to no patient harm and/or injury to a patient.

[0694] In at least one instance, bounding of the adaptive control program can occur across a series of parameters. For example, a control circuit, such as the control circuit **208103**, for example, can permit only a percentage of adjustment to the control program over a certain period of time. For example, an adaptive firing control program for a surgical stapling end effector, such as end effector **208109**, for example, may be limited to adjusting the firing speed of the control program $\pm 10\%$ of the firing speed over a week of time. Any suitable percentage restriction can be employed with any suitable time interval. Such an arrangement may eliminate drastic adaptiveness during a certain time period. For example, an end effector may possibly undergo a break-in period and have some abnormal sequence during the beginning of its usable life. Thus, it may be desirable in such an instance to limit the adaptiveness of the control program for that end effector over its break-in period. Another advantage may include eliminating drastic adaptiveness across multiple users which have different operating behaviors.

[0695] In at least one instance, maximum and minimum program limits can be specific to a given user. In such a scenario, the user may be able to set these for a lifetime cumulative effect. In at least one instance, the user may be able to select maximum and minimum program limits specific to another user. In at least one instance, where the user is using limits specific to another user, the user may not be able to adjust the limits specific to another user nor will those limits be able to be adjusted by the adaptive control program because the user specific to those limits is not employing them.

[0696] In at least one instance, limits placed on adaptive control programs could be transferred into a database and/or hub (e.g. **102**, **202**) and that control program would be reset to a nominal target value. In such an instance, limits may need to be re-learned and/or re-adjusted. In another instance, a surgeon can be given the option to reset the limits to the nominal value or to set the control program where the surgeon left off at the end of the last use. This would allow systems to benefit from improved control programs and perhaps not need the same adjustment if the program operates differently.

[0697] In at least one instance, limits placed on adaptive control programs can be based on a per-use basis. In at least one instance, the adaptiveness of the control program can be isolated to a single procedure and/or a lifetime use of the specific end effector which the control program is controlling.

[0698] In at least one instance, adaptive control programs can be limited to a predefined adjustability envelope. Adaptive algorithms and/or techniques can be used to locally adjust overall control schemes of the adaptive control programs and/or surgical instrument systems generally. Adjustments to the control program can also be based on neural networks including inputs from the surgical hub (e.g. **102**, **202**) and any other information that may be desirable to input into the neural networks when making adjustments to the control program.

[0699] FIG. **61** depicts logic **208060** of a control circuit such as those described herein. The logic **208060** comprising controlling **208061** a parameter of an end effector, adjusting **208063** the control of the parameter, and limiting **208065** the adjustment of the control of the parameter. Controlling **208061** a parameter of the end effector may include running a control program for operating a motor operatively coupled with a tissue-cutting knife, for example. The control program may be able to cause the motor to advance the knife distally, retract the knife proximally, and/or pause actuation of the knife. Speed and acceleration of the tissue-cutting knife may also be varied by the

control program. Adjusting **208063** the control of the parameter may include automatically and/or manually modifying and/or adapting the control program or control **208061** of the parameter to perform better during a use and/or for each subsequent use. This is referred to as an adaptive control program that is capable of using machine learning, for example, to cause better operation of the parameter that is being controlled. Limiting **208065** the adjustment of the control of the parameter may comprise manually setting an adjustment window or range of values that the adaptive control program is permitted to vary itself within. For example, a range of firing speeds may be defined manually and/or automatically to constrain an adaptive control program to stay within the set range of firing speeds.

[0700] FIG. **59** depicts a GUI displaying a series of menus comprising selectable options to aid a clinician in operating a particular surgical instrument, such as the instrument **208100**, for example. In the illustrated example, a first series of displays **208010** depict multiple selectable menu options where, in this instance, a specific surgeon is selected, a specific instrument is selected, and a specific function is selected. In such an instance, a specific surgeon can be selected so that a control circuit, such as the control circuit **208103**, for example, may load particular settings, such as learned adaptive limits, for example, for that particular surgeon. A specific instrument, such as the instrument **208100**, for example, can be selected so as to allow the control circuit to load a specific control program to operate that instrument. This may include a specific adaptive-limiting program corresponding to a specific instrument and a specific surgeon. All of the selected options can be taken into account by the control circuit so as to load the correct control program(s) and/or settings for operating the desired device. In the illustrated example, the firing function of STAPLER 2 for Dr. Jones has been selected. These options may be automatically sensed by the control circuit and, in at least one instance, are not selected. For example, the information may already be delivered to the control circuit in a package corresponding to the particular procedure by a surgical hub (e.g. **102, 202**), for example. In another instance, a surgeon may wear an identifier chip that a component of the control circuit can sense, a surgical robot, such as the surgical robot **110**, for example, to which the instrument is attached may be able to automatically identify what instrument is attached to the operating arm of the robot **110**, and/or the firing setting of the particular instrument may be identified by the robot based on an indirect input from the surgeon on a surgical robot control interface, for example.

[0701] Still referring to FIG. **59**, two displays **208020** are depicted showing selectable, in at least one instance, options for Dr. Jones for the firing function of STAPLER 2. As can be seen in these displays **208020**, firing time and clamp force are displayed and can be related to the overall firing speed of the instrument, such as the instrument **208100**, for example. In this instance, Dr. Jones may have limited experience. Such experience can be known by the control circuit, such as the control circuit **208103**, for example, based on information stored about Dr. Jones. In such an instance, the range of permitted values for the firing speed, whether they be selectable learned limits and/or selectable direct function parameters, may be larger than a range of permitted values allowed for an experienced surgeon. For example, a display **208030** is illustrated where Dr. Smith, a more experienced surgeon than Dr. Jones, is provided tighter default settings. This may occur due to the amount of repetitions a surgeon has with a particular instrument, such as the instrument **208100**, for example. In at least one instance, a permitted value range indicating safer operation of a particular instrument may be provided to a surgeon with less experience where more a permitted value range indicating riskier operation of a particular instrument may be provided to a surgeon with more experience.

User Customizable Performance and Program Behaviors

[0702] In various aspects, the control program behaviors of a smart surgical device (e.g., a stapling device) could be customizable with user interaction in order to customize the performance of the device.

[0703] In one aspect, a surgical device, such as the instrument **208100**, for example, could be

controlled via user adjustable controls with adjustable algorithms. In at least one instance, a GUI for controlling adaptive parameters of a surgical device is disclosed. A stapler uses an adaptive firing speed algorithm that adjusts firing speed based on the resistance to firing provided by the tissue. Variables in the algorithm include the min/max speed, the number of speed intervals in the range, and the duration of the pause in firing when force parameters exceed safety thresholds. These variables are scalable or are able to be changed by the user, such as via the GUI. These inputs inform system thresholds for the subsequent firing response of the stapler.

[0704] In one aspect, local instrument controls could allow the user to adjust their function. A control can have, for example, scalable sensitivity to link an actuation control to a powered actuation movement. In one aspect, the local instrument controls can be reclassified from one function to another by the user (i.e., controls can be mapped from a first or default function to a second function).

[0705] In one aspect, trained learning (e.g., machine learning) can be utilized to assist users in customizing the performance of a device (e.g., a surgical instrument, such as the instrument **208100**, for example, or hub (e.g. **102**, **202**)). For example, a user could input their personal opinion of the output the device has provided in its most recent uses. The device could then use this additional information to better adjust the performance of controlled functions of the device. Further, the user could then have the ability to input an opinion on the relative performance of the second use of the device to the first use of the device. This trained behavior would allow the device to personally tune not only its behavior, but the desired outcomes. For example, one of the more skilled people in the practice could input their opinions on the performance/functions of the device to tune the performance/functions and then allow the device to present this improved output behavior to all the other users of the device.

[0706] In at least one instance, there can be provided a control interface, such as a graphical user interface or any suitable control interface, to allow a clinician to choose if they want to override the learned or set limitations to a nominal value. In other words, the user may be prompted and asked if they would like to reset the adaptive control program before using the end effector, such as end effector **208111**, for example. Such a reset may set the device to a manufacturer's suggested default state. In at least one instance, the current state of the adaptive control program is shown to a user as well as its learned or set limits. A user may then be able to choose whether or not they would like to utilize this customized performance. In at least one instance, a brief history of the current state of the adaptive control program can be shown to the user. For example, what surgeons have used and contributed to the adaptive control program and its limits and/or what operating room staff were involved during the data aggregation to arrive at the current state of the adaptive control program may be shown to the next user to allow the next user to decide if the adaptive control program is in a desirable state for use in their procedure. Such an override can be selected before, after, and/or during use of the end effector **208111**. More specifically, such an override can be selected during actuation of an end effector function itself. In such an instance, a surgeon may have second thoughts about the state of the adaptive control program during firing based on real-time events and/or behavior of the end effector and would like to override the adaptive control program and/or limits set on the adaptive control program.

[0707] In at least one instance, a user may be provided the ability to completely disable the ability for limits to be set on the adaptive control program. Further to this, the user may be provided the ability to completely disable the adaptiveness of the adaptive control program such that controlling the function of the end effector may be entirely manually operated in a sense that machine learning will not affect the way that the end effector function is actuated and/or controlled, for example.

[0708] In at least one instance, a control circuit, such as the control circuit **208103**, for example, can be configured to identify a user of the end effector, such as end effector **208111**, for example, based on the behavior of the user using the end effector. In such an instance, an adaptive control program can adapt as described above and limits can be learned and/or set on the adaptive control

program as described above. If the control circuit determines that a different user is using the end effector **208111**, the new user may be made aware of the adaptive control program set in place on the current end effector and can be asked if the new user would like to continue with the current adaptive control program. In at least one instance, if the control circuit determines that a different user is using the end effector, the adaptive control program may exclude the use of the end effector under the new user from affecting the adaptive control program and/or the limits of the adaptive control program of the end effector.

[0709] An example of the benefit of user detection will now be described. For example, thicker tissue than expected and an uneven distribution of the tissue where the tissue skews to a tissue stop end of an anvil may be detected. This could be adjusted for by slowing down the firing speed of the firing member in the beginning of the firing stroke and increasing the stabilization wait period.

Waiting for tissue to regulate and flatten out within the jaws can aid in advancing a firing member through thicker tissue. If a similar event occurs but the control circuit detects that a different user is using the instrument, such as the instrument **208100**, for example, during the same scenario, the control circuit could ask if the user if they want to use the improved performance program, or adaptive control program with its learned and/or set limitations, or if the user wants to use the standard adaptive control program rather than merely adjusting the parameters automatically as it would for the first user using the device in a subsequent procedure. This can provide an advantage in a scenario where different users have different preferences when performing similar procedures.

[0710] In at least one instance, a user may be able to define and/or select a range and/or window of values to which an adaptive control program may be able to adapt within. Referring to FIG. **60**, a display is illustrated where a user is provided the options of fine tuning the permitted adjustments that an adaptive control program of an end effector, such as end effector **208111**, for example, they are about to use or are using is permitted to make during the use of the end effector. In the illustrated example, a stapler uses an adaptive firing speed algorithm or program that adjusts firing speed based on the resistance experienced by the firing member provided by the tissue. The GUI illustrated in FIG. **60** and corresponding control circuit, such as the control circuit **208103**, for example, permits the customizability of performance of the end effector with which it is used. Limits can be placed on various variables in the algorithm. Such variables include the minimum and maximum speed adjustments, the number of speed intervals in the range, and the duration of the pause in firing when force parameters exceed safety thresholds. These variables are scalable and/or are able to be changed by the user such that the user can manually define the window with which the adaptive firing speed algorithm or control program is permitted to make adjustments within. In at least one instance these inputs inform system thresholds for the subsequent firing response. In the illustrated example, the displays **208040** and **208050** depict a first slider for adjusting the range of firing speeds for an adaptive firing speed program to operate within, a second slider for adjusting the duration of a pause that a user would like the adaptive firing speed to pause for, and a selectable number of speeds option where a user can define the amount of speed intervals desired within a set range. Different settings are selected on each display **208040** and **208050**.

[0711] In at least one instance, a user could be able to input their opinion of the output of the device and thus the performance of the adaptive control program and its learned limits, for example. Such a survey could take place after an entire procedure is complete and/or after a week's use of a device. In another instance, such a survey could take place after the lifetime use of the device such that machine learning can use this surveyed data in the control programs of the next device to better adjust the performance of the control functions. Such opinions could correspond to the device's performance from one use to the next use and/or from one procedure to the next procedure, for example. This trained behavior would allow the device to personally tune not only its behavior but the desired outcomes. This could be done by one of the more skilled people in the practice and then allow the device to present this improved output and behavior to all of the other users of the device.

[0712] FIG. 62 depicts a surgical instrument **208100** comprising a user interface **208101** and a control circuit **208103** configured to receive inputs from at the user interface **208101**. The surgical instrument **208100** further comprises a motor driver **208105**, a motor **208107** configured to be driven by the motor driver **208105** and controlled by the control circuit **208103**, and an end effector **208109** comprising a firing member **208111** configured to be driven by the motor **208107**. In at least one instance, various components of the surgical instrument **208100** may be substituted for an energy-based surgical instrument such as, for example, an ultrasonic surgical instrument. The control circuits described herein, such as the control circuit **208103**, are configured to control any suitable end effector function, or parameter, powered by any suitable device. In at least one instance, the user interface **208101** comprises computer-based inputs rather than human-based inputs. For example, such computer-based inputs may originate from a surgical hub (e.g. **102**, **202**), for example. The surgical instrument **208100** can be employed with any of the systems, devices, and/or control circuits described herein. Various systems, devices, and/or control circuits described herein can be used for treating surgical patients. In the illustrated example, a surgical stapler can utilize a firing member, such as the firing member **208111**, to cut the tissue of a patient and/or drive staples through tissue to fasten tissue during a surgical procedure. In such an instance, it can be advantageous to provide a control circuit capable of providing improved operation of the firing member. Any of the control circuits herein may provide such an advantage. In at least one instance, the firing member **208111** includes a firing assembly extending between the motor **208107** and the staples, for example, configured to be ejected by a sled. In at least one instance, the firing member **208111** includes one or more components of a firing assembly extending between the motor **208107** and the staples, for example, configured to be ejected by a sled.

Examples

[0713] Various aspects of the subject matter described herein are set out in the following numbered examples:

[0714] Example 1: A method for adaptive control of surgical network control and interaction, the surgical network comprising a surgical feedback system, the surgical feedback system comprising a surgical instrument, a data source, and a surgical hub configured to communicably couple to the data source and the surgical instrument, the surgical hub comprising a control circuit, method comprising: receiving, by the control circuit, information related to devices communicatively coupled to the surgical network; and adaptively controlling, by the control circuit, the surgical network based on the received information.

[0715] Example 2: The method of Example 1, further comprising transmuting, by the control circuit, to a display coupled to the surgical hub, the received information based on presence of an actionable aspect of a task.

[0716] Example 3: The method of any one of Examples 1-2, further comprising: automatically scaling, aligning, and organizing data received by the surgical hub, by the control circuit, based on predefined parameters within the surgical network before transmission of the information; and adaptively controlling, by the control circuit, the surgical network based on the received information.

[0717] Example 4: The method of any one of Examples 1-3, further comprising: validating, by the control circuit, the received information; authenticating, by the control circuit, a source and integrity of the information; and adaptively controlling, by the control circuit, the surgical network based on surgical network, instrument, and cloud responses based on validation of the received information and authentication of the source and the integrity of the information.

[0718] Example 5: The method of any one of Examples 1-4, further comprising: interpreting, the control circuit, the received information based on at least one function of at least one device including at least one data source not originating within the device; and transmitting, by the control circuit, to a display coupled to the surgical hub, the interpreted information.

[0719] Example 6: The method of any one of Examples 1-5, further comprising modifying, by the

control circuit, the surgical network or a device communicatively coupled to the surgical network based on machine learning analysis of performance and outcomes recorded by the surgical network over more than one surgical procedure.

[0720] Example 7: The method of any one of Examples 1-6, further comprising, adjusting, by the control circuit, control programs of devices communicatively coupled to the surgical network based on stratified contextual data in addition to the received information.

[0721] Example 8: The method of any one of Examples 1-7, further comprising, adjusting, by the control circuit, a response of the surgical network to a sensed parameter.

[0722] Example 9: The method of Example 8, further comprising, adjusting, by the control circuit, a response of the surgical network to an event based on a second pre-existing sensed step, situation, or parameter, or combinations thereof.

[0723] Example 10: The method of any one of Examples 1-9, further comprising detecting, by the control circuit, security threat to the surgical network.

[0724] Example 11: The method of Example 10, further comprising, escalating, by the control circuit, security responses to the security threat based on increasing severity levels of the security threat.

[0725] Example 12: A method for adaptive feedback and control of a surgical system, the surgical system comprising a surgical instrument, a data source, and a surgical hub configured to communicably couple to the data source and the surgical instrument, the surgical hub comprising a control circuit, the method comprising: receiving, by the control circuit, information related to devices communicatively coupled to the surgical system; and adaptively adjusting, by the control circuit, an operating parameter of a device communicatively coupled the surgical system based on the received communicated recommendation.

[0726] Example 13: The method of Example 12, further comprising, updating, by the control circuit, a control program of a device communicatively coupled the surgical system based on the received communicated recommendation.

[0727] Example 14: The method of any one of Examples 12-13, further comprising, analyzing, by the control circuit, the received information data using a supervised learning technique.

[0728] Example 15: The method of any one of Examples 12-14, further comprising, analyzing, by the control circuit, the received information data using an unsupervised learning technique.

[0729] Example 16: A method for adaptively controlling a surgical network based on validating data purportedly generated in a surgical procedure, the surgical network comprising a medical hub, at least one remote server communicatively coupled to the medical hub, and a medical instrument communicatively coupled to the medical hub, the system is configured to access the data, validate the data to determine if the data is validly generated by the surgical procedure, determine that the data contains at least one flaw or error, and improve data integrity by preventing the at least one flaw or error from being integrated into a larger dataset associated with the at least one remote server, the method comprising: receiving, by the server, information related to a surgical procedure from a device communicatively coupled to the surgical network; validating, by the server, the received information; and adaptively adjusting, by the server, the surgical network based on the received information.

[0730] Example 17: The method of Example 16, further comprising, determining, by the server, the presence of a sequential trend or pattern in the received information that is common to surgical procedures.

[0731] Example 18: The method of any one of Examples 16-17, further comprising, identifying, by the server, an encrypted validation key associated with a device communicatively coupled to the surgical network.

[0732] Example 19: The method of any one of Examples 16-18, further comprising, analyzing, by the server, the received information to determine the presence of a sequential trend or pattern in the received information that is unique to a type of surgical procedure that purportedly occurred when

the received information was generated.

[0733] Example 20: The method of any one of Examples 16-19, further comprising, receiving, by the server, a report of a malicious actor from another medical hub.

[0734] Various additional aspects of the subject matter described herein are set out in the following numbered examples:

[0735] Example 1: A system for validating data purportedly generated in a medical procedure, the system comprising a medical hub, at least one remote server communicatively coupled to the medical hub, and a medical instrument communicatively coupled to the medical hub. The system is configured to access the data, validate the data to determine if the data is validly generated by the medical procedure, determine that the data contains at least one flaw or error, and improve data integrity by preventing the at least one flaw or error from being integrated into a larger dataset associated with the at least one remote server.

[0736] Example 2: The system of Example 1, wherein the system is further configured to analyze the data to determine if there is a sequential trend or pattern in the data that is common to surgical procedures, wherein determining that the data contains the at least one flaw or error comprises determining that the data does not contain the sequential trend or pattern.

[0737] Example 3: The system of Examples 1 or 2, wherein the system is further configured to identify an encrypted validation key associated with the medical instrument, wherein determining that the data contains the at least one flaw or error comprises determining that at least part of the data does not match with the encrypted validation key.

[0738] Example 4: The system of any one of Examples 1-3, wherein the system is further configured to analyze the data to determine if there is a sequential trend or pattern in the data that is unique to a type of surgical procedure that purportedly occurred when the data was generated, wherein determining that the data contains the at least one flaw or error comprises determining that the data does not contain the unique sequential trend or pattern.

[0739] Example 5: The system of any one of Examples 1-4, wherein the system is further configured to receive a report of a malicious actor from another medical hub, wherein determining that the data contains the at least one flaw or error comprises determining that the data contains a characteristic consistent with the report of the malicious actor.

[0740] Example 6: The system of any one of Examples 1-5, wherein preventing the at least one flaw or error from being integrated into the larger dataset comprises isolating the at least one flaw or error from the larger dataset and integrating the remainder of the data into the larger dataset.

[0741] Example 7: The system of any one of Examples 1-5, wherein preventing the at least one flaw or error from being integrated into the larger dataset comprises determining that the at least one flaw or error is a result of the data being altered and removing the data from being integrated into the larger dataset.

[0742] Example 8: The system of any one of Examples 1-7, wherein the system is further configured to determine that the data is generated from a validated medical instrument, wherein preventing the at least one flaw or error from being integrated into the larger dataset comprises flagging the at least one flaw or error in the data and identifying the validated medical instrument as containing a systematic flaw or error.

[0743] Example 9: The system of Example 8, wherein identifying the validated medical instrument as containing the systematic flaw or error comprises configuring the validated medical instrument into a controlled situation, running a predefined routine on the validated medical instrument during the controlled situation, and comparing a response by the validated medical instrument during the predefined routine to an expected result.

[0744] Example 10: A method for validating data purportedly generated in a medical procedure, the method comprising accessing the data through a processor of at least one remote medical server, validating the data to determine if the data is validly generated by the medical procedure, determining that the data contains at least one flaw or error, and improving data integrity by

preventing the at least one flaw or error from being integrated into a larger dataset associated with the at least one remote medical server.

[0745] Example 11: The method of Example 10, further comprising analyzing the data to determine if there is a sequential trend or pattern in the data that is common to surgical procedures, wherein determining that the data contains the at least one flaw or error comprises determining that the data does not contain the sequential trend or pattern.

[0746] Example 12: The method of Examples 10 or 11, further comprising identifying an encrypted validation key associated with a medical instrument, wherein determining that the data contains the at least one flaw or error comprises determining that at least part of the data does not match with the encrypted validation key.

[0747] Example 13: The method of any one of Examples 10-12, further comprising analyzing the data to determine if there is a sequential trend or pattern in the data that is unique to a type of surgical procedure that purportedly occurred when the data was generated, wherein determining that the data contains the at least one flaw or error comprises determining that the data does not contain the unique sequential trend or pattern.

[0748] Example 14: The method of any one of Examples 10-13, further comprising, receiving a report of a malicious actor from a medical hub, wherein determining that the data contains the at least one flaw or error comprises determining that the data contains a characteristic consistent with the report of the malicious actor.

[0749] Example 15: The method of any one of Examples 10-14, wherein preventing the at least one flaw or error from being integrated into the larger dataset comprises isolating the at least one flaw or error from the larger dataset and integrating the remainder of the data into the larger dataset.

[0750] Example 16: The method of any one of Examples 10-14, wherein preventing the at least one flaw or error from being integrated into the larger dataset comprises determining that the at least one flaw or error is a result of the data being altered and removing the data from being integrated into the larger dataset.

[0751] Example 17: The method of any one of Examples 10-16, further comprising determining that the data is generated from a validated medical instrument, wherein preventing the at least one flaw or error from being integrated into the larger dataset comprises flagging the at least one flaw or error in the data and identifying the validated medical instrument as containing a systematic flaw or error.

[0752] Example 18: The method of Example 17, wherein identifying the validated medical instrument as containing the systematic flaw or error comprises configuring the validated medical instrument into a controlled situation, running a predefined routine on the validated medical instrument during the controlled situation, and comparing a response by the validated medical instrument during the predefined routine to an expected result.

[0753] Example 19: A computer readable medium comprising no transitory signals and comprising instructions that, when executed by a processor, cause the processor to perform operations. The operations comprise accessing data through a processor of at least one remote medical server, validating the data to determine if the data is validly generated by a medical procedure, determining that the data contains at least one flaw or error, and improving data integrity by preventing the at least one flaw or error from being integrated into a larger dataset associated with the at least one remote medical server.

[0754] Example 20: The computer readable medium of Example 19, wherein the operations further comprise analyzing the data to determine if there is a sequential trend or pattern in the data that is common to surgical procedures, wherein determining that the data contains the at least one flaw or error comprises determining that the data does not contain the sequential trend or pattern.

[0755] Various additional aspects of the subject matter described herein are set out in the following numbered examples:

[0756] Example 1: A surgical instrument comprising an end effector configured to deploy staples

into tissue grasped by the end effector and cut the grasped tissue during a firing stroke. The surgical instrument further comprises a user interface and a control circuit. The control circuit is configured to cause at least one parameter setting associated with the firing stroke to be displayed on the user interface and cause interpreted information relevant to the firing stroke to be displayed concurrently with the at least one parameter setting on the user interface, wherein the interpreted information is based on external data.

[0757] Example 2: The surgical instrument of Example 1, wherein the external data originated with a measurement device that is separate from the surgical instrument.

[0758] Example 3: The surgical instrument of Examples 1 or 2, wherein the external data is transmitted to the surgical instrument through a wireless communication link.

[0759] Example 4: The surgical instrument of any one of Examples 1-3, wherein the interpreted information is updated in real time.

[0760] Example 5: The surgical instrument of any one of Examples 1-3, wherein the interpreted information is updated at a predetermined update rate.

[0761] Example 6: The surgical instrument of any one of Examples 1-5, wherein the interpreted information relates to tissue hemostasis.

[0762] Example 7: The surgical instrument of any one of Examples 1-6, wherein the interpreted information relates to hemostasis of tissue previously treated with the end effector.

[0763] Example 8: The surgical instrument of any one of Examples 1-7, wherein the interpreted information relates to blood pressure of a selected blood vessel.

[0764] Example 9: The surgical instrument of any one of Examples 1-8, wherein the at least one parameter setting comprises a speed setting of the firing stroke.

[0765] Example 10: The surgical instrument of any one of Examples 1-9, wherein the at least one parameter setting is a wait-time setting before beginning the firing stroke.

[0766] Example 11: A surgical instrument comprising an end effector configured to deploy staples into tissue grasped by the end effector and cut the grasped tissue during a firing stroke. The surgical instrument further comprises a user interface and a control circuit. The control circuit is configured to cause at least one parameter setting associated with the firing stroke to be displayed on the user interface and cause interpreted information relevant to the firing stroke to be displayed concurrently with the at least one parameter setting on the user interface, wherein the interpreted information is based on imaging data.

[0767] Example 12: The surgical instrument of Example 11, wherein the interpreted information is updated in real time.

[0768] Example 13: The surgical instrument of Example 11, wherein the interpreted information is updated at a predetermined update rate.

[0769] Example 14: The surgical instrument of any one of Examples 11-13, wherein the interpreted information relates to tissue hemostasis.

[0770] Example 15: The surgical instrument of any one of Examples 11-14, wherein the interpreted information relates to hemostasis of tissue previously treated with the end effector.

[0771] Example 16: The surgical instrument of any one of Examples 11-15, wherein the interpreted information relates to blood pressure of a selected blood vessel.

[0772] Example 17: The surgical instrument of any one of Examples 11-16, wherein the at least one parameter setting comprises a speed setting of the firing stroke.

[0773] Example 18: The surgical instrument of any one of Examples 11-17, wherein the at least one parameter setting is a wait-time setting before beginning the firing stroke.

[0774] Example 19: A surgical instrument comprising an end effector configured to perform a function to treat tissue grasped by the end effector. The surgical instrument further comprises a user interface and a control circuit. The control circuit is configured to cause at least one parameter setting associated with the function to be displayed on the user interface and cause interpreted information relevant to the function to be displayed concurrently with the at least one parameter

setting on the user interface, wherein the interpreted information is based on external data.

[0775] Example 20: The surgical instrument of Example 19, wherein the external data originated with a measurement device that is separate from the surgical instrument.

[0776] Various additional aspects of the subject matter described herein are set out in the following numbered examples:

[0777] Example 1: A surgical feedback system is disclosed. The surgical feedback system comprises a surgical instrument, a data source, and a surgical hub configured to communicably couple to the data source and the surgical instrument. The surgical hub comprises a control circuit, wherein the control circuit is configured to receive an input from the data source, analyze the received data against a stored set of data to optimize an outcome of a surgical procedure, and communicate a recommendation based on the analyzed data.

[0778] Example 2: The surgical feedback system of Example 1, wherein the control circuit is further configured to adjust an operating parameter of the surgical instrument based on the communicated recommendation.

[0779] Example 3: The surgical feedback system of any one of Examples 1 and 2, wherein the control circuit is further configured to update a control program of the surgical instrument based on the communicated recommendation.

[0780] Example 4: The surgical feedback system of any one of Examples 1-3, wherein the control circuit is configured to communicate the recommendation by suggesting a procedural modification to a user.

[0781] Example 5: The surgical feedback system of any one of Examples 1-4, wherein the control circuit is configured to analyze the received data and the stored set of data using a supervised learning technique.

[0782] Example 6: The surgical feedback system of any one of Examples 1-5, wherein the control circuit is configured to analyze the received data and the stored set of data using an unsupervised learning technique.

[0783] Example 7: The surgical feedback system of any one of Examples 1-6, wherein the control circuit is configured to analyze the received data within a local network.

[0784] Example 8: The surgical feedback system of any one of Examples 1-7, wherein the control circuit is configured to analyze the received data by exporting the data to a remote location for compilation.

[0785] Example 9: The surgical feedback system of any one of Examples 1-8, wherein the stored set of data comprises data gathered during previous surgical procedures.

[0786] Example 10: The surgical feedback system of any one of Examples 1-9, wherein the data source comprises data specific to a particular patient.

[0787] Example 11: The surgical feedback system of any one of Examples 1-10, wherein the control circuit is further configured to update the stored set of data with the communicated recommendation.

[0788] Example 12: A surgical feedback system is disclosed. The surgical feedback system comprises a data source and a surgical hub comprising a control circuit. The control circuit is configured to receive an input from the data source, analyze the received data against a stored set of data to optimize an outcome of a surgical procedure, wherein the stored set of data comprises data collected during previous surgical procedures, and communicate a recommendation based on the analyzed data.

[0789] Example 13: The surgical feedback system of Example 12, wherein the communicated recommendation is based on a particular surgical instrument positioned within a surgical site.

[0790] Example 14: The surgical feedback system of any one of Examples 12 and 13, wherein the control circuit is further configured to adjust an operating parameter of the surgical instrument based on the communicated recommendation.

[0791] Example 15: The surgical feedback system of any one of Examples 12-14, wherein the

control circuit is configured to analyze the received data against the stored set of data using a supervised learning technique.

[0792] Example 16: The surgical feedback system of any one of Examples 12-15, wherein the control circuit is configured to analyze the received data against the stored set of data using an unsupervised learning technique.

[0793] Example 17: The surgical feedback system of any one of Examples 12-16, wherein the control circuit is configured to communicate the recommendation by suggesting a procedural modification to a user.

[0794] Example 18: A surgical feedback system is disclosed. The surgical feedback system comprises a data source and a surgical instrument comprising a control circuit. The control circuit is configured to receive an input from the data source, analyze the received data against a stored set of data to optimize an outcome of a surgical procedure, wherein the stored set of data comprises data collected during previous surgical procedures, and determine a recommendation based on the analyzed data.

[0795] Example 19: The surgical feedback system of Example 18, wherein the control circuit is further configured to adjust an operating parameter of the surgical instrument based on the determined recommendation.

[0796] Example 20: The surgical feedback system of any one of Examples 18 and 19, wherein the control circuit is further configured to update the stored set of data with the determined recommendation.

[0797] Various additional aspects of the subject matter described herein are set out in the following numbered examples:

[0798] Example 1: A system for adjusting a parameter of a medical instrument in a specific context of a patient in which the medical instrument is to be used on the patient, the system comprising a medical hub and the medical instrument communicatively coupled to the medical hub. The medical hub is configured to access a first contextual dataset representing a first circumstance pertaining to the specific context in which the medical instrument is to be used on the patient, access a second contextual dataset representing a second circumstance pertaining to the specific context in which the medical instrument is to be used on the patient, arrange the first and second contextual datasets in a hierarchy of priority, such that the first contextual dataset possesses a higher priority than the second contextual dataset, determine that at least a first portion of the first contextual dataset is relevant to adjust the parameter of the medical instrument, determine that at least a second portion of the second contextual dataset is relevant to adjust the parameter of the medical instrument, resolve a difference between the first portion and the second portion as it pertains to adjusting the parameter, and program the medical instrument by adjusting the parameter of the medical instrument in accordance with the resolved difference. The medical instrument is configured to be used on the patient utilizing the adjusted parameter such that performance of the medical instrument on the patient in the specific context is more effective with the adjusted parameter than without the adjusted parameter.

[0799] Example 2: The system of Example 1, wherein resolving the difference between the first portion and the second portion comprises determining that the first portion contradicts the second portion as it relates to adjusting the parameter and adjusting the parameter only according to the first portion.

[0800] Example 3: The system of Example 1, wherein resolving the difference between the first portion and the second portion comprises determining that the first portion contradicts the second portion as it relates to adjusting the parameter, determining that an exception in the first contextual dataset indicates to adhere to the second contextual dataset as it relates to the parameter, and adjusting the parameter only according to the second portion.

[0801] Example 4: The system of Example 1, wherein resolving the difference between the first portion and the second portion comprises determining that the first portion can be modified by the

second portion as it relates to adjusting the parameter and adjusting the parameter according to the first portion and the second portion.

[0802] Example 5: The system of any one of Examples 1-4, wherein the first or the second contextual dataset is derived from medical data of the patient, updated settings of the medical instrument, or information about a condition of the patient that needs treatment.

[0803] Example 6: The system of any one of Examples 1-4, wherein one of the first and the second contextual dataset is derived from non-instrument-specific contextual cues that relate to operation of any instrument, but that are not specific to any particular type of instrument, and wherein the other of the first and the second contextual dataset is derived from instrument-specific contextual cues that relate to specific operation of the medical instrument.

[0804] Example 7: The system of any one of Examples 1-4, wherein the first or the second contextual dataset is derived from medical contextual cues associated with medical complications known to occur in the specific context the medical instrument is used on the patient.

[0805] Example 8: The system of any one of Examples 1-4, wherein the first or the second contextual dataset is derived from physiologic cues of the patient comprising time since the patient last ate, fasting blood glucose level, blood pressure, macro tissue tension, tissue fluid levels, and tissue oxygenation.

[0806] Example 9: The system of any one of Examples 1-8, wherein the specific context comprises a medical procedure in which the medical instrument assists in, and wherein the first or the second contextual dataset is derived from procedure-specific contextual cues comprising a time of day the medical procedure is expected to occur, an indication of whether the medical procedure is an emergency or a planned surgery, a time duration of the medical procedure, a type of medical procedure, and an indication of whether the medical procedure is a reoperative or original procedure.

[0807] Example 10: The system of any one of Examples 1-4 or 8, wherein the first or the second contextual dataset is derived from surgeon-specific contextual cues comprising an indication of whether a surgeon using the medical instrument on the patient is a specialist or a general practitioner, a skill level of the surgeon, a number of procedures already performed that day by the surgeon, and an expected duration of a medical procedure.

[0808] Example 11: A method of a system for adjusting a parameter of a medical instrument in a specific context of a patient in which the medical instrument is to be used on the patient, the system comprising the medical instrument and a medical hub communicatively coupled to the medical instrument. The method comprises accessing, by the medical hub, a first contextual dataset representing a first circumstance pertaining to the specific context in which the medical instrument is to be used on the patient, accessing, by the medical hub, a second contextual dataset representing a second circumstance pertaining to the specific context in which the medical instrument is to be used on the patient, arranging, by the medical hub, the first and second contextual datasets in a hierarchy of priority, such that the first contextual dataset possesses a higher priority than the second contextual dataset, determining, by the medical hub, that at least a first portion of the first contextual dataset is relevant to adjust the parameter of the medical instrument, determining, by the medical hub, that at least a second portion of the second contextual dataset is relevant to adjust the parameter of the medical instrument, resolving, by the medical hub, a difference between the first portion and the second portion as it pertains to adjusting the parameter, and programing, by the medical hub, the medical instrument by adjusting the parameter of the medical instrument in accordance with the resolved difference. The medical instrument is configured to be used on the patient utilizing the adjusted parameter such that performance of the medical instrument on the patient in the specific context is more effective with the adjusted parameter than without the adjusted parameter.

[0809] Example 12: The method of Example 11, wherein resolving the difference between the first portion and the second portion comprises determining that the first portion contradicts the second

portion as it relates to adjusting the parameter and adjusting the parameter only according to the first portion.

[0810] Example 13: The method of Example 11, wherein resolving the difference between the first portion and the second portion comprises determining that the first portion contradicts the second portion as it relates to adjusting the parameter, determining that an exception in the first contextual dataset indicates to adhere to the second contextual dataset as it relates to the parameter, and adjusting the parameter only according to the second portion.

[0811] Example 14: The method of Example 11, wherein resolving the difference between the first portion and the second portion comprises determining that the first portion can be modified by the second portion as it relates to adjusting the parameter and adjusting the parameter according to the first portion and the second portion.

[0812] Example 15: The method of any one of Examples 11-14, wherein the first or the second contextual dataset is derived from medical data of the patient, updated settings of the medical instrument, or information about a condition of the patient that needs treatment.

[0813] Example 16: The method of any one of Examples 11-14, wherein one of the first and the second contextual dataset is derived from non-instrument-specific contextual cues that relate to operation of any instrument, but that are not specific to any particular type of instrument, and wherein the other of the first and the second contextual dataset is derived from instrument-specific contextual cues that relate to specific operation of the medical instrument.

[0814] Example 17: The method of any one of Examples 11-14, wherein the first or the second contextual dataset is derived from medical contextual cues associated with medical complications known to occur in the specific context the medical instrument is used on the patient.

[0815] Example 18: The method of any one of Examples 11-14, wherein the first or the second contextual dataset is derived from physiologic cues of the patient comprising time since the patient last ate, fasting blood glucose level, blood pressure, macro tissue tension, tissue fluid levels, and tissue oxygenation.

[0816] Example 19: The method of any one of Examples 11-18, wherein the specific context comprises a medical procedure in which the medical instrument assists in, and the first or the second contextual dataset is derived from procedure-specific contextual cues comprising a time of day the medical procedure is expected to occur, an indication of whether the medical procedure is an emergency or a planned surgery, a time duration of the medical procedure, a type of medical procedure, and an indication of whether the medical procedure is a reoperative or original procedure.

[0817] Example 20: A computer readable medium having no transitory signals and comprising instructions that, when executed by a processor, cause the processor to perform operations. The operations comprise accessing a first contextual dataset representing a first circumstance pertaining to a specific context in which a medical instrument is to be used on a patient, accessing a second contextual dataset representing a second circumstance pertaining to the specific context in which the medical instrument is to be used on the patient, arranging the first and second contextual datasets in a hierarchy of priority, such that the first contextual dataset possesses a higher priority than the second contextual dataset, determining that at least a first portion of the first contextual dataset is relevant to adjust a parameter of the medical instrument, determining that at least a second portion of the second contextual dataset is relevant to adjust the parameter of the medical instrument, resolving a difference between the first portion and the second portion as it pertains to adjusting the parameter, and programing the medical instrument by adjusting the parameter of the medical instrument in accordance with the resolved difference. The medical instrument is configured to be used on the patient utilizing the adjusted parameter such that performance of the medical instrument on the patient in the specific context is more effective with the adjusted parameter than without the adjusted parameter.

[0818] Various additional aspects of the subject matter described herein are set out in the following

numbered examples:

[0819] Example 1: A surgical system for use in a surgical procedure, wherein the surgical system comprises a modular device, at least one data source, and a surgical hub configured to communicably couple to the at least one data source and the modular device. The surgical hub comprises a control circuit configured to receive data from the at least one data source, wherein the data is determinative of a progress status the surgical procedure. The control circuit is further configured to adjust a response to a sensed parameter based on the progress status.

[0820] Example 2: The surgical system of Example 1, wherein the at least one data source comprises a patient monitoring device.

[0821] Example 3: The surgical system of Example 1 or 2, wherein the at least one data source comprises a surgical staff detection device.

[0822] Example 4: The surgical system of any one of Examples 1-3, wherein the progress status comprises a preoperative status while the surgical procedure is in preoperative steps.

[0823] Example 5: The surgical system of any one of Examples 1-4, wherein the progress status comprises an intraoperative status while the surgical procedure is in intraoperative steps.

[0824] Example 6: The surgical system of any one of Examples 1-5, wherein the sensed parameter comprises a fault detection parameter.

[0825] Example 7: The surgical system of any one of Examples 1-6, wherein the sensed parameter comprises a surgeon detection parameter of the modular device.

[0826] Example 8: The surgical system of any one of Examples 1-7, wherein the sensed parameter comprises a security-threat detection parameter.

[0827] Example 9: A surgical hub for use in a surgical procedure, wherein the surgical hub is configured to communicably couple to at least one data source. The surgical hub comprises a control circuit configured to receive data from the at least one data source, wherein the data is determinative of a progress status the surgical procedure. The control circuit is further configured to adjust a response to a sensed parameter based on the progress status.

[0828] Example 10: The surgical hub of Example 9, wherein the at least one data source comprises a patient monitoring device.

[0829] Example 11: The surgical hub of Example 9 or 10, wherein the at least one data source comprises a surgical staff detection device.

[0830] Example 12: The surgical hub of any one of Examples 9-11, wherein the progress status comprises a preoperative status while the surgical procedure is in preoperative steps.

[0831] Example 13: The surgical hub of any one of Examples 9-12, wherein the progress status comprises an intraoperative status while the surgical procedure is in intraoperative steps.

[0832] Example 14: The surgical hub of any one of Examples 9-13, wherein the sensed parameter comprises a modular device fault-detection parameter.

[0833] Example 15: The surgical hub of any one of Examples 9-14, wherein the sensed parameter comprises a surgeon detection parameter.

[0834] Example 16: The surgical hub of any one of Examples 9-15, wherein the sensed parameter comprises a security-threat detection parameter.

[0835] Example 17: A surgical hub for use in a surgical procedure, wherein the surgical hub is configured to communicably couple to at least one data source. The surgical hub comprises a control circuit configured to receive data from the at least one data source, wherein the data is determinative of a situational parameter of the surgical procedure. The control circuit is further configured to adjust a response to a sensed parameter based on the situational parameter.

[0836] Example 18: The surgical hub of Example 17, wherein the sensed parameter comprises a modular device fault-detection parameter.

[0837] Example 19: The surgical hub of Examples 17 or 18, wherein the sensed parameter comprises a surgeon detection parameter.

[0838] Example 20: The surgical hub of any one of Examples 17-19, wherein the sensed parameter

comprises a security-threat detection parameter.

[0839] Various additional aspects of the subject matter described herein are set out in the following numbered examples:

[0840] Example 1: A surgical instrument for use with a surgical hub in a surgical procedure, wherein the surgical instrument comprises a surgical end effector, a communication module configured to wirelessly pair the surgical instrument to the surgical hub, and a control circuit. The control circuit is configured to detect a first security violation, cause the surgical instrument to generate a first response to the first security violation, store a record of the first security violation, detect a second security violation, and cause the surgical instrument to generate a second response to the second security violation, wherein the second response is escalated from the first response.

[0841] Example 2: The surgical instrument of Example 1, wherein the second security violation is greater in severity than the first security violation.

[0842] Example 3: The surgical instrument of Examples 1 or 2, wherein the first response comprises issuance of a warning.

[0843] Example 4: The surgical instrument of any one of Examples 1-3, wherein the second response comprises deactivating the communication module.

[0844] Example 5: The surgical instrument of any one of Examples 1-4, wherein the second response comprises activating an autonomous usage mode of the surgical instrument.

[0845] Example 6: A surgical instrument for use with a surgical hub in a surgical procedure, wherein the surgical instrument comprises a surgical end effector, a communication module configured to wirelessly pair the surgical instrument to the surgical hub, and a control circuit. The control circuit is configured to detect a first unauthorized interaction, cause the surgical instrument to generate a first response to the first unauthorized interaction, store a record of the first unauthorized interaction, detect a second unauthorized interaction, and cause the surgical instrument to generate a second response to the second unauthorized interaction, wherein the second response is escalated from the first response.

[0846] Example 7: The surgical instrument of Example 6, wherein the second unauthorized interaction is greater in severity than the first unauthorized interaction.

[0847] Example 8: The surgical instrument of Examples 6 or 7, wherein the first response comprises issuance of a warning.

[0848] Example 9: The surgical instrument of any one of Examples 6-8, wherein the second response comprises deactivating the communication module.

[0849] Example 10: The surgical instrument of any one of Examples 6-9, wherein the second response comprises activating an autonomous usage mode of the surgical instrument.

[0850] Example 11: A surgical instrument for use with a surgical hub in a surgical procedure, wherein the surgical instrument comprises a surgical end effector, a communication module configured to wirelessly pair the surgical instrument to the surgical hub, and a control circuit. The control circuit is configured to detect a first unauthorized activation of the surgical instrument, cause the surgical instrument to generate a first response to the first unauthorized activation of the surgical instrument, store a record of the first unauthorized activation of the surgical instrument, detect a second unauthorized activation of the surgical instrument, and cause the surgical instrument to generate a second response to the second unauthorized activation of the surgical instrument, wherein the second response is escalated from the first response.

[0851] Example 12: The surgical instrument of Example 11, wherein the second unauthorized interaction is greater in severity than the first unauthorized interaction.

[0852] Example 13: The surgical instrument of Examples 11 or 12, wherein the first response comprises issuance of a warning.

[0853] Example 14: The surgical instrument of any one of Examples 11-13, wherein the second response comprises deactivating the communication module.

[0854] Example 15: The surgical instrument of any one of Examples 11-14, wherein the second

response comprises activating an autonomous usage mode of the surgical instrument.

[0855] Various additional aspects of the subject matter described herein are set out in the following numbered examples:

[0856] Example 1: A surgical instrument comprising an end effector configured to deploy staples into tissue grasped by the end effector and cut the grasped tissue during a firing stroke. The surgical instrument further comprises a user interface and a control circuit. The control circuit is configured to cause at least one parameter setting associated with the firing stroke to be displayed on the user interface, cause interpreted information relevant to the firing stroke to be displayed concurrently with the at least one parameter setting on the user interface, wherein the interpreted information is based on external data, recommend an adjustment of the at least one parameter setting through the user interface, wherein the recommended adjustment is based on the interpreted information.

[0857] Example 2: The surgical instrument of Example 1, wherein the external data originated with a measurement device that is separate from the surgical instrument.

[0858] Example 3: The surgical instrument of Examples 1 or 2, wherein the external data is transmitted to the surgical instrument through a wireless communication link.

[0859] Example 4: The surgical instrument of any one of Examples 1-3, wherein the interpreted information is updated in real time.

[0860] Example 5: The surgical instrument of any one of Examples 1-3, wherein the interpreted information is updated at a predetermined update rate.

[0861] Example 6: The surgical instrument of any one of Examples 1-5, wherein the interpreted information relates to tissue hemostasis.

[0862] Example 7: The surgical instrument of any one of Examples 1-6, wherein the interpreted information relates to hemostasis of tissue previously treated with the end effector.

[0863] Example 8: The surgical instrument of any one of Examples 1-7, wherein the at least one parameter setting comprises a speed setting of the firing stroke.

[0864] Example 9: The surgical instrument of any one of Examples 1-8, wherein the at least one parameter setting comprises a wait-time before beginning the firing stroke.

[0865] Example 10: A surgical instrument comprising an end effector configured to perform a function to treat tissue grasped by the end effector. The surgical instrument further comprises a user interface and a control circuit. The control circuit is configured to cause at least one parameter setting associated with the function to be displayed on the user interface, cause interpreted information relevant to the function to be displayed concurrently with the at least one parameter setting on the user interface, wherein the interpreted information is based on external data, and recommend an adjustment of the at least one parameter setting through the user interface, wherein the recommended adjustment is based on the interpreted information.

[0866] Example 11: The surgical instrument of Example 10, wherein the interpreted information is updated in real time.

[0867] Example 12: The surgical instrument of Example 10, wherein the interpreted information is updated at a predetermined update rate.

[0868] Example 13: The surgical instrument of any one of Examples 10-12, wherein the interpreted information relates to tissue hemostasis.

[0869] Example 14: The surgical instrument of any one of Examples 10-13, wherein the interpreted information relates to hemostasis of tissue previously treated with the end effector.

[0870] Example 15: The surgical instrument of any one of Examples 10-14, wherein the interpreted information relates to blood pressure of a selected blood vessel.

[0871] Example 16: The surgical instrument of any one of Examples 10-15, wherein the at least one parameter setting comprises a speed setting of the firing stroke.

[0872] Example 17: The surgical instrument of any one of Examples 10-16, wherein the at least one parameter setting comprises a wait-time before beginning the firing stroke.

[0873] Example 18: A surgical instrument for use with a medical imaging device and a surgical hub

including a visualization module in communication with the medical imaging device. The surgical instrument comprises an end effector configured to perform a function to treat tissue grasped by the end effector. The surgical instrument further comprises a user interface and a control circuit. The control circuit is configured to receive an input from the surgical hub indicative of a position of a critical structure with respect to a current field of view of the medical imaging device as determined by the visualization module and cause the user interface to recommend an adjustment that changes the position of the critical structure with respect to the current field of view of the medical imaging based on the received input.

[0874] Example 19: The surgical instrument of Example 18, wherein the critical structure is the end effector.

[0875] Example 20: The surgical instrument of Examples 18 or 19, wherein the adjustment comprises selecting an auto-centering mode.

[0876] Various additional aspects of the subject matter described herein are set out in the following numbered examples:

[0877] Example 1: A system for automatically fusing data from a medical procedure. The system comprises a medical hub comprising at least one processor and at least one memory, and a remote server communicatively coupled to the medical hub. The at least one processor is configured to access a first dataset comprising data sampled at a first data sampling rate recorded during a sampling time period, access a second dataset comprising data sampled at a second data sampling rate that is slower than the first data sampling rate and is recorded during the sampling time period, scale the second dataset to match the first data sampling rate, fuse the first dataset and the second dataset into a composite dataset, align the first dataset and the second dataset in the composite dataset, such that data from both the first dataset and the second dataset is sequentially ordered in the composite dataset in an order in which the data was recorded, cause display of the composite dataset, generate a graphical overlay on top of the display of the composite dataset that provides an interpretation of the composite dataset, and transmit the composite dataset to the remote server.

[0878] Example 2: The system of Example 1, wherein the first or the second dataset comprises one or more error data points, and wherein the processor is further configured to smooth out the one or more error data points.

[0879] Example 3: The system of Examples 1 or 2, wherein the graphical overlay comprises a horizontal axis and a vertical axis, and wherein the composite dataset is displayed in a graph form according to the horizontal and vertical axes.

[0880] Example 4: The system of Example 3, wherein the graphical overlay further comprises visual boundaries that indicate visual limits of the composite dataset.

[0881] Example 5: The system of Examples 1 or 4, wherein the graphical overlay comprises a horizontal axis, a first vertical axis and a second vertical axis, wherein the first dataset comprises data related to a first measurement that is expressed by the first vertical axis over the horizontal axis, and wherein the second dataset comprises data related to a second measurement different than the first measurement that is expressed by the second vertical axis over the horizontal axis.

[0882] Example 6: The system of any one of Examples 1-5, wherein the processor is further configured to access first metadata associated with the first dataset and recorded during the sampling time period, access second metadata associated with the second dataset and recorded during the sampling time period, transmit the first and second metadata to an offsite repository, and store the first and second datasets in the memory of the system.

[0883] Example 7: The system of any one of Examples 1-6, wherein the first dataset is recorded in a first format, wherein the second dataset is recorded in a second format different from the first format, and wherein the processor is further configured to convert the first and second datasets into a common format.

[0884] Example 8: The system of any one of Examples 1-7, wherein the processor is further configured to determine duplicate data from the first and the second datasets and remove all copies

of the duplicate data before fusing the first and the second datasets into the composite dataset.

[0885] Example 9: The system of any one of Examples 1-8, wherein the first dataset is generated by a first device having a first internal clock, the second dataset is generated by a second device having a second internal clock, and the first dataset and the second dataset do not have a common time period due to the first and the second datasets being recorded by their respective internal clocks. The processor is further configured to access a synchronizer signal between the first and second device and align the first dataset and the second dataset using the synchronizer signal to interrelate the first dataset and the second dataset.

[0886] Example 10: The system of any one of Examples 1-9, wherein the processor is further configured to access first metadata associated with the first dataset and recorded during the sampling time period, access second metadata associated with the second dataset and recorded during the sampling time period, transform the first dataset into first related aspect data using the first metadata, and transform the second dataset into second related aspect data using the second metadata, wherein fusing the first dataset and the second dataset into the composite dataset comprises fusing the first related aspect data with the second related aspect data.

[0887] Example 11: The system of any one of Examples 1-10, wherein the remote server is configured to access updated parameters from one or more other medical hubs communicatively coupled to the remote server and propagate the updated parameters to the medical hub, wherein the medical hub is configured to adjust the composite dataset according to the updated parameters.

[0888] Example 12: A method of a system for automatically fusing data from a medical procedure. the system comprising a medical hub comprising at least one processor and at least one memory. The method comprises accessing a first dataset comprising data sampled at a first data sampling rate recorded during a sampling time period, accessing a second dataset comprising data sampled at a second data sampling rate that is slower than the first data sampling rate and is recorded during the sampling time period, scaling the second dataset to match the first data sampling rate, fusing the first dataset and the second dataset into a composite dataset, aligning the first dataset and the second dataset in the composite dataset, such that data from both the first dataset and the second dataset is sequentially ordered in the composite dataset in an order in which the data was recorded, causing display of the composite dataset, generating a graphical overlay on top of the display of the composite dataset that provides an interpretation of the composite dataset, and transmitting the composite dataset to a remote server.

[0889] Example 13: The method of Example 12, wherein the first or the second dataset comprises one or more error data points, and wherein the method further comprises smoothing out the error data points.

[0890] Example 14: The method of Examples 12 or 13, wherein the graphical overlay comprises a horizontal axis, a first vertical axis and a second vertical axis, wherein the first dataset comprises data related to a first measurement that is expressed by the first vertical axis over the horizontal axis, and wherein the second dataset comprises data related to a second measurement different than the first measurement that is expressed by the second vertical axis over the horizontal axis.

[0891] Example 15: The method of any one of Examples 12-14, wherein the first dataset is recorded in a first format, wherein the second dataset is recorded in a second format different from the first format, and wherein the method further comprises converting the first and second datasets into a common format.

[0892] Example 16: The method of any one of Examples 12-15, further comprising determining duplicate data from the first and the second datasets and removing all copies of the duplicate data before fusing the first and the second datasets into the composite dataset.

[0893] Example 17: The method of any one of Examples 12-16, wherein the first dataset is generated by a first device having a first internal clock, the second dataset is generated by a second device having a second internal clock, and the first dataset and the second dataset do not have a common time period due to the first and the second datasets being recorded by their respective

internal clocks. The method further comprises accessing a synchronizer signal between the first and second device and aligning the first dataset and the second dataset using the synchronizer signal to interrelate the first dataset and the second dataset.

[0894] Example 18: The method of any one of Examples 12-17, further comprising accessing first metadata associated with the first dataset and recorded during the sampling time period, accessing second metadata associated with the second dataset and recorded during the sampling time period, transforming the first dataset into first related aspect data using the first metadata, and transforming the second dataset into second related aspect data using the second metadata, wherein fusing the first dataset and the second dataset into the composite dataset comprises fusing the first related aspect data with the second related aspect data.

[0895] Example 19: The method of any one of Examples 12-18, further comprising accessing, by the remote server, updated parameters from one or more other medical hubs communicatively coupled to the remote server, propagating, by the remote server, the updated parameters to the medical hub, and adjusting, by the medical hub, the composite dataset according to the updated parameters.

[0896] Example 20: A computer readable medium comprising no transitory signals and comprising instructions that, when executed by a processor, cause the processor to perform operations. The operations comprise accessing a first dataset comprising data sampled at a first data sampling rate recorded during a sampling time period, accessing a second dataset comprising data sampled at a second data sampling rate that is slower than the first data sampling rate and is recorded during the sampling time period, scaling the second dataset to match the first data sampling rate, fusing the first dataset and the second dataset into a composite dataset, aligning the first dataset and the second dataset in the composite dataset, such that data from both the first dataset and the second dataset is sequentially ordered in the composite dataset in an order in which the data was recorded, causing display of the composite dataset, generating a graphical overlay on top of the display of the composite dataset that provides an interpretation of the composite dataset, and transmitting the composite dataset to a remote server.

[0897] Various additional aspects of the subject matter described herein are set out in the following numbered examples:

[0898] Example 1: A surgical system comprising: a monopolar return pad; and a surgical hub communicatively coupled to the monopolar return pad, the surgical hub comprising a control circuit configured to determine a presence and a position of a patient on the monopolar return pad according to data received from the monopolar return pad.

[0899] Example 2: The surgical system of Example 1, wherein the control circuit is configured to control a visualization means according to the determined presence and/or the determined position of the patient on the monopolar return pad.

[0900] Example 3: The surgical system of any one of Example 1 or 2, wherein the control circuit is further configured to: control an electrosurgical generator to provide a varying range of electrosurgical frequencies to the patient; and monitor a response to the varying range of electrosurgical frequencies by the monopolar return pad to determine the position of the patient.

[0901] Example 4: The surgical system of any one of Examples 1-3, wherein the control circuit is further configured to: determine a maximum power of a generator according to capacitive coupling variations of the monopolar return pad; and adjust a power of the generator accordingly.

[0902] Example 5: The surgical system of any one of Examples 1-4, wherein the control circuit is further configured to monitor radiative resistance of the monopolar return pad to determine the presence or the position of the patient on the monopolar return pad.

[0903] Example 6: The surgical system of any one of Examples 1-5, wherein the control circuit is further configured to utilize situational awareness in combination with the monitored radiative resistance of the monopolar return pad to determine the presence or the position of the patient, by comparing the monitored radiative resistance to previous radiative resistance data obtained in

similarly situated events of other patients on similarly situated monopolar return pads.

[0904] Example 7: The surgical system of any one of Examples 1-6, wherein the control circuit is further configured to monitor parasitic loading of the monopolar return pad to determine the presence or the position of the patient on the monopolar return pad.

[0905] Example 8: The surgical system of Example 7, wherein the control circuit is further configured to utilize situational awareness in combination with the monitored parasitic loading of the monopolar return pad to determine the presence or the position of the patient, by comparing the monitored parasitic loading to previous parasitic loading data obtained in similarly situated events of other patients on similarly situated monopolar return pads.

[0906] Example 9: The surgical system of any one of Examples 1-8, further comprising a monopolar surgical device configured to stimulate a nerve using RF energy at a surgical site of the patient, wherein the control circuit is further configured to monitor patient movement based on the nerve stimulation to determine the presence or the position of the patient on the monopolar return pad.

[0907] Example 10: A surgical system comprising: an electrosurgical instrument; a generator coupled to the electrosurgical instrument; and a surgical hub communicatively coupled to the generator, the surgical hub comprising a control circuit configured to modulate a nerve detection waveform and/or power supplied by the generator to the electrosurgical instrument based on situational awareness of the electrosurgical instrument and/or the generator.

[0908] Example 11: The surgical system of Example 10, wherein the situational awareness is based on a surgery type, an anatomic location, an activation state of the electrosurgical instrument, previous detections of nerves due to previous signals at a surgical site, continuity of a return pad, and/or proximity to critical structures at the surgical site.

[0909] Example 12: The surgical system of any one of Examples 10 or 11, wherein: the situational awareness comprises knowledge of previous nerve stimulation measurements; and the control circuit is configured to adjust the nerve detection waveform or an amplitude of the generator as the electrosurgical instrument approaches or moves away from a detected nerve.

[0910] Example 13: The surgical system of any one of Examples 10-12, wherein: the situational awareness comprises knowledge of a surgery type of a surgery being performed and/or an anatomic location of the surgery; and the control circuit is configured to adjust the nerve detection waveform accordingly.

[0911] Example 14: The surgical system of any one of Examples 10-13, wherein the control circuit is configured to adjust the nerve detection waveform according to a power level of the electrosurgical instrument.

[0912] Example 15: A surgical system comprising: a monopolar return pad; and a surgical hub communicatively coupled to the monopolar return pad; and a monopolar surgical instrument communicatively coupled to the surgical hub and configured to supply energy to a patient on the monopolar return pad; the surgical hub comprising a compensation circuit configured to adjust power to the monopolar surgical instrument to maintain a peak applied power at the monopolar surgical instrument while the patient is on the monopolar return pad.

[0913] Example 16: The surgical system of Example 15, wherein the compensation circuit comprises a plurality of binary compensation relays.

[0914] Example 17: The surgical system of Example 16, wherein adjusting power to the monopolar surgical instrument comprises: measuring a power level of the monopolar surgical instrument; incrementing the power supplied to the monopolar surgical instrument using the plurality of compensation relays by one power unit; measuring the power level of the monopolar surgical instrument after the power is incremented; and comparing the power level before the power was incremented to the power level after the power was incremented.

[0915] Example 18: The surgical system of Example 17, wherein adjusting power to the monopolar surgical instrument further comprises: determining that the power level before the power was

incremented is higher than the power level after the power was incremented; and maintaining the power level accordingly.

[0916] Example 19: The surgical system of any one of Examples 15-18, wherein the control circuit is further configured to: determine a presence and a position of a patient on the monopolar return pad according to data received from the monopolar return pad; and automatically halt power supplied to the surgical instrument after it is determined that the patient is out of position or off the monopolar return pad.

[0917] Example 20: The surgical system of Example 19, wherein the control circuit is further configured to utilize situational awareness to determine that the patient is out of position or off the monopolar return pad.

[0918] Various additional aspects of the subject matter described herein are set out in the following numbered examples:

[0919] Example 1: A surgical system comprising a surgical instrument comprising an end effector, wherein the end effector is configured to perform an end effector function and a control circuit configured to control the end effector function and automatically adapt the control of the end effector function over time and limit the automatic adaptation of the control of the end effector function.

[0920] Example 2: The surgical system of Example 1, wherein the control circuit is further configured to automatically adapt the control of the end effector function using machine learning.

[0921] Example 3: The surgical system of Examples 1 or 2, wherein the automatic adaptation is dependent on a sensed parameter in the surgical instrument.

[0922] Example 4: The surgical system of Example 3, wherein the sensed parameter comprises a set of previously-sensed parameters from previous uses of the surgical instrument.

[0923] Example 5: The surgical system of Example 4, wherein the set of previously-sensed parameters comprises parameters sensed during uses of the surgical instrument by a specific user.

[0924] Example 6: The surgical system of Examples 4 or 5, wherein the set of previously-sensed parameters comprises parameters sensed during uses of the surgical instrument in a specific location.

[0925] Example 7: The surgical system of Examples 1-6, wherein the control circuit is further configured to limit the automatic adaptation of the control of the end effector function to a specific range of adjustments.

[0926] Example 8: The surgical system of Example 7, wherein the specific range of adjustments is predefined.

[0927] Example 9: The surgical system of Examples 7 or 8, wherein the specific range of adjustments is manually adjustable.

[0928] Example 10: The surgical system of any one of Examples 7-9, wherein the specific range of adjustments is automatically adjusted by the control circuit based on machine learning.

[0929] Example 11: The surgical system of any one of Examples 1-10, wherein the control circuit is further configured to limit the automatic adaptation of the control of the end effector function to a maximum threshold adjustment.

[0930] Example 12: The surgical system of any one of Examples 1-11, wherein limiting the automatic adaptation of the control of the end effector function is based on a per-use basis.

[0931] Example 13: The surgical system of any one of Examples 1-12, wherein limiting the automatic adaptation of the control of the end effector function is based on a specific user.

[0932] Example 14: The surgical system of any one of Examples 1-13, wherein the limiting the automatic adaptation of the control of the end effector function is based on a specific location of the surgical instrument.

[0933] Example 15: A surgical system comprising a surgical instrument comprising an end effector and a control circuit configured to control a parameter of the end effector, automatically adjust the control of the parameter, and limit the automatic adjustment of the control of the parameter to an

adjustability window.

[0934] Example 16: The surgical system of Example 15, wherein the control circuit is configured to automatically adjust the control of the parameter using machine learning.

[0935] Example 17: The surgical system of Examples 15 or 16, wherein the adjustability window is manually selectable by a clinician.

[0936] Example 18: The surgical system of any one of Examples 15-17, wherein the adjustability window is automatically selected based on machine learning.

[0937] Example 19: A surgical system comprising a surgical instrument comprising an end effector and a control circuit configured to receive information about a sensed parameter, control an end effector function, adapt the control of the end effector function over time based on the sensed parameter, and limit the adaptation of the control of the end effector function.

[0938] Example 20: The surgical system of Example 19, wherein limiting the adaptation of the control of the end effector function comprises limiting the adaptation of the control of the end effector function to a range of adaptability.

[0939] While several forms have been illustrated and described, it is not the intention of Applicant to restrict or limit the scope of the appended claims to such detail. Numerous modifications, variations, changes, substitutions, combinations, and equivalents to those forms may be implemented and will occur to those skilled in the art without departing from the scope of the present disclosure. Moreover, the structure of each element associated with the described forms can be alternatively described as a means for providing the function performed by the element. Also, where materials are disclosed for certain components, other materials may be used. It is therefore to be understood that the foregoing description and the appended claims are intended to cover all such modifications, combinations, and variations as falling within the scope of the disclosed forms. The appended claims are intended to cover all such modifications, variations, changes, substitutions, modifications, and equivalents.

[0940] The foregoing detailed description has set forth various forms of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples contain one or more functions and/or operations, it will be understood by those within the art that each function and/or operation within such block diagrams, flowcharts, and/or examples can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or virtually any combination thereof. Those skilled in the art will recognize that some aspects of the forms disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and or firmware would be well within the skill of one of skill in the art in light of this disclosure. In addition, those skilled in the art will appreciate that the mechanisms of the subject matter described herein are capable of being distributed as one or more program products in a variety of forms, and that an illustrative form of the subject matter described herein applies regardless of the particular type of signal bearing medium used to actually carry out the distribution.

[0941] Instructions used to program logic to perform various disclosed aspects can be stored within a memory in the system, such as dynamic random access memory (DRAM), cache, flash memory, or other storage. Furthermore, the instructions can be distributed via a network or by way of other computer readable media. Thus a machine-readable medium may include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computer), but is not limited to, floppy diskettes, optical disks, compact disc, read-only memory (CD-ROMs), and magneto-optical disks, read-only memory (ROMs), random access memory (RAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only

memory (EEPROM), magnetic or optical cards, flash memory, or a tangible, machine-readable storage used in the transmission of information over the Internet via electrical, optical, acoustical or other forms of propagated signals (e.g., carrier waves, infrared signals, digital signals, etc.). Accordingly, the non-transitory computer-readable medium includes any type of tangible machine-readable medium suitable for storing or transmitting electronic instructions or information in a form readable by a machine (e.g., a computer).

[0942] As used in any aspect herein, the term “control circuit” may refer to, for example, hardwired circuitry, programmable circuitry (e.g., a computer processor including one or more individual instruction processing cores, processing unit, processor, microcontroller, microcontroller unit, controller, digital signal processor (DSP), programmable logic device (PLD), programmable logic array (PLA), or field programmable gate array (FPGA)), state machine circuitry, firmware that stores instructions executed by programmable circuitry, and any combination thereof. The control circuit may, collectively or individually, be embodied as circuitry that forms part of a larger system, for example, an integrated circuit (IC), an application-specific integrated circuit (ASIC), a system on-chip (SoC), desktop computers, laptop computers, tablet computers, servers, smart phones, etc. Accordingly, as used herein “control circuit” includes, but is not limited to, electrical circuitry having at least one discrete electrical circuit, electrical circuitry having at least one integrated circuit, electrical circuitry having at least one application specific integrated circuit, electrical circuitry forming a general purpose computing device configured by a computer program (e.g., a general purpose computer configured by a computer program which at least partially carries out processes and/or devices described herein, or a microprocessor configured by a computer program which at least partially carries out processes and/or devices described herein), electrical circuitry forming a memory device (e.g., forms of random access memory), and/or electrical circuitry forming a communications device (e.g., a modem, communications switch, or optical-electrical equipment). Those having skill in the art will recognize that the subject matter described herein may be implemented in an analog or digital fashion or some combination thereof.

[0943] As used in any aspect herein, the term “logic” may refer to an app, software, firmware and/or circuitry configured to perform any of the aforementioned operations. Software may be embodied as a software package, code, instructions, instruction sets and/or data recorded on non-transitory computer readable storage medium. Firmware may be embodied as code, instructions or instruction sets and/or data that are hard-coded (e.g., nonvolatile) in memory devices.

[0944] As used in any aspect herein, the terms “component,” “system,” “module” and the like can refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution.

[0945] As used in any aspect herein, an “algorithm” refers to a self-consistent sequence of steps leading to a desired result, where a “step” refers to a manipulation of physical quantities and/or logic states which may, though need not necessarily, take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It is common usage to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like. These and similar terms may be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities and/or states.

[0946] A network may include a packet switched network. The communication devices may be capable of communicating with each other using a selected packet switched network communications protocol. One example communications protocol may include an Ethernet communications protocol which may be capable permitting communication using a Transmission Control Protocol/Internet Protocol (TCP/IP). The Ethernet protocol may comply or be compatible with the Ethernet standard published by the Institute of Electrical and Electronics Engineers (IEEE) titled “IEEE 802.3 Standard”, published in December, 2008 and/or later versions of this standard. Alternatively or additionally, the communication devices may be capable of communicating with each other using an X.25 communications protocol. The X.25 communications protocol may

comply or be compatible with a standard promulgated by the International Telecommunication Union-Telecommunication Standardization Sector (ITU-T). Alternatively or additionally, the communication devices may be capable of communicating with each other using a frame relay communications protocol. The frame relay communications protocol may comply or be compatible with a standard promulgated by Consultative Committee for International Telegraph and Telephone (CCITT) and/or the American National Standards Institute (ANSI). Alternatively or additionally, the transceivers may be capable of communicating with each other using an Asynchronous Transfer Mode (ATM) communications protocol. The ATM communications protocol may comply or be compatible with an ATM standard published by the ATM Forum titled “ATM-MPLS Network Interworking 2.0” published August 2001, and/or later versions of this standard. Of course, different and/or after-developed connection-oriented network communication protocols are equally contemplated herein.

[0947] Unless specifically stated otherwise as apparent from the foregoing disclosure, it is appreciated that, throughout the foregoing disclosure, discussions using terms such as “processing,” “computing,” “calculating,” “determining,” “displaying,” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

[0948] One or more components may be referred to herein as “configured to,” “configurable to,” “operable/operative to,” “adapted/adaptable,” “able to,” “conformable/conformed to,” etc. Those skilled in the art will recognize that “configured to” can generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

[0949] The terms “proximal” and “distal” are used herein with reference to a clinician manipulating the handle portion of the surgical instrument. The term “proximal” refers to the portion closest to the clinician and the term “distal” refers to the portion located away from the clinician. It will be further appreciated that, for convenience and clarity, spatial terms such as “vertical”, “horizontal”, “up”, and “down” may be used herein with respect to the drawings. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and/or absolute.

[0950] Those skilled in the art will recognize that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

[0951] In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers,

typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase “A or B” will be typically understood to include the possibilities of “A” or “B” or “A and B.”

[0952] With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flow diagrams are presented in a sequence(s), it should be understood that the various operations may be performed in other orders than those which are illustrated, or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like “responsive to,” “related to,” or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

[0953] It is worthy to note that any reference to “one aspect,” “an aspect,” “an exemplification,” “one exemplification,” and the like means that a particular feature, structure, or characteristic described in connection with the aspect is included in at least one aspect. Thus, appearances of the phrases “in one aspect,” “in an aspect,” “in an exemplification,” and “in one exemplification” in various places throughout the specification are not necessarily all referring to the same aspect. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner in one or more aspects.

[0954] Any patent application, patent, non-patent publication, or other disclosure material referred to in this specification and/or listed in any Application Data Sheet is incorporated by reference herein, to the extent that the incorporated materials is not inconsistent herewith. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing definitions, statements, or other disclosure material set forth herein will only be incorporated to the extent that no conflict arises between that incorporated material and the existing disclosure material.

[0955] In summary, numerous benefits have been described which result from employing the concepts described herein. The foregoing description of the one or more forms has been presented for purposes of illustration and description. It is not intended to be exhaustive or limiting to the precise form disclosed. Modifications or variations are possible in light of the above teachings. The one or more forms were chosen and described in order to illustrate principles and practical application to thereby enable one of ordinary skill in the art to utilize the various forms and with various modifications as are suited to the particular use contemplated. It is intended that the claims submitted herewith define the overall scope.

Claims

- 1.** A method for adaptive control of surgical network control and interaction, the surgical network comprising a surgical feedback system, the surgical feedback system comprising a surgical instrument, a data source, and a surgical hub configured to communicably couple to the data source and the surgical instrument, the surgical hub comprising a control circuit, method comprising: receiving, by the control circuit, information related to devices communicatively coupled to the surgical network; and adaptively controlling, by the control circuit, the surgical network based on the received information.
- 2.** The method of claim 1, further comprising transmuting, by the control circuit, to a display coupled to the surgical hub, the received information based on presence of an actionable aspect of a task.
- 3.** The method of claim 1, further comprising: automatically scaling, aligning, and organizing data received by the surgical hub, by the control circuit, based on predefined parameters within the surgical network before transmission of the information; and adaptively controlling, by the control circuit, the surgical network based on the received information.
- 4.** The method of claim 1, further comprising: validating, by the control circuit, the received information; authenticating, by the control circuit, a source and integrity of the information; and adaptively controlling, by the control circuit, the surgical network based on surgical network, instrument, and cloud responses based on validation of the received information and authentication of the source and the integrity of the information.
- 5.** The method of claim 1, further comprising: interpreting, the control circuit, the received information to based on at least one function of at least one device including at least one data source not originating within the device; and transmitting, by the control circuit, to a display coupled to the surgical hub, the interpreted information.
- 6.** The method of claim 1, further comprising modifying, by the control circuit, the surgical network or a device communicatively coupled to the surgical network based on machine learning analysis of performance and outcomes recorded by the surgical network over more than one surgical procedure.
- 7.** The method of claim 1, further comprising, adjusting, by the control circuit, control programs of devices communicatively coupled to the surgical network based on stratified contextual data in addition to the received information.
- 8.** The method of claim 1, further comprising, adjusting, by the control circuit, a response of the surgical network to a sensed parameter.
- 9.** The method of claim 1, further comprising, adjusting, by the control circuit, a response of the surgical network to an event based on a second pre-existing sensed step, situation, or parameter, or combinations thereof.
- 10.** The method of claim 1, further comprising detecting, by the control circuit, security threat to the surgical network.
- 11.** The method of claim 10, further comprising, escalating, by the control circuit, security responses to the security threat based on increasing severity levels of the security threat.
- 12.** A method for adaptive feedback and control of a surgical system, the surgical system comprising a surgical instrument, a data source, and a surgical hub configured to communicably couple to the data source and the surgical instrument, the surgical hub comprising a control circuit, the method comprising: receiving, by the control circuit, information related to devices communicatively coupled to the surgical system; and adaptively adjusting, by the control circuit, an operating parameter of a device communicatively coupled the surgical system based on the received communicated recommendation.
- 13.** The method of claim 12, further comprising, updating, by the control circuit, a control program of a device communicatively coupled the surgical system based on the received communicated recommendation.

- 14.** The method of claim 12, further comprising, analyzing, by the control circuit, the received information data using a supervised learning technique.
- 15.** The method of claim 12, further comprising, analyzing, by the control circuit, the received information data using an unsupervised learning technique.
- 16.** A method for adaptively controlling a surgical network based on validating data purportedly generated in a surgical procedure, the surgical network comprising a medical hub, at least one remote server communicatively coupled to the medical hub, and a medical instrument communicatively coupled to the medical hub, the system is configured to access the data, validate the data to determine if the data is validly generated by the surgical procedure, determine that the data contains at least one flaw or error, and improve data integrity by preventing the at least one flaw or error from being integrated into a larger dataset associated with the at least one remote server, the method comprising: receiving, by the server, information related to a surgical procedure from a device communicatively coupled to the surgical network; validating, by the server, the received information; and adaptively adjusting, by the server, the surgical network based on the received information.
- 17.** The method of claim 16, further comprising, determining, by the server, the presence of a sequential trend or pattern in the received information that is common to surgical procedures.
- 18.** The method of claim 16, further comprising, identifying, by the server, an encrypted validation key associated with a device communicatively coupled to the surgical network.
- 19.** The method of claim 16, further comprising, analyzing, by the server, the received information to determine the presence of a sequential trend or pattern in the received information that is unique to a type of surgical procedure that purportedly occurred when the received information was generated.
- 20.** The method of claim 16, further comprising, receiving, by the server, a report of a malicious actor from another medical hub.
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